

*Supporting Information for*  
*Direct, Convergent  $\alpha$ -Amidation of  $\beta$ -Keto Esters via Nickelocene Catalysis*

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## Materials and Methods

Unless otherwise stated, reactions were performed in flame-dried glassware under an argon or nitrogen atmosphere using dry, deoxygenated solvents. Solvents were dried by passage through an activated alumina column under argon.<sup>1</sup> Reaction progress was monitored by thin-layer chromatography (TLC) or Agilent 1290 UHPLC-MS. TLC was performed using E. Merck silica gel 60 F254 precoated glass plates (0.25 mm) and visualized by UV fluorescence quenching, *p*-anisaldehyde, or KMnO<sub>4</sub> staining. Silicycle SiliaFlash® P60 Academic Silica gel (particle size 40–63  $\mu$ m) was used for silica gel flash chromatography. <sup>1</sup>H NMR spectra were recorded on Bruker 400 MHz spectrometers and are reported relative to residual CHCl<sub>3</sub> ( $\delta$  7.26 ppm). <sup>13</sup>C NMR spectra were recorded on a Bruker 400 MHz spectrometer (100 MHz) and are reported relative to residual CDCl<sub>3</sub> ( $\delta$  77.16 ppm). <sup>19</sup>F NMR spectra were recorded on a Varian Mercury 300 MHz spectrometer (282 MHz). Data for <sup>1</sup>H NMR are reported as follows: chemical shift ( $\delta$  ppm) (multiplicity, coupling constant (Hz), integration). Multiplicities are reported as follows: s = singlet, d = doublet, t = triplet, q = quartet, p = pentet, sept = septuplet, m = multiplet, br s = broad singlet. Data for <sup>13</sup>C NMR, and <sup>19</sup>F NMR are reported in terms of chemical shifts ( $\delta$  ppm). IR spectra were obtained by use of a Perkin Elmer Spectrum BXII spectrometer using thin films deposited on NaCl plates and reported in frequency of absorption (cm<sup>-1</sup>). High resolution mass spectra (HRMS) were obtained from the Caltech Mass Spectral Facility using either a JEOL JMS-T2000 AccuTOF GC-Alpha time-of-flight mass spectrometer using Field Desorption (FD) ionization (ions detected are M<sup>+</sup>), an Agilent 6200 Series TOF with an Agilent G1978A Multimode source in mixed ionization mode (MM: ESI-APCI), or an LC-MS using a Waters LCT Premier XE Electrospray (ES) TOF mass spectrometer interfaced with a Waters UPLC chromatography system (negative ion mode, ions detected are M<sup>-</sup>). The Caltech Chemistry Division Mass Spectrometry laboratory acknowledges DOW Chemical Company (DOW Next Generation Instrumentation Grant) and the NSF CRIF program for providing funds that enabled the purchase of this instrumentation.

Reagents were purchased from commercial sources and used as received unless otherwise stated. 1*R*,2*S*,5*R*)-5-methyl-2-(2-phenylpropan-2-yl)cyclohexyl 2-methyl-3-oxobutanoate (**1f**)<sup>2</sup> *N*-(benzoyloxy)benzamide (**2a**),<sup>3</sup> *N*-(benzoyloxy)-4-methylbenzamide (**2b**),<sup>3</sup> *N*-(benzoyloxy)-4-methoxybenzamide (**2c**),<sup>3</sup> methyl 4-((benzoyloxy)carbamoyl)benzoate (**2d**),<sup>3</sup> *N*-(benzoyloxy)-4-chlorobenzamide (**2e**),<sup>3</sup> *N*-(benzoyloxy)-4-(trifluoromethyl)benzamide (**2f**),<sup>3</sup> *N*-

(benzoyloxy)cyclopent-1-ene-1-carboxamide (**2h**),<sup>3</sup> *N*-(benzoyloxy)cyclopropanecarboxamide (**2k**),<sup>3</sup> *tert*-butyl (3-((benzoyloxy)carbamoyl)bicyclo[1.1.1]pentan-1-yl)carbamate (**2m**)<sup>3</sup> [Ni(Cp)(NHC-*i*Pr)(Cl)] (**4a**),<sup>3</sup> [Ni(Cp)(IPr)(Cl)] (**4b**),<sup>3</sup> and 1,3-diisopropyl-1,3-dihydro-2*H*-imidazole-2-thione (**S1**)<sup>4</sup> were prepared according to literature procedures.

### List of Abbreviations

TLC – thin-layer chromatography, EtOAc – ethyl acetate, THF – tetrahydrofuran, KHMDS – potassium hexamethyldisilazane, AcOH – acetic acid, CH<sub>2</sub>Cl<sub>2</sub> – dichloromethane, DMF – dimethylformamide, TFA – trifluoroacetic acid

**Additional Optimization Data****Table S1.** Evaluation of additional solvents in amidation reaction.<sup>a</sup>

CC(=O)C(C)C(=O)OCC + CC(=O)Nc1ccccc1
 $\xrightarrow[\text{solvent, 30 } ^\circ\text{C, 4 h}]{\begin{array}{l} [\text{Ni}] \text{ 4b (10 mol\%)} \\ \text{LiO}^t\text{Bu (2.0 equiv)} \\ \text{PhSiH}_3 \text{ (1.0 equiv)} \end{array}}$ 
CC(=O)C(C)(C(=O)OCC)Nc1ccccc1

**1a** (2.2 equiv)                      **2a**                      **3a**

| entry | solvent           | yield (%) |
|-------|-------------------|-----------|
| 1     | THF               | 33        |
| 2     | TBME              | 38        |
| 3     | MeCN              | 5         |
| 4     | DMF               | 19        |
| 5     | CpME              | <1        |
| 6     | DME               | <1        |
| 7     | 2-MeTHF           | 38        |
| 8     | Et <sub>2</sub> O | 40        |

<sup>a</sup>Isolated yields.**Table S2.** Evaluation of bases in amidation reaction.<sup>a</sup>

CC(=O)C(C)C(=O)OCC + CC(=O)Nc1ccccc1
 $\xrightarrow[\text{1,4-dioxane, 30 } ^\circ\text{C, 4 h}]{\begin{array}{l} [\text{Ni}] \text{ 4b (10 mol\%)} \\ \text{base (2.0 equiv)} \\ \text{PhSiH}_3 \text{ (1.0 equiv)} \end{array}}$ 
CC(=O)C(C)(C(=O)OCC)Nc1ccccc1

**1a** (2.0 equiv)                      **2a**                      **3a**

| entry | base                | yield (%) |
|-------|---------------------|-----------|
| 1     | LiO <sup>t</sup> Bu | 77        |
| 2     | LiO <sup>i</sup> Pr | —         |
| 3     | LiOMe               | —         |
| 4     | LiOAc               | <1        |
| 5     | LiHMDS              | 48        |
| 6     | NaO <sup>t</sup> Bu | 2         |
| 7     | KO <sup>t</sup> Bu  | 5         |
| 8     | CsOAc               | 5         |
| 9     | TBD                 | 29        |
| 10    | MTBD                | —         |
| 11    | DBU                 | —         |
| 12    | DABCO               | —         |
| 13    | pempidine           | —         |
| 14    | DMAP                | —         |

<sup>a</sup>Isolated yields.

**Table S3.** Evaluation of additives.<sup>a</sup>

| entry | additive                         | yield (%) |
|-------|----------------------------------|-----------|
| 1     | PhSiH <sub>3</sub>               | 73        |
| 2     | Li(Et) <sub>3</sub> BH           | 14        |
| 3     | HBPIn                            | <1        |
| 4     | HBCat                            | 7         |
| 5     | PMHS                             | —         |
| 6     | TMDS                             | 14        |
| 7     | Ph <sub>2</sub> SiH <sub>2</sub> | <1        |
| 8     | Ph <sub>3</sub> SiH              | 7         |

<sup>a</sup>Isolated yields.**Table S4.** Evaluation of phenylsilane equivalents<sup>a</sup>

| entry | silane equivalents | yield (%) |
|-------|--------------------|-----------|
| 1     | 10 mol%            | 2         |
| 2     | 25 mol%            | 2         |
| 3     | 50 mol%            | 47        |
| 4     | 1.0 equiv          | 69        |

<sup>a</sup>Isolated yields**Table S5.** Evaluation of Lewis acids<sup>a</sup>

| entry | Lewis Acid                         | yield (%) |
|-------|------------------------------------|-----------|
| 1     | MgCl <sub>2</sub>                  | —         |
| 2     | Ca(O <sup>i</sup> Pr) <sub>2</sub> | —         |
| 3     | AlCl <sub>3</sub>                  | —         |
| 4     | La(O <sup>i</sup> Pr) <sub>3</sub> | —         |

<sup>a</sup>Isolated yields

**Table S6.** Investigation of alternate leaving groups on hydroxamate<sup>a</sup>

| entry | Hydroxamate Leaving Group (R) | yield (%) |
|-------|-------------------------------|-----------|
| 1     | Bz                            | 82        |
| 2     | Piv                           | 71        |
| 3     | Ac                            | 47        |
| 4     | Me                            | —         |
| 5     | Boc                           | —         |
| 6     | H                             | —         |

<sup>a</sup>Isolated yields**Table S7.** Control studies.<sup>a</sup>

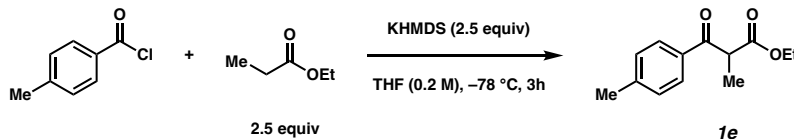
| entry | deviation              | yield (%) |
|-------|------------------------|-----------|
| 1     | no NiCp <sub>2</sub>   | —         |
| 2     | no PhSiH <sub>3</sub>  | —         |
| 3     | no LiO <sup>t</sup> Bu | —         |
| 4     | 10 uL air              | 77        |
| 5     | open to air            | <1        |

<sup>a</sup>Isolated yields.**Scheme S1.** Substrate scope with nickel half-sandwich complex<sup>a</sup>

| Substrate | [Ni(Cp)(Cl)(IPr)] (10 mol%): | NiCp <sub>2</sub> (5 mol%): |
|-----------|------------------------------|-----------------------------|
|           | 44% (24 h)                   | 34% (2 h)                   |
|           | 65% (72 h)                   | 64% (16 h)                  |
|           | 72% (72 h)                   | 71% (16 h)                  |

<sup>a</sup>Isolated yields.

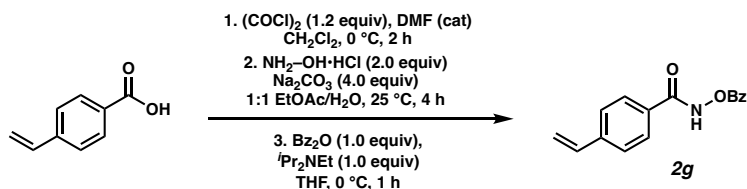
### Preparation of $\beta$ -Keto Esters



#### Ethyl 2-methyl-3-oxo-3-(*p*-tolyl)propanoate (**1e**)

To a flame-dried 50 mL round bottom flask was added KHMDS (10 mmol, 2.5 equiv) followed by THF (20 mL, 0.2 M). After cooling to  $-78^\circ\text{C}$ , the ethyl propionate (10 mmol, 2.5 equiv) was added dropwise. After stirring for 10 minutes, the 4-methylbenzoyl chloride (4.0 mmol, 1.0 equiv) was added dropwise. The reaction was then allowed to stir for 3 hours at the same temperature, after which the reaction mixture was quenched with sat.  $\text{NH}_4\text{Cl}$ , then extracted three times with ethyl acetate. The combined organics were washed with brine, dried with sodium sulfate, then concentrated. Purification by column chromatography (1% AcOH/2.5% EtOAc in  $\text{CH}_2\text{Cl}_2$ ) provided **1e** (239.9 mg, 27% yield) as a colorless oil;  $^1\text{H NMR}$  (400 MHz,  $\text{CDCl}_3$ )  $\delta$  7.90 (d,  $J = 8.3$  Hz, 2H), 7.29 (d,  $J = 7.7$  Hz, 2H), 4.37 (q,  $J = 7.1$  Hz, 1H), 4.16 (qd,  $J = 7.1, 0.7$  Hz, 2H), 2.44 (s, 3H), 1.50 (d,  $J = 7.2$  Hz, 3H), 1.19 (t,  $J = 7.1$  Hz, 3H); all characterization data match those reported in the literature.<sup>5</sup>

### Preparation of Hydroxamic Esters



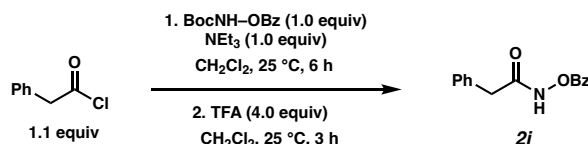
#### N-(benzoyloxy)-4-vinylbenzamide (**2g**)

4-vinylbenzoic acid (3.0 mmol, 1.0 equiv) was added to a flame-dried 50 mL round bottom flask with a stir bar, followed by dichloromethane (12 mL, 0.25 M) and DMF (2 drops, cat.). The solution was cooled to  $0^\circ\text{C}$ , then oxalyl chloride (3.6 mmol, 1.2 equiv) was added dropwise before the reaction was allowed to stir for 2 hours. The crude acid chloride was concentrated under reduced pressure.

Hydroxylamine hydrochloride (6.0 mmol, 2.0 equiv) and sodium carbonate (12.0 mmol, 4.0 equiv) were added to a 25 mL round bottom flask, followed by water and ethyl acetate (1:1, 12 mL, 0.2 M). This biphasic solution was then cooled to  $0^\circ\text{C}$ , then crude acid chloride was added dropwise, at which point the solution was allowed to warm slowly over 2 hours. The organic layer was then

separated, and the aqueous layer was extracted three times with ethyl acetate. The combined organics were washed with brine, dried with sodium sulfate, then concentrated under reduced pressure, and used without further purification.

To a flame-dried 50 mL round bottom flask was added the crude hydroxamic acid followed by THF (30 mL, 0.1 M) and diisopropyl ethyl amine (3.0 mmol, 1.0 equiv). The mixture was cooled to 0 °C, then benzoic anhydride (3.0 mmol, 1.0 equiv) was added portionwise over 10 minutes. After stirring for 1 hour, the reaction was quenched with sat. NaHCO<sub>3</sub> then extracted three times with dichloromethane. The organic layers were combined, washed with brine, dried with sodium sulfate, then concentrated. Purification by trituration (30% EtOAc/hexanes) yielded **2g** (341.1 mg, 43% yield) as a white solid; <sup>1</sup>H NMR (400 MHz, CDCl<sub>3</sub>) δ 9.70 (bs, 1H), 8.26 – 8.06 (m, 2H), 7.89 – 7.81 (m, 2H), 7.74 – 7.61 (m, 1H), 7.52 (ddd, *J* = 9.0, 4.7, 2.1 Hz, 4H), 6.76 (ddd, *J* = 17.6, 10.9, 2.0 Hz, 1H), 5.88 (dd, *J* = 17.6, 0.7 Hz, 1H), 5.41 (dt, *J* = 10.8, 1.1 Hz, 1H); <sup>13</sup>C NMR (100 MHz, CDCl<sub>3</sub>) δ 166.4, 165.5, 142.1, 135.9, 134.4, 130.5, 130.2, 129.9, 128.9, 128.0, 126.7, 116.9; IR (Thin Film, NaCl) 3159, 2974, 2358, 1756, 1654, 1531, 1252, 1050, 702 cm<sup>-1</sup>; HRMS (ESI<sup>-</sup>) *m/z* calc'd for C<sub>16</sub>H<sub>12</sub>NO<sub>3</sub> [M-H]<sup>-</sup>: 266.0823, found 266.0821.

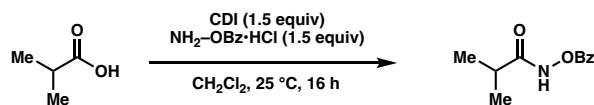


#### N-(benzoyloxy)-2-phenylacetamide (**2i**)

To flame-dried 100 mL round bottom flask equipped with a stir bar was added *tert*-butyl (benzoyloxy)carbamate (3.0 mmol, 1.0 equiv), followed by dichloromethane (30 mL, 0.1 M) and triethylamine (3.0 mmol, 1.0 equiv). 2-phenylacetyl chloride (3.3 mmol, 1.1 equiv) was then added dropwise. After stirring for 6 hours, the reaction was quenched with a saturated solution of NaHCO<sub>3</sub>, extracted three times with dichloromethane, washed with brine, then dried with Na<sub>2</sub>SO<sub>4</sub>, filtered, then concentrated under reduced pressure.

The crude material was dissolved in dichloromethane (20 mL, 0.15 M), then TFA (12.0 mmol, 4.0 equiv) was added. After stirring for 3 hours, the reaction mixture was quenched with sat. NaHCO<sub>3</sub> then extracted three times with dichloromethane. The combined organics were washed with brine, then dried with Na<sub>2</sub>SO<sub>4</sub>, filtered, and concentrated under reduced pressure. Purification by column chromatography (10–30% EtOAc in hexanes) provided **2i** (3.4157g, 45% yield) as a white solid; <sup>1</sup>H NMR (400 MHz, CDCl<sub>3</sub>) δ 8.95 (bs, 1H), 8.11 – 8.04 (m, 2H), 7.67 – 7.55 (m, 1H), 7.47 (t, *J* =

7.8 Hz, 2H), 7.43–7.31(m, 5H), 3.74 (s, 2H);  $^{13}\text{C}$  NMR (100 MHz,  $\text{CDCl}_3$ )  $\delta$  168.9, 164.9, 134.4, 133.2, 130.1, 129.5, 129.1, 128.8, 127.7, 126.6, 40.8; IR (Thin Film, NaCl) 3167, 2960, 2354, 1769, 1681, 1515, 1240, 1175, 1053, 994, 834, 701  $\text{cm}^{-1}$ ; HRMS (ESI $^{+}$ )  $m/z$  calc'd for  $\text{C}_{15}\text{H}_{14}\text{NO}_3$   $[\text{M}+\text{H}]^{+}$ : 256.0968, found 256.0968.

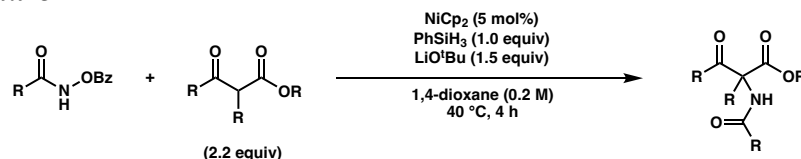


**N-(benzyloxy)isobutyramide (2j)**

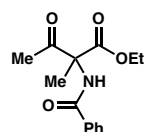
To a flame-dried 50 mL round bottom flask was added carbonyl diimidazole (6.0 mmol, 1.5 equiv) followed by THF (12 mL, 0.33 M) then isobutyric acid (4.0 mmol, 1.0 equiv). After stirring for 1 hour, the *O*-benzoylhydroxylamine hydrochloride (6.0 mmol, 1.5 equiv) was added, at which point the reaction was stirred for 16 hours. The reaction mixture was then quenched with sat.  $\text{NaHCO}_3$  and extracted three times with dichloromethane. The combined organics were washed with brine, dried with  $\text{Na}_2\text{SO}_4$ , then concentrated. Purification by column chromatography (0–50% EtOAc in hexanes) followed by recrystallization from hot EtOAc provided **2j** (296.6 mg, 36% yield) as a white solid;  $^1\text{H}$  NMR (400 MHz,  $\text{CDCl}_3$ )  $\delta$  9.13 (bs, 1H), 8.20 – 8.06 (m, 2H), 7.68 – 7.57 (m, 1H), 7.55 – 7.43 (m, 2H), 2.70 – 2.46 (m, 1H), 1.27 (d,  $J$  = 6.9 Hz, 6H);  $^{13}\text{C}$  NMR (100 MHz,  $\text{CDCl}_3$ )  $\delta$  175.6, 165.2, 134.3, 130.1, 128.8, 126.8, 32.8, 19.3; IR (Thin Film, NaCl) 3152, 2974, 1767, 1667, 1452, 1242, 1096, 1047, 708, 682  $\text{cm}^{-1}$ ; HRMS (ESI $^{+}$ )  $m/z$  calc'd for  $\text{C}_{11}\text{H}_{14}\text{NO}_3$   $[\text{M}+\text{H}]^{+}$ : 208.0968, found 208.0964.

## Ni-Catalyzed Amidation Reactions

### General Procedure A

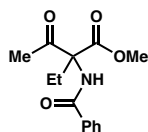


In a nitrogen-filled glovebox, a 1-dram vial was charged with hydroxamic ester (0.1 mmol, 1.0 equiv). 1,4-dioxane was added (0.1 mL) followed sequentially by a stock solution of nickelocene (0.005 mmol, 5 mol%) and phenylsilane (0.1 mmol, 1.0 equiv) in 1,4-dioxane (0.1 mL), then a stock solution of  $\beta$ -keto ester (0.22 mmol, 2.2 equiv) and lithium *tert*-butoxide (0.15 mmol, 1.5 equiv) in 1,4-dioxane (0.3 mL). The latter solution was prepared by first preparing a solution of the  $\beta$ -keto ester in 1,4-dioxane, then adding this solution to solid lithium *tert*-butoxide and allowing to dissolve. The reaction vial was then sealed with a Teflon-lined cap, removed from the glovebox, and heated at 40 °C in a heating mantle for 4 h unless otherwise noted. Reactions were typically green at the beginning of the reaction and brown upon completion. Following reaction completion, the reaction was quenched with sat.  $\text{NaHCO}_3$  (1 mL), extracted five times with ethyl acetate, dried with a plug of  $\text{Na}_2\text{SO}_4$ , concentrated, and purified by preparative TLC.



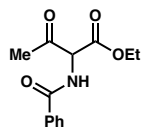
### Ethyl 2-benzamido-2-methyl-3-oxobutanoate (**3a**)

Prepared according to general procedure A using *N*-(benzoyloxy)benzamide (**2a**) (0.1 mmol) and ethyl 2-methyl-3-oxobutanoate (**1a**) (0.22 mmol). Purification by preparatory TLC (5%  $\text{NEt}_3$ / 20%  $\text{CH}_2\text{Cl}_2$ / 75% hexanes) provided **3a** (21.6 mg, 82% yield) as a colorless oil;  $^1\text{H NMR}$  (400 MHz,  $\text{CDCl}_3$ )  $\delta$  7.87 – 7.80 (m, 2H), 7.75 (s, 1H), 7.58 – 7.50 (m, 1H), 7.50 – 7.42 (m, 2H), 4.36 – 4.17 (m, 2H), 2.26 (s, 3H), 1.84 (s, 3H), 1.26 (t,  $J$  = 7.2 Hz, 3H);  $^{13}\text{C NMR}$  (100 MHz,  $\text{CDCl}_3$ )  $\delta$  200.5, 169.3, 166.0, 133.6, 132.1, 128.8, 127.2, 68.8, 62.9, 24.2, 20.3, 14.1; **IR** (Thin Film, NaCl) 3405, 3061, 2985, 2940, 1725, 1665, 1602, 1579, 1508, 1479, 1442, 1369, 1290, 1262, 1201, 1119, 1019, 918, 858, 802, 778, 715, 692, 632  $\text{cm}^{-1}$ ; **HRMS** (ESI+)  $m/z$  calc'd for  $\text{C}_{14}\text{H}_{18}\text{NO}_4$   $[\text{M}+\text{H}]^+$ : 264.1230, found 264.1237.



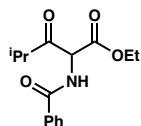
**Methyl 2-benzamido-2-ethyl-3-oxobutanoate (3b)**

Prepared according to general procedure A using *N*-(benzoyloxy)benzamide (**2a**) (0.1 mmol) and ethyl methyl 2-ethyl-3-oxobutanoate (**1b**) (0.22 mmol). Purification by preparatory TLC (5% NEt<sub>3</sub>/ 20% CH<sub>2</sub>Cl<sub>2</sub>/75% hexanes) provided **3b** (21.6 mg, 82% yield) as a colorless oil; <sup>1</sup>H NMR (400 MHz, CDCl<sub>3</sub>) δ 7.89 – 7.80 (m, 2H), 7.68 (s, 1H), 7.58 – 7.50 (m, 1H), 7.46 (ddt, *J* = 8.2, 6.6, 1.3 Hz, 2H), 3.79 (s, 3H), 2.62 (dq, *J* = 15.2, 7.6 Hz, 1H), 2.43 (dq, *J* = 14.9, 7.5 Hz, 1H), 2.24 (s, 3H), 0.76 (t, *J* = 7.6 Hz, 3H); <sup>13</sup>C NMR (100 MHz, CDCl<sub>3</sub>) δ 200.2, 169.4, 166.0, 133.5, 132.2, 128.8, 127.3, 72.9, 53.6, 25.0, 24.5, 7.8; IR (Thin Film, NaCl) 3408, 2961, 1727, 1666, 1581, 1506, 1481, 1248, 1185, 1118, 802, 716 cm<sup>-1</sup>; HRMS (ESI+) *m/z* calc'd for C<sub>14</sub>H<sub>18</sub>NO<sub>4</sub> [M+H]<sup>+</sup>: 264.1230, found 264.1235.



**Ethyl 2-benzamido-3-oxobutanoate (3c)**

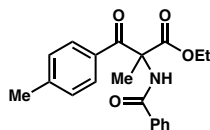
Prepared according to general procedure A using *N*-(benzoyloxy)benzamide (**2a**) (0.1 mmol) and ethyl 3-oxobutanoate (**1c**) (0.22 mmol). Purification by preparatory TLC (5% NEt<sub>3</sub>/ 20% CH<sub>2</sub>Cl<sub>2</sub>/75% hexanes) provided **3c** (11.9 mg, 48% yield) as a colorless oil; <sup>1</sup>H NMR (400 MHz, CDCl<sub>3</sub>) δ 7.91 – 7.78 (m, 2H), 7.60 – 7.49 (m, 1H), 7.46 (tt, *J* = 7.5, 0.7 Hz, 2H), 7.31 (d, *J* = 6.2 Hz, 1H), 5.43 (d, *J* = 6.3 Hz, 1H), 4.39 – 4.24 (m, 2H), 2.46 (s, 3H), 1.33 (t, *J* = 7.1 Hz, 3H); all characterization data match those reported in the literature.<sup>6</sup>



**Ethyl 2-benzamido-4-methyl-3-oxopentanoate (3d)**

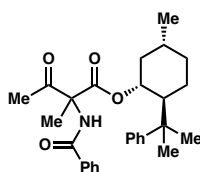
Prepared according to general procedure A using *N*-(benzoyloxy)benzamide (**2a**) (0.1 mmol) and ethyl 4-methyl-3-oxopentanoate (**1d**) (0.22 mmol). Purification by preparatory TLC (5% NEt<sub>3</sub>/ 20% CH<sub>2</sub>Cl<sub>2</sub>/75% hexanes) provided **3d** (11.9 mg, 63% yield) as a colorless oil; <sup>1</sup>H NMR (400 MHz, CDCl<sub>3</sub>) δ 7.89 – 7.81 (m, 2H), 7.60 – 7.49 (m, 1H), 7.46 (t, *J* = 7.7 Hz, 2H), 7.31 (d, *J* = 6.5

Hz, 1H), 5.60 (dd,  $J = 6.7, 0.9$  Hz, 1H), 4.35 – 4.35 (m, 2H), 3.16 (hept,  $J = 6.9$  Hz, 1H), 1.32 (td,  $J = 7.1, 0.9$  Hz, 3H), 1.28 – 1.24 (m, 3H), 1.16 (dd,  $J = 6.7, 0.9$  Hz, 3H); all characterization data match those reported in the literature.<sup>7</sup>



***Ethyl 2-benzamido-2-methyl-3-oxo-3-(p-tolyl)propanoate (3e)***

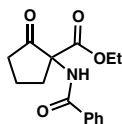
Prepared according to general procedure A using *N*-(benzoyloxy)benzamide (**2a**) (0.1 mmol) and ethyl 3-oxo-3-(p-tolyl)propanoate (**1e**) (0.22 mmol). Purification by preparatory TLC (5% NEt<sub>3</sub>/20% CH<sub>2</sub>Cl<sub>2</sub>/75% hexanes) provided **3e** (30.9 mg, 91% yield) as a colorless oil; <sup>1</sup>H NMR (400 MHz, CDCl<sub>3</sub>) δ 8.04 (s, 1H), 7.95 – 7.87 (m, 2H), 7.81 – 7.74 (m, 2H), 7.56 – 7.46 (m, 1H), 7.46 – 7.38 (m, 2H), 7.19 (dt,  $J = 8.5, 1.6$  Hz, 2H), 4.25 (q,  $J = 7.1$  Hz, 2H), 2.36 (s, 3H), 1.98 (s, 3H), 1.15 (t,  $J = 7.1$  Hz, 3H); <sup>13</sup>C NMR (100 MHz, CDCl<sub>3</sub>) δ 191.8, 170.9, 165.5, 144.4, 133.8, 131.9, 131.5, 129.4, 129.1, 128.7, 127.2, 66.9, 63.0, 22.3, 21.8, 14.0; IR (Thin Film, NaCl) 3390, 2931, 1733, 1697, 1664, 1606, 1506, 1477, 1362, 1287, 1227, 1121, 1019, 712 cm<sup>-1</sup>; HRMS (ESI+) *m/z* calc'd for C<sub>20</sub>H<sub>22</sub>NO<sub>4</sub> [M+H]<sup>+</sup>: 340.1543, found 340.1556.



***(1R,2S,5R)-5-methyl-2-(2-phenylpropan-2-yl)cyclohexyl 2-methyl-3-oxobutanoate (3f)***

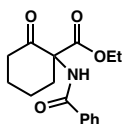
Prepared according to general procedure A using *N*-(benzoyloxy)benzamide (**2a**) (0.1 mmol) and (1*R*,2*S*,5*R*)-5-methyl-2-(2-phenylpropan-2-yl)cyclohexyl 2-methyl-3-oxobutanoate (**1f**) (0.22 mmol). Purification by preparatory TLC (5% NEt<sub>3</sub>/20% CH<sub>2</sub>Cl<sub>2</sub>/75% hexanes) provided **3f** (27.4 mg, 61% yield, 2.4:1 d.r) as a colorless oil; <sup>1</sup>H NMR (400 MHz, CDCl<sub>3</sub>) δ 7.90 – 7.77 (m, 2H), 7.60 – 7.52 (m, 2H), 7.51 – 7.44 (m, 2H), 7.35 – 7.26 (m, 2H), 7.25 – 7.21 (m, 2H), 7.12 (ddd,  $J = 8.4, 5.0, 3.3$  Hz, 1H), 4.98 (td,  $J = 10.7, 4.3$  Hz, 1H), 2.27 (s, 3H), 2.07 – 2.02 (m, 1H), 1.96 (ddd,  $J = 12.2, 10.5, 3.6$  Hz, 1H), 1.83 (s, 3H), 1.61 – 1.42 (m, 2H), 1.41 – 1.30 (m, 3H), 1.24 (s, 3H), 1.21 (s, 3H), 1.10 – 1.01 (m, 1H), 0.90 (d,  $J = 6.4$  Hz, 3H); <sup>13</sup>C NMR (100 MHz, CDCl<sub>3</sub>) δ 200.1, 168.6, 166.1, 150.4, 133.6, 132.1, 128.7, 128.3, 127.4, 125.8, 125.6, 78.9, 69.0, 50.3, 41.2,

40.3, 34.4, 31.5, 29.7, 27.5, 25.2, 23.9, 21.9, 20.1; **IR** (Thin Film, NaCl) 3408, 2956, 2923, 1718, 1666, 1601, 1580, 1506, 1479, 1442, 1370, 1287, 1262, 1203, 1117, 1094, 977, 914, 765, 702, 634  $\text{cm}^{-1}$ ; **HRMS** (ESI+)  $m/z$  calc'd for  $\text{C}_{28}\text{H}_{36}\text{NO}_4$   $[\text{M}+\text{H}]^+$ : 450.2639, found 450.2647.



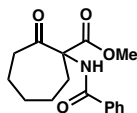
***Ethyl 1-benzamido-2-oxocyclopentane-1-carboxylate (3g)***

Prepared according to general procedure A using *N*-(benzoyloxy)benzamide (**2a**) (0.1 mmol) and ethyl 2-oxocyclopentane-1-carboxylate (**1g**) (0.22 mmol). Purification by preparatory TLC (5%  $\text{NEt}_3$ / 20%  $\text{CH}_2\text{Cl}_2$  /75% hexanes) provided **3g** (19.5 mg, 71% yield) as a white solid;  **$^1\text{H}$  NMR** (400 MHz,  $\text{CDCl}_3$ )  $\delta$  7.84 – 7.77 (m, 2H), 7.52 (tt,  $J$  = 6.5, 1.3 Hz, 1H), 7.47 – 7.40 (m, 3H), 4.29 (qd,  $J$  = 7.1, 3.3 Hz, 2H), 2.86 (dt,  $J$  = 19.2, 9.8 Hz, 1H), 2.68 (ddd,  $J$  = 13.6, 10.4, 8.5 Hz, 1H), 2.62 – 2.46 (m, 2H), 2.36 – 2.18 (m, 2H), 1.31 (t,  $J$  = 7.1 Hz, 3H); all characterization data match those reported in the literature.<sup>8</sup>



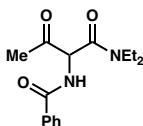
***Ethyl 1-benzamido-2-oxocyclohexane-1-carboxylate (3h)***

Prepared according to general procedure A using *N*-(benzoyloxy)benzamide (**2a**) (0.1 mmol) and ethyl 2-oxocyclohexane-1-carboxylate (**1h**) (0.22 mmol). Purification by preparatory TLC (5%  $\text{NEt}_3$ / 20%  $\text{CH}_2\text{Cl}_2$  /75% hexanes) provided **3h** (19.7 mg, 68% yield) as a white solid;  **$^1\text{H}$  NMR** (400 MHz,  $\text{CDCl}_3$ )  $\delta$  7.85 – 7.78 (m, 3H), 7.59 – 7.47 (m, 1H), 7.47 – 7.38 (m, 2H), 4.33 – 4.11 (m, 2H), 3.30 (ddt,  $J$  = 13.8, 5.0, 2.5 Hz, 1H), 2.79 – 2.45 (m, 2H), 2.45 – 2.23 (m, 1H), 2.26 – 2.08 (m, 1H), 1.88 – 1.67 (m, 3H), 1.21 (t,  $J$  = 7.1 Hz, 3H);  **$^{13}\text{C}$  NMR** (100 MHz,  $\text{CDCl}_3$ )  $\delta$  203.2, 168.0, 166.3, 133.8, 132.0, 128.7, 127.3, 69.0, 62.4, 39.4, 37.7, 27.7, 22.3, 14.1; **IR** (Thin Film, NaCl) 3388, 2941, 2867, 1722, 1666, 1602, 1579, 1525, 1478, 1446, 1305, 1244, 1205, 1134, 1090, 1023, 862, 713  $\text{cm}^{-1}$ ; **HRMS** (ESI+)  $m/z$  calc'd for  $\text{C}_{16}\text{H}_{20}\text{NO}_4$   $[\text{M}+\text{H}]^+$ : 290.1387, found 290.1394.



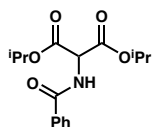
***Ethyl 1-benzamido-2-oxocycloheptane-1-carboxylate (3i)***

Prepared according to general procedure A using *N*-(benzoyloxy)benzamide (**2a**) (0.1 mmol) and ethyl 2-oxocycloheptane-1-carboxylate (**1i**) (0.22 mmol). Purification by preparatory TLC (5% NEt<sub>3</sub>/ 20% CH<sub>2</sub>Cl<sub>2</sub>/75% hexanes) provided **3i** (12.7 mg, 42% yield) as a white solid; <sup>1</sup>H NMR (400 MHz, CDCl<sub>3</sub>) δ 7.83 – 7.72 (m, 3H), 7.53 – 7.43 (m, 1H), 7.42 – 7.35 (m, 2H), 3.65 (s, 3H), 2.85 (ddd, *J* = 12.7, 11.1, 3.0 Hz, 1H), 2.78 – 2.52 (m, 2H), 2.41 (ddd, *J* = 10.9, 6.2, 2.2 Hz, 1H), 2.04 – 1.89 (m, 1H), 1.87 – 1.71 (m, 2H), 1.55 – 1.35 (m, 2H), 1.08 – 0.92 (m, 1H); <sup>13</sup>C NMR (100 MHz, CDCl<sub>3</sub>) δ 204.7, 168.3, 166.5, 133.8, 132.0, 128.7, 127.3, 73.1, 53.6, 38.5, 31.0, 30.3, 27.8, 23.4; IR (Thin Film, NaCl) 3399, 2931, 2858, 1745, 1715, 1666, 1601, 1580, 1506, 1477, 1323, 1266, 1219, 1153, 1134, 1071, 1045, 1006, 938, 914, 891, 832, 802, 779, 761, 714 cm<sup>-1</sup>; HRMS (ESI<sup>+</sup>) *m/z* calc'd for C<sub>16</sub>H<sub>20</sub>NO<sub>4</sub> [M+H]<sup>+</sup>: 290.1387, found 290.1397.



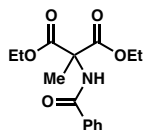
***N-(1-(diethylamino)-1,3-dioxobutan-2-yl)benzamide (3j)***

Prepared according to general procedure A using *N*-(benzoyloxy)benzamide (**2a**) (0.1 mmol) and *N,N*-diethyl-3-oxobutanamide (**1j**) (0.22 mmol). Purification by preparatory TLC (5% NEt<sub>3</sub>/ 20% CH<sub>2</sub>Cl<sub>2</sub>/75% hexanes) provided **3j** (15.5 mg, 56% yield) as a white solid; <sup>1</sup>H NMR (400 MHz, CDCl<sub>3</sub>) δ 7.92 – 7.84 (m, 2H), 7.69 (d, *J* = 6.8 Hz, 1H), 7.59 – 7.51 (m, 1H), 7.50 – 7.42 (m, 2H), 5.56 (d, *J* = 6.8 Hz, 1H), 3.72 (dq, *J* = 14.5, 7.2 Hz, 1H), 3.62 – 3.46 (m, 2H), 3.43 – 3.25 (m, 1H), 2.23 (s, 3H), 1.26 (t, *J* = 7.2 Hz, 3H), 1.16 (t, *J* = 7.1 Hz, 3H); <sup>13</sup>C NMR (100 MHz, CDCl<sub>3</sub>) δ 202.2, 167.0, 165.2, 133.2, 132.3, 128.8, 127.4, 61.1, 42.6, 41.1, 26.3, 14.3, 12.8; IR (Thin Film, NaCl) 3307, 2974, 1717, 1631, 1462, 1357, 777 cm<sup>-1</sup>; HRMS (ESI<sup>+</sup>) *m/z* calc'd for C<sub>15</sub>H<sub>21</sub>N<sub>2</sub>O<sub>3</sub> [M+H]<sup>+</sup>: 277.1547, found 277.1547.



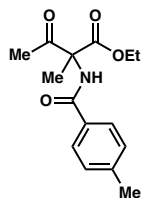
### Diisopropyl 2-benzamidomalonate (**3k**)

Prepared according to general procedure A using *N*-(benzoyloxy)benzamide (**2a**) (0.1 mmol) and diisopropyl malonate (**1k**) (0.22 mmol). Purification by preparatory TLC (5% NEt<sub>3</sub>/ 20% CH<sub>2</sub>Cl<sub>2</sub> /75% hexanes) provided **3k** (13.2 mg, 43% yield) as a white solid; <sup>1</sup>H NMR (400 MHz, CDCl<sub>3</sub>) δ 7.87 – 7.83 (m, 2H), 7.54 (ddt, *J* = 8.2, 6.5, 1.3 Hz, 1H), 7.46 (ddd, *J* = 8.2, 6.9, 1.2 Hz, 2H), 7.11 (d, *J* = 6.8 Hz, 1H), 5.27 (d, *J* = 6.8 Hz, 1H), 5.14 (hept, *J* = 6.3 Hz, 2H), 1.33 (d, *J* = 6.3 Hz, 6H), 1.28 (d, *J* = 6.3 Hz, 6H); <sup>13</sup>C NMR (100 MHz, CDCl<sub>3</sub>) δ 167.0, 166.1, 133.3, 132.2, 128.8, 127.4, 70.8, 57.3, 21.8, 21.7; IR (Thin Film, NaCl) 3362, 2981, 1739, 1666, 1518, 1482, 1225, 1166, 1100, 682 cm<sup>-1</sup>; HRMS (ESI+) *m/z* calc'd for C<sub>16</sub>H<sub>22</sub>NO<sub>5</sub> [M+H]<sup>+</sup>: 308.1492, found 308.1491.



### Diethyl 2-benzamido-2-methylmalonate (**3l**)

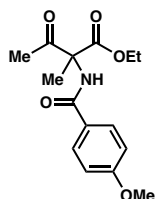
Prepared according to general procedure A using *N*-(benzoyloxy)benzamide (**2a**) (0.1 mmol) and diethyl 2-methylmalonate (**1l**) (0.22 mmol). Purification by preparatory TLC (5% NEt<sub>3</sub>/ 20% CH<sub>2</sub>Cl<sub>2</sub> /75% hexanes) provided **3l** (13.5 mg, 46% yield) as a white solid; <sup>1</sup>H NMR (400 MHz, CDCl<sub>3</sub>) δ 7.80 – 7.72 (m, 2H), 7.52 – 7.43 (m, 2H), 7.42 – 7.35 (m, 2H), 4.22 (qd, *J* = 7.1, 2.2 Hz, 4H), 1.80 (s, 3H), 1.20 (t, *J* = 7.1 Hz, 6H); <sup>13</sup>C NMR (100 MHz, CDCl<sub>3</sub>) δ 169.0, 166.1, 133.7, 132.0, 128.8, 127.3, 63.3, 62.8, 21.1, 14.1; IR (Thin Film, NaCl) 3418, 2977, 1739, 1667, 1511, 1482, 1442, 1373, 1295, 1219, 1108, 1020, 714 cm<sup>-1</sup>; HRMS (ESI+) *m/z* calc'd for C<sub>15</sub>H<sub>20</sub>NO<sub>5</sub> [M+H]<sup>+</sup>: 294.1336, found 294.1336.



### Ethyl 2-methyl-2-(4-methylbenzamido)-3-oxobutanoate (**6a**)

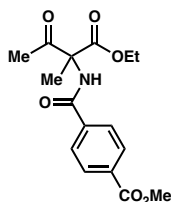
Prepared according to general procedure A using *N*-(benzoyloxy)-4-methylbenzamide (0.1 mmol) (**2b**) and ethyl 2-methyl-3-oxobutanoate (**1a**) (0.22 mmol). Purification by preparatory TLC (5%

NEt<sub>3</sub>/ 20% CH<sub>2</sub>Cl<sub>2</sub> /75% hexanes) provided **6a** (22.7 mg, 82% yield) as a colorless oil; **<sup>1</sup>H NMR** (400 MHz, CDCl<sub>3</sub>) δ 7.76 – 7.68 (m, 3H), 7.26 – 7.22 (m, 2H), 4.34 – 4.17 (m, 2H), 2.40 (s, 3H), 2.25 (s, 3H), 1.83 (s, 3H), 1.25 (t, *J* = 7.1 Hz, 3H); **<sup>13</sup>C NMR** (100 MHz, CDCl<sub>3</sub>) δ 200.5, 169.4, 166.0, 142.6, 130.7, 129.4, 127.3, 68.7, 62.8, 24.3, 21.6, 20.3 14.1; **IR** (Thin Film, NaCl) 3408, 2985, 1726, 1663, 1612, 1522, 1485, 1443, 1370, 1285, 1192, 1119, 1020, 857, 752 cm<sup>-1</sup>; **HRMS** (ESI+) *m/z* calc'd for C<sub>15</sub>H<sub>20</sub>NO<sub>4</sub> [M+H]<sup>+</sup>: 278.1387, found 278.1396.



***Ethyl 2-(4-methoxybenzamido)-2-methyl-3-oxobutanoate (6b)***

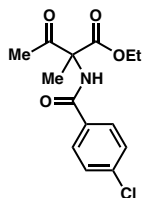
Prepared according to general procedure A using *N*-(benzoyloxy)-4-methoxybenzamide (**2c**) (0.1 mmol) and ethyl 2-methyl-3-oxobutanoate (**1a**) (0.22 mmol). Purification by preparatory TLC (5% NEt<sub>3</sub>/ 20% CH<sub>2</sub>Cl<sub>2</sub> /75% hexanes) provided **6b** (20.8 mg, 71% yield) as a colorless oil; **<sup>1</sup>H NMR** (400 MHz, CDCl<sub>3</sub>) δ 7.84 – 7.75 (m, 2H), 7.65 (s, 1H), 6.98 – 6.90 (m, 2H), 4.34 – 4.18 (m, 2H), 3.85 (s, 3H), 2.24 (s, 3H), 1.82 (s, 3H), 1.25 (t, *J* = 7.2 Hz, 3H); **<sup>13</sup>C NMR** (100 MHz, CDCl<sub>3</sub>) δ 200.6, 169.4, 165.6, 162.7, 129.1, 125.8, 114.0, 68.7, 62.8, 55.6, 24.3, 20.4, 14.1; **IR** (Thin Film, NaCl) 3410, 2983, 2940, 2841, 2358, 2047, 1729, 1662, 1607, 1577, 1526, 1486, 1452, 1370, 1258, 1186, 1120, 1026, 918, 876, 847, 814, 768, 698, 663, 635 cm<sup>-1</sup>; **HRMS** (ESI+) *m/z* calc'd for C<sub>15</sub>H<sub>20</sub>NO<sub>5</sub> [M+H]<sup>+</sup>: 294.1336, found 294.1348.



***Methyl 4-((1-ethoxy-2-methyl-1,3-dioxobutan-2-yl)carbamoyl)benzoate (6c)***

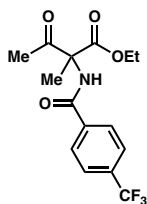
Prepared according to general procedure A using methyl 4-((benzoyloxy)carbamoyl)benzoate (**2d**) (0.1 mmol) and ethyl 2-methyl-3-oxobutanoate (**1a**) (0.22 mmol). Purification by preparatory TLC (5% NEt<sub>3</sub>/ 20% CH<sub>2</sub>Cl<sub>2</sub> /75% hexanes) provided **6c** (26.3 mg, 82% yield) as a colorless oil; **<sup>1</sup>H NMR** (400 MHz, CDCl<sub>3</sub>) δ 8.11 (d, *J* = 8.5 Hz, 2H), 7.87 (d, *J* = 8.5 Hz, 2H), 7.80 (s, 1H), 4.35 – 4.18 (m, 2H), 3.94 (s, 3H), 2.26 (s, 3H), 1.84 (s, 3H), 1.24 (t, *J* = 7.1 Hz, 3H); **<sup>13</sup>C NMR** (100

MHz, CDCl<sub>3</sub>)  $\delta$  200.3, 169.0, 166.3, 165.2, 137.4, 133.2, 130.0, 127.3, 68.8, 62.9, 52.6, 24.2, 20.2, 14.1; **IR** (Thin Film, NaCl) 3398, 2992, 1728, 1668, 1518, 1486, 1442, 1372, 1280, 1194, 1108, 1018, 965, 917, 872, 826, 731, 544 cm<sup>-1</sup>; **HRMS** (ESI+)  $m/z$  calc'd for C<sub>16</sub>H<sub>20</sub>NO<sub>6</sub> [M+H]<sup>+</sup>: 322.1285, found 322.1294.



**Ethyl 2-(4-chlorobenzamido)-2-methyl-3-oxobutanoate (6d)**

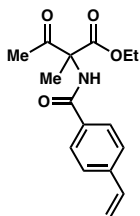
Prepared according to general procedure A using *N*-(benzoyloxy)-4-chlorobenzamide (**2e**) (0.1 mmol) and ethyl 2-methyl-3-oxobutanoate (**1a**) (0.22 mmol). Purification by preparatory TLC (5% NEt<sub>3</sub>/ 20% CH<sub>2</sub>Cl<sub>2</sub>/75% hexanes) provided **6d** (17.6 mg, 59% yield) as a colorless oil; **<sup>1</sup>H NMR** (400 MHz, CDCl<sub>3</sub>)  $\delta$  7.79 – 7.73 (m, 2H), 7.72 (s, 1H), 7.47 – 7.39 (m, 2H), 4.33 – 4.18 (m, 2H), 2.26 (s, 3H), 1.83 (s, 3H), 1.25 (t,  $J$  = 7.1 Hz, 3H); **<sup>13</sup>C NMR** (100 MHz, CDCl<sub>3</sub>)  $\delta$  200.4, 169.1, 165.0, 138.4, 132.0, 129.1, 128.7, 68.8, 62.9, 24.2, 20.3, 14.1; **IR** (Thin Film, NaCl) 3403, 2984, 1725, 1665, 1595, 1509, 1477, 1365, 1275, 1192, 1093 cm<sup>-1</sup>; **HRMS** (ESI+)  $m/z$  calc'd for C<sub>14</sub>H<sub>17</sub>ClNO<sub>4</sub> [M+H]<sup>+</sup>: 298.0841, found 298.0847.



**Ethyl 2-methyl-3-oxo-2-(4-(trifluoromethyl)benzamido)butanoate (6e)**

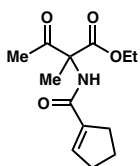
Prepared according to general procedure A using *N*-(benzoyloxy)-4-(trifluoromethyl)benzamide (**2f**) (0.1 mmol) and ethyl 2-methyl-3-oxobutanoate (**1a**) (0.22 mmol). Purification by preparatory TLC (5% NEt<sub>3</sub>/ 20% CH<sub>2</sub>Cl<sub>2</sub>/75% hexanes) provided **6e** (22.0 mg, 66% yield) as a colorless oil; **<sup>1</sup>H NMR** (400 MHz, CDCl<sub>3</sub>)  $\delta$  7.95 (ddt,  $J$  = 7.5, 1.6, 0.8 Hz, 2H), 7.80 (s, 1H), 7.76 – 7.68 (m, 2H), 4.36 – 4.18 (m, 2H), 2.27 (s, 3H), 1.86 (s, 3H), 1.25 (t,  $J$  = 7.1 Hz, 3H); **<sup>13</sup>C NMR** (100 MHz, CDCl<sub>3</sub>)  $\delta$  200.3, 168.9, 164.8, 133.8 (q,  $J$  = 32.9 Hz), 127.8, 125.9 (q,  $J$  = 3.7 Hz), 125.1, 122.4, 68.9, 63.0, 24.2, 20.2, 14.1; **<sup>19</sup>F NMR** (282 MHz, CDCl<sub>3</sub>)  $\delta$  -63.0; **IR** (Thin Film, NaCl) 3399,

2989, 1727, 1667, 1581, 1523, 1490, 1326, 1167, 1129, 1067, 1017, 859  $\text{cm}^{-1}$ ; **HRMS** (ESI+)  $m/z$  calc'd for  $\text{C}_{15}\text{H}_{17}\text{F}_3\text{NO}_4$   $[\text{M}+\text{H}]^+$ : 332.1104, found 332.1109.



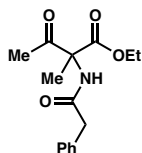
***Ethyl 2-methyl-3-oxo-2-(4-vinylbenzamido)butanoate (6f)***

Prepared according to general procedure A using *N*-(benzoyloxy)-4-vinylbenzamide (**2g**) (0.1 mmol) and ethyl 2-methyl-3-oxobutanoate (**1a**) (0.22 mmol). Purification by preparatory TLC (5%  $\text{NEt}_3$ / 20%  $\text{CH}_2\text{Cl}_2$  /75% hexanes) provided **6f** (11.9 mg, 41% yield) as a colorless oil;  **$^1\text{H}$  NMR** (400 MHz,  $\text{CDCl}_3$ )  $\delta$  7.80 (d,  $J$  = 8.4 Hz, 2H), 7.74 (s, 1H), 7.48 (d,  $J$  = 8.2 Hz, 2H), 6.75 (dd,  $J$  = 17.6, 10.9 Hz, 1H), 5.85 (dd,  $J$  = 17.6, 0.8 Hz, 1H), 5.37 (dd,  $J$  = 10.9, 0.8 Hz, 1H), 4.35 – 4.18 (m, 2H), 2.26 (s, 3H), 1.84 (s, 3H), 1.25 (t,  $J$  = 7.1 Hz, 3H);  **$^{13}\text{C}$  NMR** (100 MHz,  $\text{CDCl}_3$ )  $\delta$  200.4, 141.1, 135.9, 132.5, 127.5, 126.4, 116.2, 68.6, 62.7, 24.1, 20.2, 14.0; **IR** (Thin Film, NaCl) 3404, 2980, 1725, 1661, 1519, 1485, 1286, 1192, 1121, 1017, 917, 857  $\text{cm}^{-1}$ ; **HRMS** (ESI+)  $m/z$  calc'd for  $\text{C}_{16}\text{H}_{20}\text{NO}_4$   $[\text{M}+\text{H}]^+$ : 290.1387, found 290.1387.

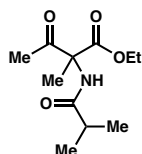


***Ethyl 2-(cyclopent-1-ene-1-carboxamido)-2-methyl-3-oxobutanoate (6g)***

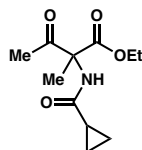
Prepared according to general procedure A using *N*-(benzoyloxy)cyclopent-1-ene-1-carboxamide (**2h**) (0.1 mmol) and ethyl 2-methyl-3-oxobutanoate (**1a**) (0.22 mmol). Purification by preparatory TLC (5%  $\text{NEt}_3$ / 20%  $\text{CH}_2\text{Cl}_2$  /75% hexanes) provided **6g** (8.4 mg, 33% yield) as a colorless oil;  **$^1\text{H}$  NMR** (400 MHz,  $\text{CDCl}_3$ )  $\delta$  7.18 (s, 1H), 6.63 – 6.57 (m, 1H), 4.24 (q,  $J$  = 7.1 Hz, 2H), 2.64 – 2.45 (m, 4H), 2.21 (s, 3H), 2.00 (p,  $J$  = 7.6 Hz, 2H), 1.75 (s, 3H), 1.25 (t,  $J$  = 7.1 Hz, 3H);  **$^{13}\text{C}$  NMR** (100 MHz,  $\text{CDCl}_3$ )  $\delta$  200.5, 169.3, 164.3, 139.7, 138.7, 68.3, 62.8, 33.4, 31.5, 24.3, 23.5, 20.3, 14.1; **IR** (Thin Film, NaCl) 3408, 2946, 1727, 1664, 1490, 1368, 1264, 1205, 1118, 1019  $\text{cm}^{-1}$ ; **HRMS** (ESI+)  $m/z$  calc'd for  $\text{C}_{13}\text{H}_{20}\text{NO}_4$   $[\text{M}+\text{H}]^+$ : 254.1387, found 254.1398.

**Ethyl 2-methyl-3-oxo-2-(2-phenylacetamido)butanoate (6h)**

Prepared according to general procedure A using *N*-(benzoyloxy)-2-phenylacetamide (**2i**) (0.1 mmol) and ethyl 2-methyl-3-oxobutanoate (**1a**) (0.22 mmol). Purification by preparatory TLC (5% NEt<sub>3</sub>/ 20% CH<sub>2</sub>Cl<sub>2</sub>/75% hexanes) provided **6h** (18.0 mg, 65% yield) as a colorless oil; **<sup>1</sup>H NMR** (400 MHz, CDCl<sub>3</sub>) δ 7.40 – 7.32 (m, 2H), 7.33 – 7.24 (m, 3H), 7.01 (s, 1H), 4.18 (qd, *J* = 7.1, 5.8 Hz, 2H), 3.58 (s, 2H), 2.13 (s, 3H), 1.69 (s, 3H), 1.19 (t, *J* = 7.2 Hz, 3H); **<sup>13</sup>C NMR** (100 MHz, CDCl<sub>3</sub>) δ 200.2, 170.0, 169.0, 134.6, 129.3, 129.1, 127.5, 68.5, 62.7, 43.6, 24.1, 20.2, 14.0; **IR** (Thin Film, NaCl) 3273, 2991, 1745, 1723, 1641, 1510, 1371, 1263, 1210, 1112, 1030, 965, 732 cm<sup>-1</sup>; **HRMS** (ESI+) *m/z* calc'd for C<sub>15</sub>H<sub>20</sub>NO<sub>4</sub> [M+H]<sup>+</sup>: 278.1387, found 278.1394.

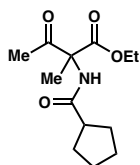
**Ethyl 2-isobutyramido-2-methyl-3-oxobutanoate (6i)**

Prepared according to general procedure A using *N*-(benzoyloxy)isobutyramide (**2j**) (0.1 mmol) and ethyl 2-methyl-3-oxobutanoate (**1a**) (0.22 mmol). Purification by preparatory TLC (5% NEt<sub>3</sub>/ 20% CH<sub>2</sub>Cl<sub>2</sub>/75% hexanes) provided **6i** (13.3 mg, 58% yield) as a colorless oil; **<sup>1</sup>H NMR** (400 MHz, CDCl<sub>3</sub>) δ 7.01 (s, 1H), 4.30 – 4.13 (m, 2H), 2.43 (hept, *J* = 6.9 Hz, 1H), 2.19 (s, 3H), 1.71 (s, 3H), 1.24 (t, *J* = 7.1 Hz, 3H), 1.15 (d, *J* = 6.9 Hz, 6H); **<sup>13</sup>C NMR** (100 MHz, CDCl<sub>3</sub>) δ 200.6, 175.9, 169.3, 68.3, 62.7, 35.3, 24.2, 20.4, 19.4, 19.4, 14.1; **IR** (Thin Film, NaCl) 3333, 2972, 1727, 1670, 1503, 1370, 1263, 1107, 1017, 954, 861, 826, 762, 678, 648 cm<sup>-1</sup>; **HRMS** (ESI+) *m/z* calc'd for C<sub>11</sub>H<sub>20</sub>NO<sub>4</sub> [M+H]<sup>+</sup>: 230.1387, found 230.1389.

**Ethyl 2-(cyclopropanecarboxamido)-2-methyl-3-oxobutanoate (6j)**

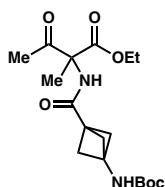
Prepared according to general procedure A using *N*-(benzoyloxy)cyclopropanecarboxamide (**2k**) (0.1 mmol) and ethyl 2-methyl-3-oxobutanoate (**1a**) (0.22 mmol). Purification by preparatory TLC

(5% NEt<sub>3</sub>/ 20% CH<sub>2</sub>Cl<sub>2</sub> /75% hexanes) provided **6j** (18.0 mg, 79% yield) as a colorless oil; <sup>1</sup>H NMR (400 MHz, CDCl<sub>3</sub>) δ 4.32 – 4.13 (m, 2H), 2.18, (s, 3H), 1.71 (s, 3H), 1.48 (tt, *J* = 7.9, 4.6 Hz, 1H), 1.24 (t, *J* = 7.1 Hz, 3H), 1.00 – 0.85 (m, 2H), 0.82 – 0.70 (m, 2H); <sup>13</sup>C NMR (100 MHz, CDCl<sub>3</sub>) δ 200.5, 172.5, 169.3, 68.5, 62.7, 24.3, 20.5, 14.6, 14.1, 7.5, 7.4; IR (Thin Film, NaCl) 3378, 2986, 2938, 1726, 1668, 1504, 1443, 1393, 1369, 1267, 1235, 1097, 1019, 968, 893, 859 cm<sup>-1</sup>; HRMS (ESI+) *m/z* calc'd for C<sub>11</sub>H<sub>18</sub>NO<sub>4</sub> [M+H]<sup>+</sup>: 228.1230, found 228.1231.



***Ethyl 2-(cyclopentanecarboxamido)-2-methyl-3-oxobutanoate (6k)***

Prepared according to general procedure A using *N*-(benzoyloxy)cyclopentanecarboxamide (**2l**) (0.1 mmol) and ethyl 2-methyl-3-oxobutanoate (**1a**) (0.22 mmol). Purification by preparatory TLC (5% NEt<sub>3</sub>/ 20% CH<sub>2</sub>Cl<sub>2</sub> /75% hexanes) provided **6k** (1.8 mg, 7% yield) as a colorless oil; <sup>1</sup>H NMR (400 MHz, CDCl<sub>3</sub>) δ 6.99 (s, 1H), 4.33 – 4.13 (m, 2H), 2.68 – 2.56 (m, 1H), 2.19 (s, 3H), 1.93 – 1.82 (m, 2H), 1.80 – 1.68 (m, 6H), 1.66 – 1.50 (m, 3H), 1.24 (t, *J* = 7.1 Hz, 3H); <sup>13</sup>C NMR (100 MHz, CDCl<sub>3</sub>) δ 200.6, 175.3, 169.4, 68.4, 62.7, 45.3, 30.4, 30.2, 26.1, 24.3, 20.5, 14.1; IR (Thin Film, NaCl) 3384, 2927, 1727, 1667, 1493, 1446, 1364, 1265, 1021, 803, 741 cm<sup>-1</sup>; HRMS (ESI+) *m/z* calc'd for C<sub>13</sub>H<sub>22</sub>NO<sub>4</sub> [M+H]<sup>+</sup>: 256.1543, found 256.1546.

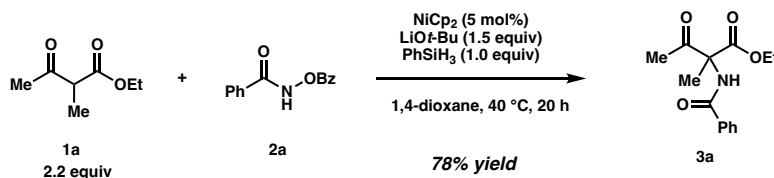


***Ethyl 2-(3-((tert-butoxycarbonyl)amino)bicyclo[1.1.1]pentane-1-carboxamido)-2-methyl-3-oxobutanoate (6l)***

Prepared according to general procedure A using *tert*-butyl (3-((benzoyloxy)carbamoyl)bicyclo[1.1.1]pentan-1-yl)carbamate (**2m**) (0.1 mmol) and ethyl 2-methyl-3-oxobutanoate (0.22 mmol). Purification by preparatory TLC (5% NEt<sub>3</sub>/ 20% CH<sub>2</sub>Cl<sub>2</sub> /75% hexanes) provided **6l** (8.5 mg, 23% yield) as a colorless oil; <sup>1</sup>H NMR (400 MHz, CDCl<sub>3</sub>) δ 7.06 (s, 1H), 4.31 – 4.12 (m, 2H), 2.28 (s, 6H), 2.19 (s, 3H), 1.72 (s, 3H), 1.45 (s, 9H), 1.23 (t, *J* =

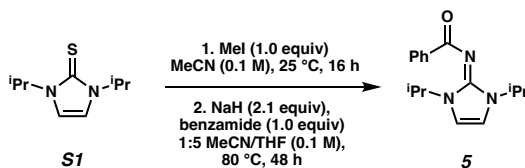
7.1 Hz, 3H);  $^{13}\text{C}$  NMR (100 MHz,  $\text{CDCl}_3$ )  $\delta$  200.2, 168.9, 168.4, 68.3, 62.8, 53.7, 45.3, 36.8,  $f/28.5$ , 24.2, 10.2, 14.1; IR (Thin Film, NaCl) 3339, 2981, 1723, 1667, 1521, 1489, 1369, 1251, 1177, 1014, 680  $\text{cm}^{-1}$ ; HRMS (ESI+)  $m/z$  calc'd for  $\text{C}_{18}\text{H}_{28}\text{N}_2\text{O}_6$   $[\text{M}+\text{H}]^+$ : 369.2020, found 369.2026.

### **Procedure for 1.0 mmol Scale Nickelocene-Catalyzed Amidation Reaction**



In a nitrogen-filled glovebox, a 2-dram vial was charged with hydroxamic ester (1.0 mmol, 1.0 equiv). 1,4-dioxane was added (1.0 mL) followed sequentially by a stock solution of nickelocene (0.05 mmol, 5 mol%) and phenylsilane (1.0 mmol, 1.0 equiv) in 1,4-dioxane (1.0 mL), then a stock solution of  $\beta$ -keto ester (2.2 mmol, 2.2 equiv) and lithium *tert*-butoxide (1.5 mmol, 1.5 equiv) in 1,4-dioxane (3.0 mL). The latter solution was prepared by first preparing a solution of the  $\beta$ -keto ester in 1,4-dioxane, then adding this solution to solid lithium *tert*-butoxide and allowing to dissolve. The reaction vial was then sealed with a Teflon-lined cap, removed from the glovebox, and stirred at 40  $^\circ\text{C}$  in a heating mantle for 20 h. Reaction was green at the beginning of the reaction and brown upon completion. Following reaction completion as determined by TLC and LCMS, the reaction was quenched with sat.  $\text{NaHCO}_3$  (5 mL), extracted five times with ethyl acetate, dried with over  $\text{Na}_2\text{SO}_4$ , concentrated, and purified by column chromatography (30% EtOAc/hexanes) to give the amidated product as a colorless oil (204.8 mg, 78% yield).

### **Synthesis of *N*-(1,3-diisopropyl-1,3-dihydro-2*H*-imidazol-2-ylidene)benzamide (5)**



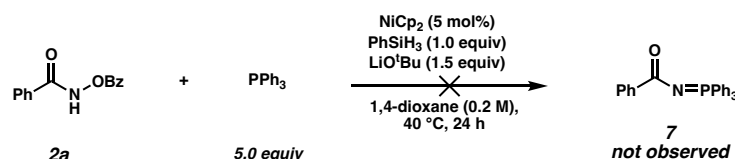
### ***N*-(1,3-diisopropyl-1,3-dihydro-2*H*-imidazol-2-ylidene)benzamide (5)**

To a 20 mL vial with a stir bar was added 1,3-diisopropyl-1,3-dihydro-2*H*-imidazole-2-thione (**S1**) (0.59 mmol, 1.0 equiv) as a solution in MeCN (6 mL, 0.1 M) followed by methyl iodide (0.59

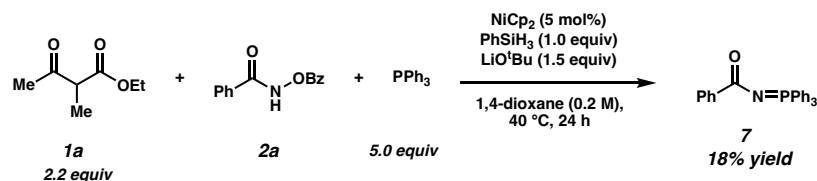
mmol, 1.0 equiv). After stirring for 16 h, the reaction mixture was concentrated under reduced pressure.

To a flame-dried 10 mL sealed tube with a stir bar was added a 60% weight dispersion of sodium hydride (1.23 mmol, 2.1 equiv) followed by THF (5.0 mL, 0.12 M). The crude reaction mixture from the previous step was dissolved in MeCN (1.0 mL, 6 M) and added to the sodium hydride dispersion. The mixture was then heated to 80 °C in an oil bath for 48 h, after which it was cooled, quenched with sat. NaHCO<sub>3</sub>, and extracted three times with ethyl acetate. The combined organics were then washed with brine, dried with Na<sub>2</sub>SO<sub>4</sub>, and concentrated under reduced pressure. Purification by column chromatography (5–10% MeOH/EtOAc) provided **5** (26.2 mg, 16% yield) as a white crystalline solid. <sup>1</sup>H NMR (400 MHz, CDCl<sub>3</sub>) δ 8.26–8.17 (m, 2H), 7.45–7.35 (m, 3H), 6.76 (s, 2H), 4.66 (hept, *J* = 6.7 Hz, 2H), 1.39 (d, *J* = 6.8 Hz, 12 H); <sup>13</sup>C NMR (100 MHz, CDCl<sub>3</sub>) δ 170.7, 150.6, 139.1, 130.1, 129.0, 127.8, 111.9, 47.9, 22.4; IR (Thin Film, NaCl) 3092, 2981, 2932, 1517, 1455, 1357, 1305, 1225, 877, 721 cm<sup>-1</sup>; HRMS (ESI+) *m/z* calc'd for C<sub>16</sub>H<sub>22</sub>N<sub>3</sub>O [M+H]<sup>+</sup>: 272.1757, found 272.1760.

### Procedures for Trapping Experiments

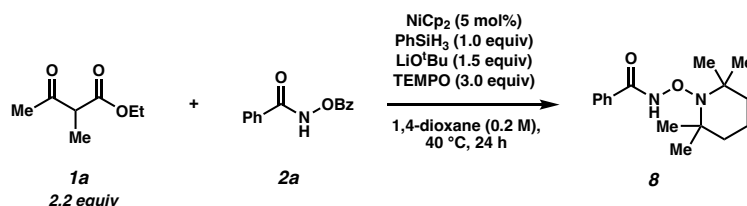


In a nitrogen-filled glovebox, a one-dram vial with a stir bar was charged with *N*-(benzoyloxy)benzamide (**2a**) (0.1 mmol, 1.0 equiv) and triphenylphosphine (0.5 mmol, 5.0 equiv). 1,4-dioxane was added (0.1 mL) followed sequentially by a stock solution of nickelocene (0.005 mmol, 5 mol%) and phenylsilane (0.1 mmol, 1.0 equiv) in 1,4-dioxane (0.1 mL), then a stock solution of lithium *tert*-butoxide (0.15 mmol, 1.5 equiv) in 1,4-dioxane (0.3 mL). The reaction vial was then sealed with a Teflon-lined cap, removed from the glovebox, and stirred at 40 °C in a heating mantle for 4 h. No product was observed by either TLC or LC-MS.



In a nitrogen-filled glovebox, a one-dram vial with a stir bar was charged with *N*-(benzoyloxy)benzamide (**2a**) (0.1 mmol, 1.0 equiv) and triphenylphosphine (0.5 mmol, 5.0 equiv).

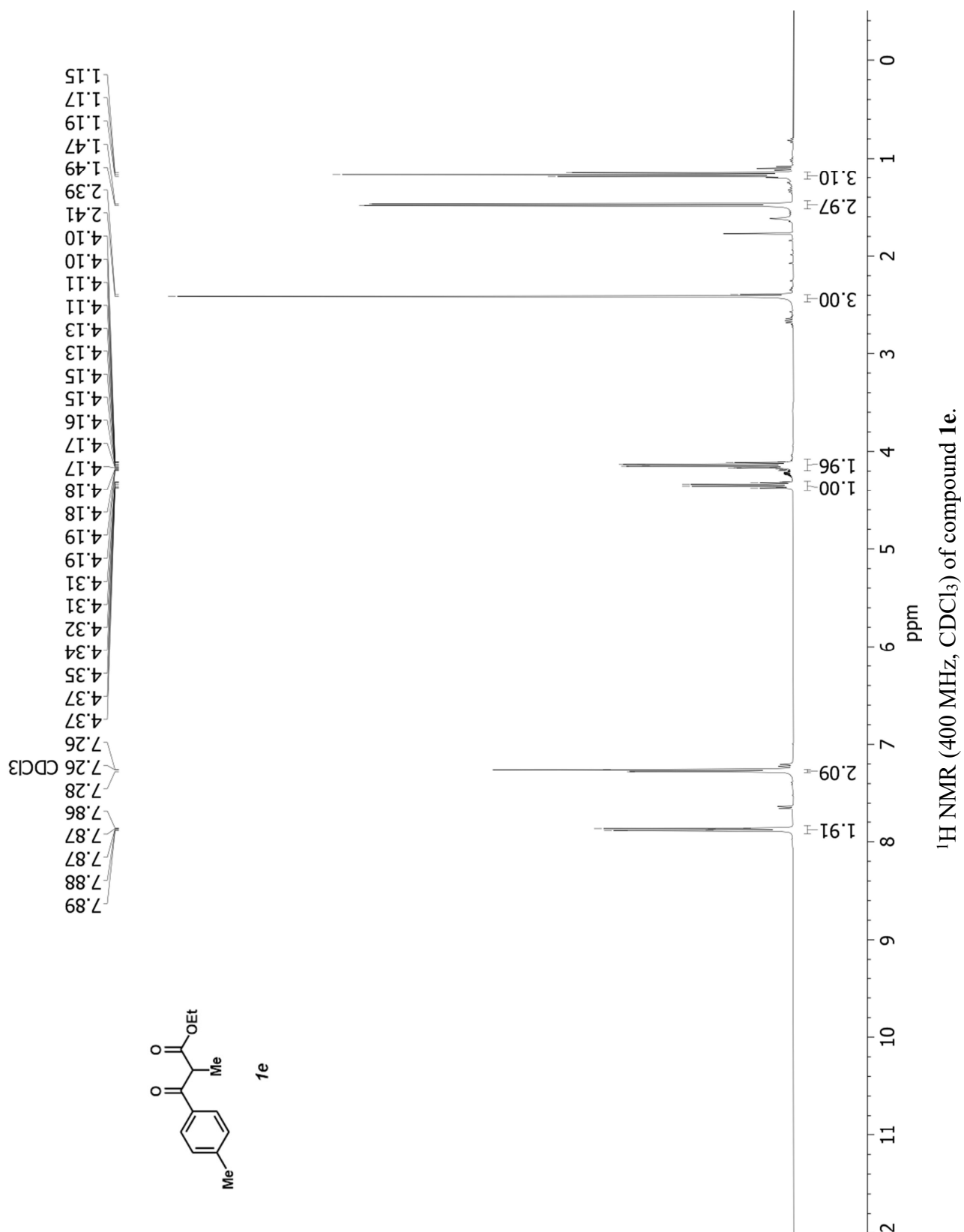
1,4-dioxane was added (0.1 mL) followed sequentially by a stock solution of nickelocene (0.005 mmol, 5 mol%) and phenylsilane (0.1 mmol, 1.0 equiv) in 1,4-dioxane (0.1 mL), then a stock solution of ethyl 2-methyl-3-oxobutanoate (**1a**) (0.22 mmol, 2.2 equiv) and lithium *tert*-butoxide (0.15 mmol, 1.5 equiv) in 1,4-dioxane (0.3 mL). The latter solution was prepared by first preparing a solution of **1a** in 1,4-dioxane, then adding this solution to solid lithium *tert*-butoxide and allowing to dissolve. The reaction vial was then sealed with a Teflon-lined cap, removed from the glovebox, and stirred at 40 °C in a heating mantle for 4 h, at which point the reaction was quenched with sat. NaHCO<sub>3</sub> (1 mL), extracted five times with ethyl acetate, dried with a plug of Na<sub>2</sub>SO<sub>4</sub> and concentrated. Purification by preparative TLC (35:60:5 hexanes/CH<sub>2</sub>Cl<sub>2</sub>/EtOAc) yielded **7** (7.0 mg, 18% yield) as a white solid; <sup>1</sup>H NMR (400 MHz, CDCl<sub>3</sub>) δ 8.39 – 8.31 (m, 2H), 7.90 – 7.79 (m, 6H), 7.60 – 7.53 (m, 3H), 7.53 – 7.35 (m, 9H); all characterization data matched those reported in the literature.<sup>9</sup>

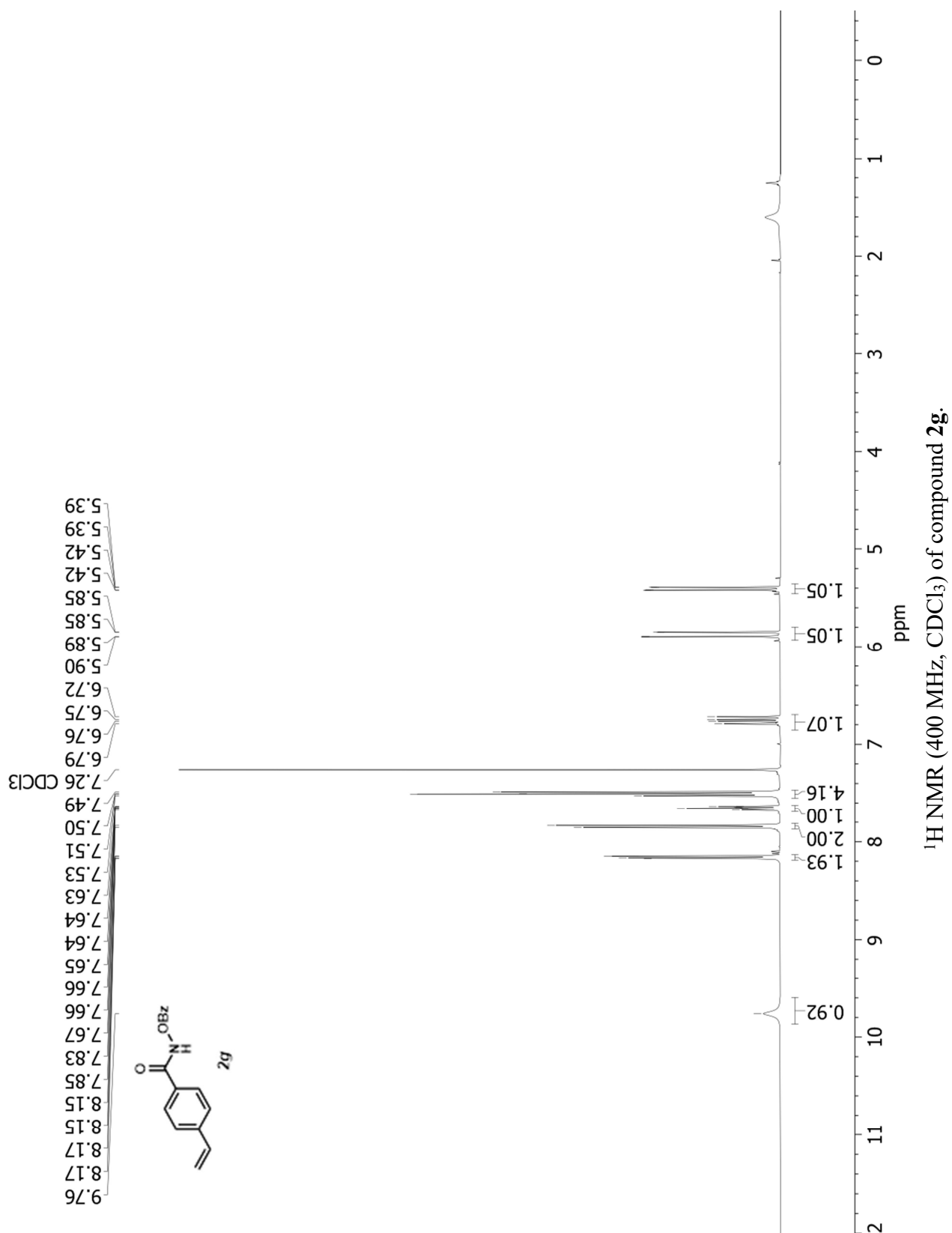


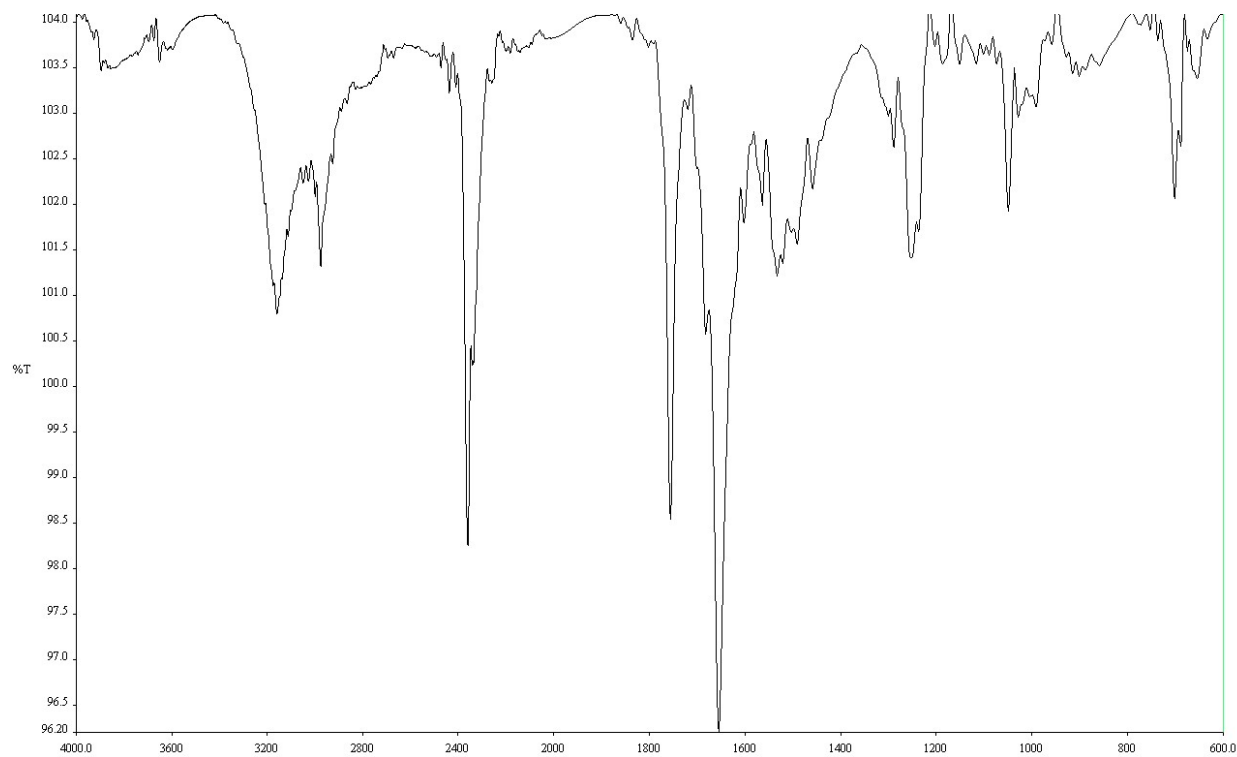
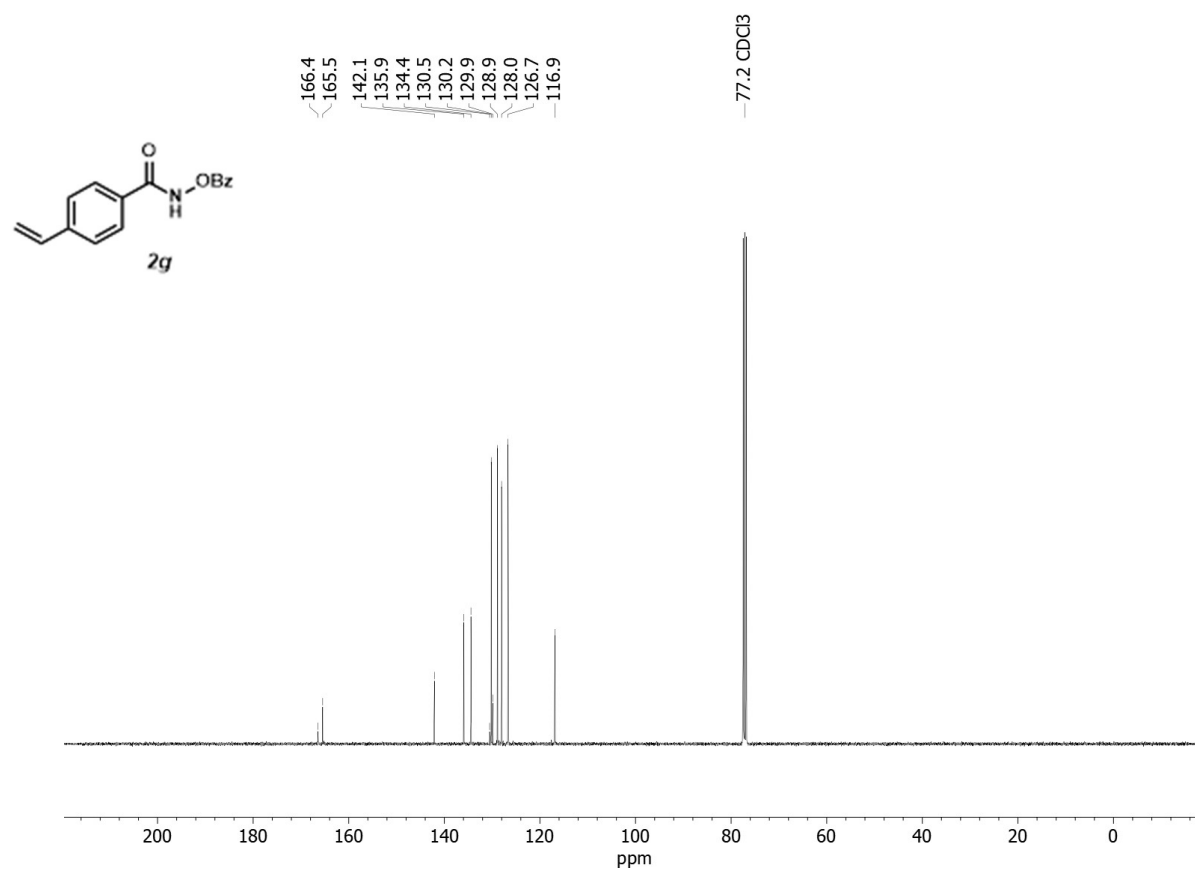
In a nitrogen-filled glovebox, a one-dram vial with a stir bar was charged with *N*-(benzoyloxy)benzamide (**2a**) (0.1 mmol, 1.0 equiv) and 2,2,6,6-tetramethylpiperidine 1-oxyl (TEMPO) (0.3 mmol, 3.0 equiv). 1,4-dioxane was added (0.1 mL) followed sequentially by a stock solution of nickelocene (0.005 mmol, 5 mol%) and phenylsilane (0.1 mmol, 1.0 equiv) in 1,4-dioxane (0.1 mL), then a stock solution of ethyl 2-methyl-3-oxobutanoate (**1a**) (0.22 mmol, 2.2 equiv) and lithium *tert*-butoxide (0.15 mmol, 1.5 equiv) in 1,4-dioxane (0.3 mL). The latter solution was prepared by first preparing a solution of **1a** in 1,4-dioxane, then adding this solution to solid lithium *tert*-butoxide and allowing to dissolve. The reaction vial was then sealed with a Teflon-lined cap, removed from the glovebox, and stirred at 40 °C in a heating mantle for 4 h, after which a sample from the crude reaction mixture was analyzed; HRMS (ESI<sup>+</sup>) *m/z* calc'd for C<sub>16</sub>H<sub>25</sub>N<sub>2</sub>O<sub>2</sub> [M+H]<sup>+</sup>: 277.1911, found 277.1908.

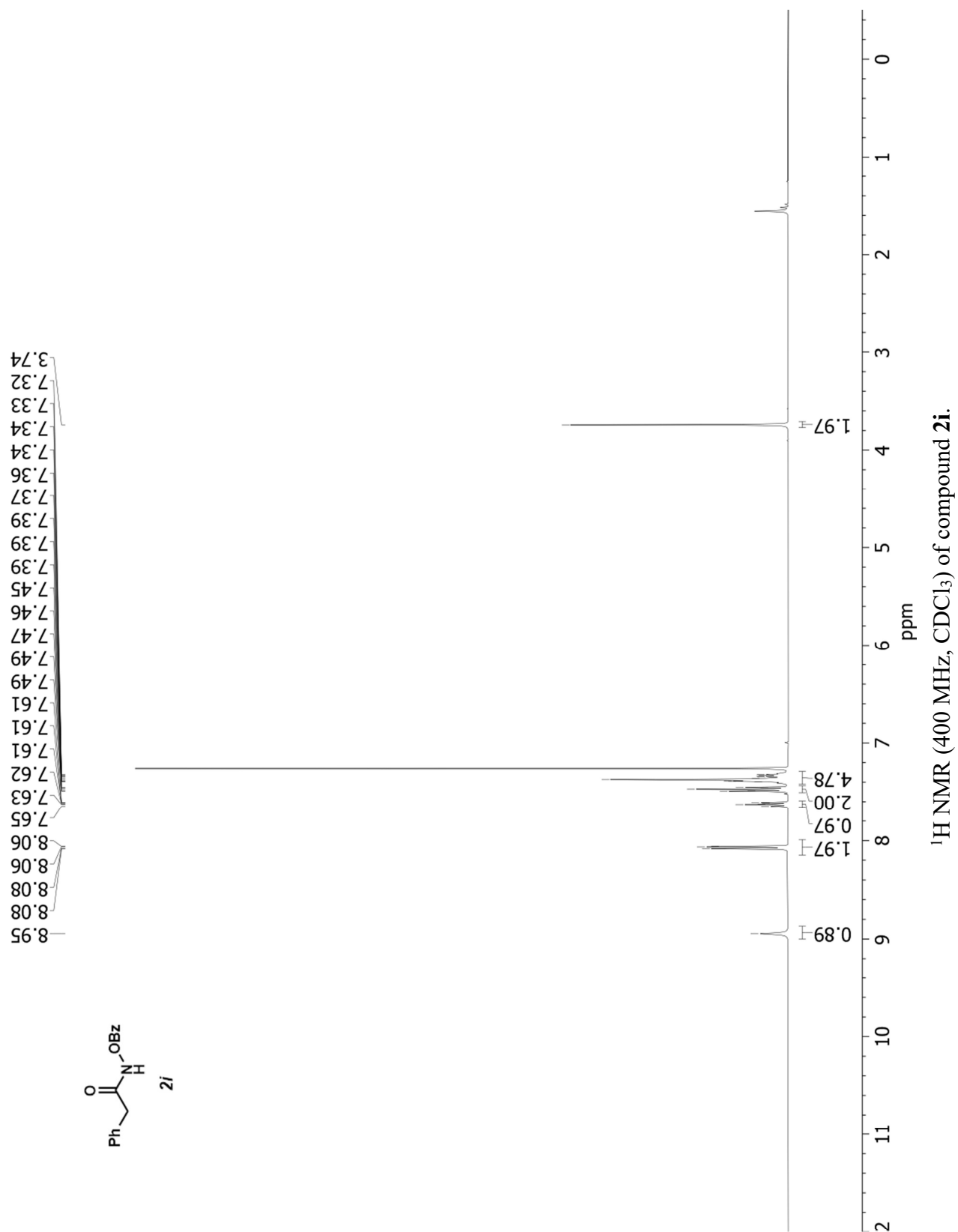
## References

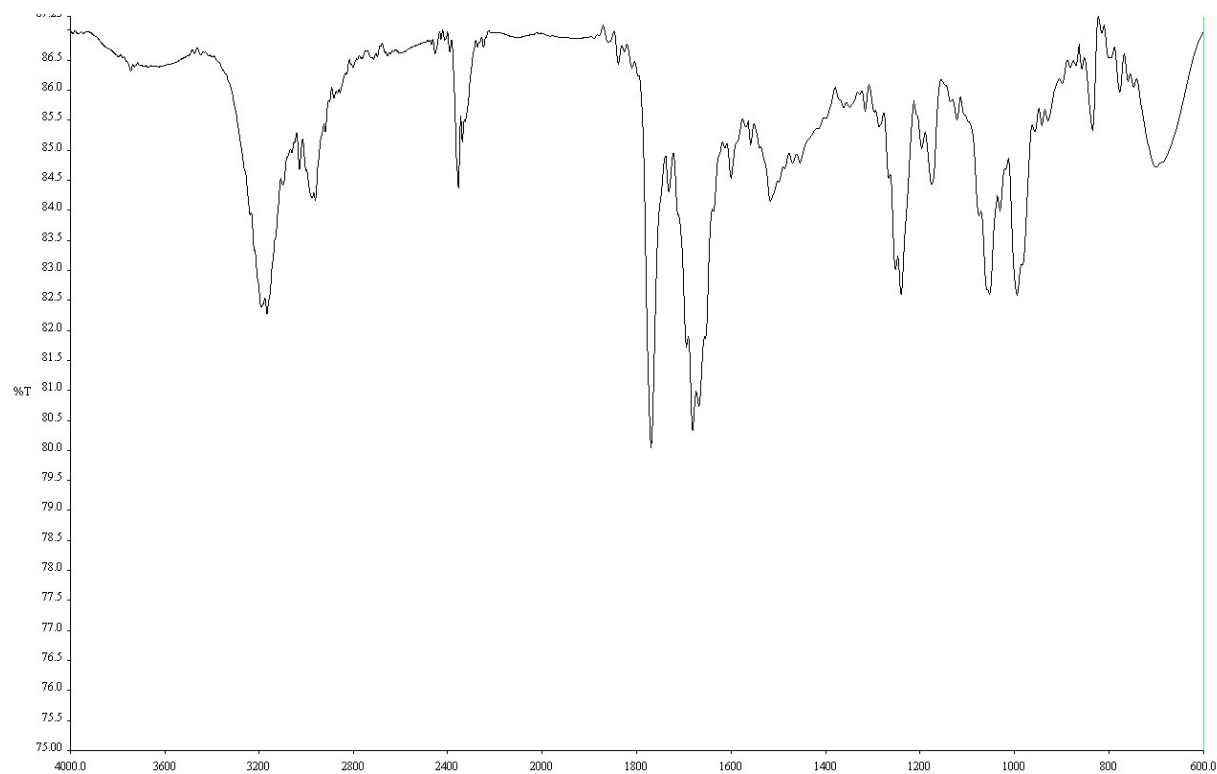
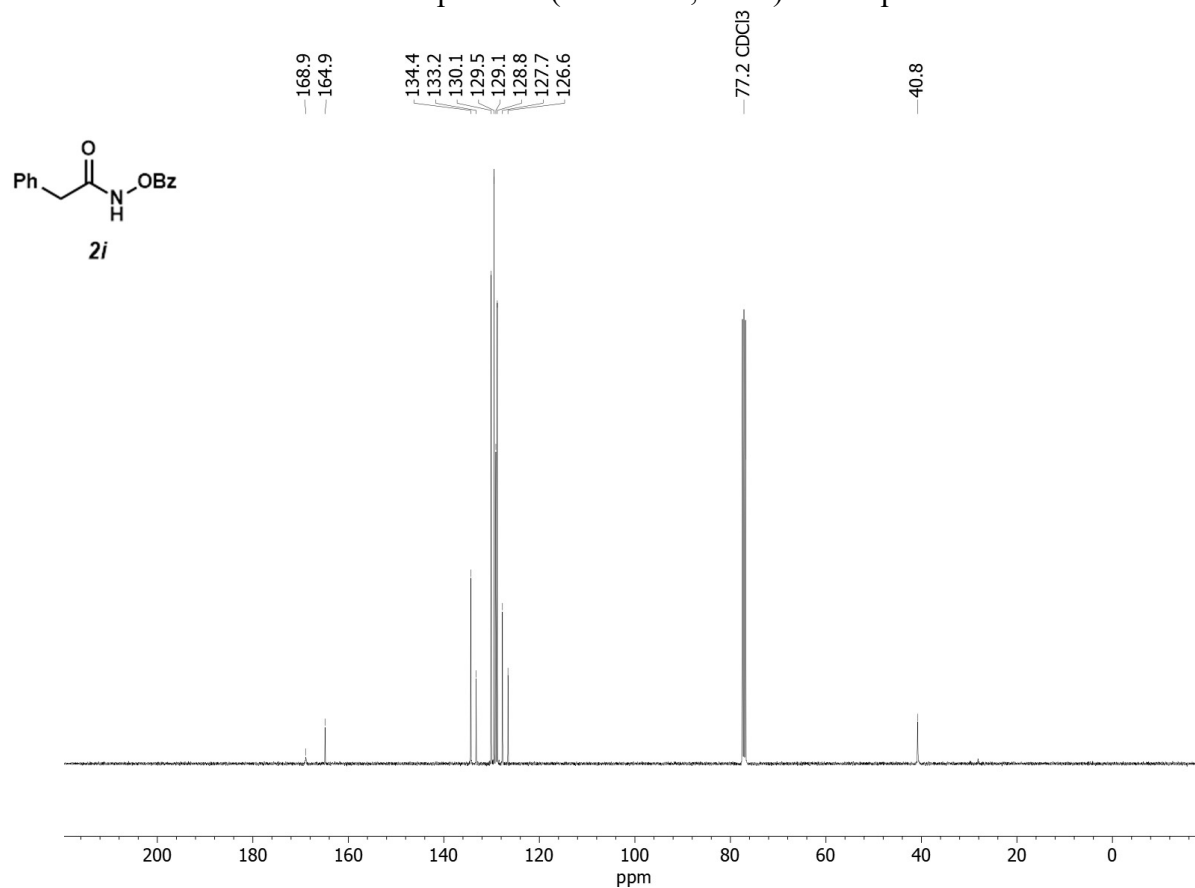
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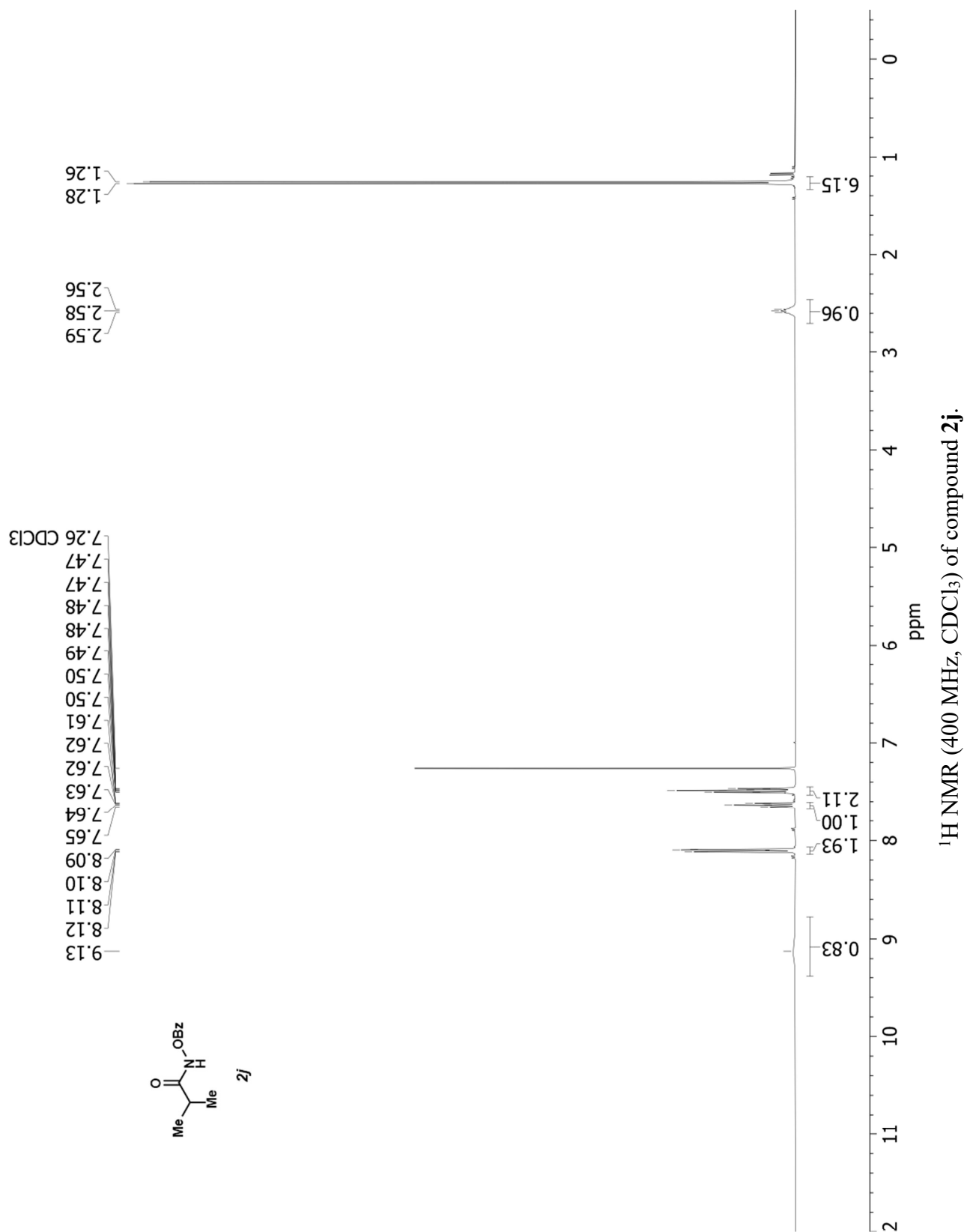


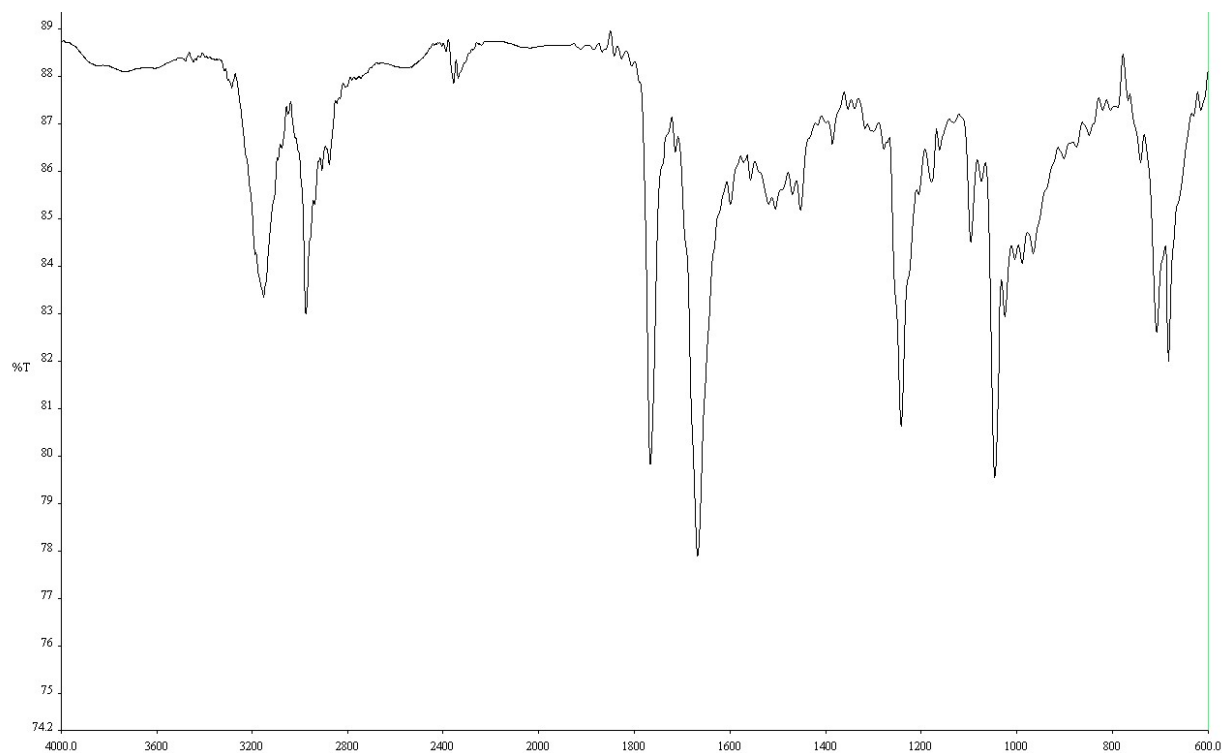
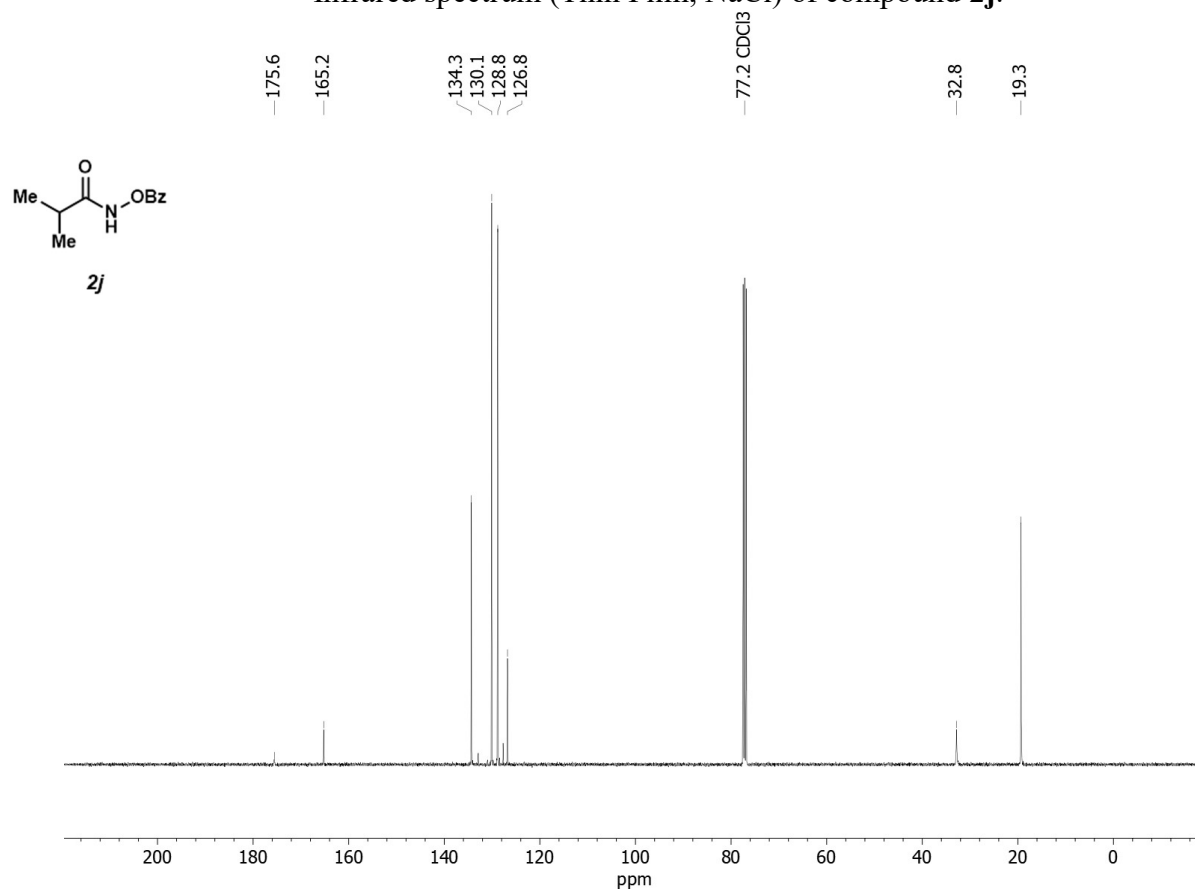


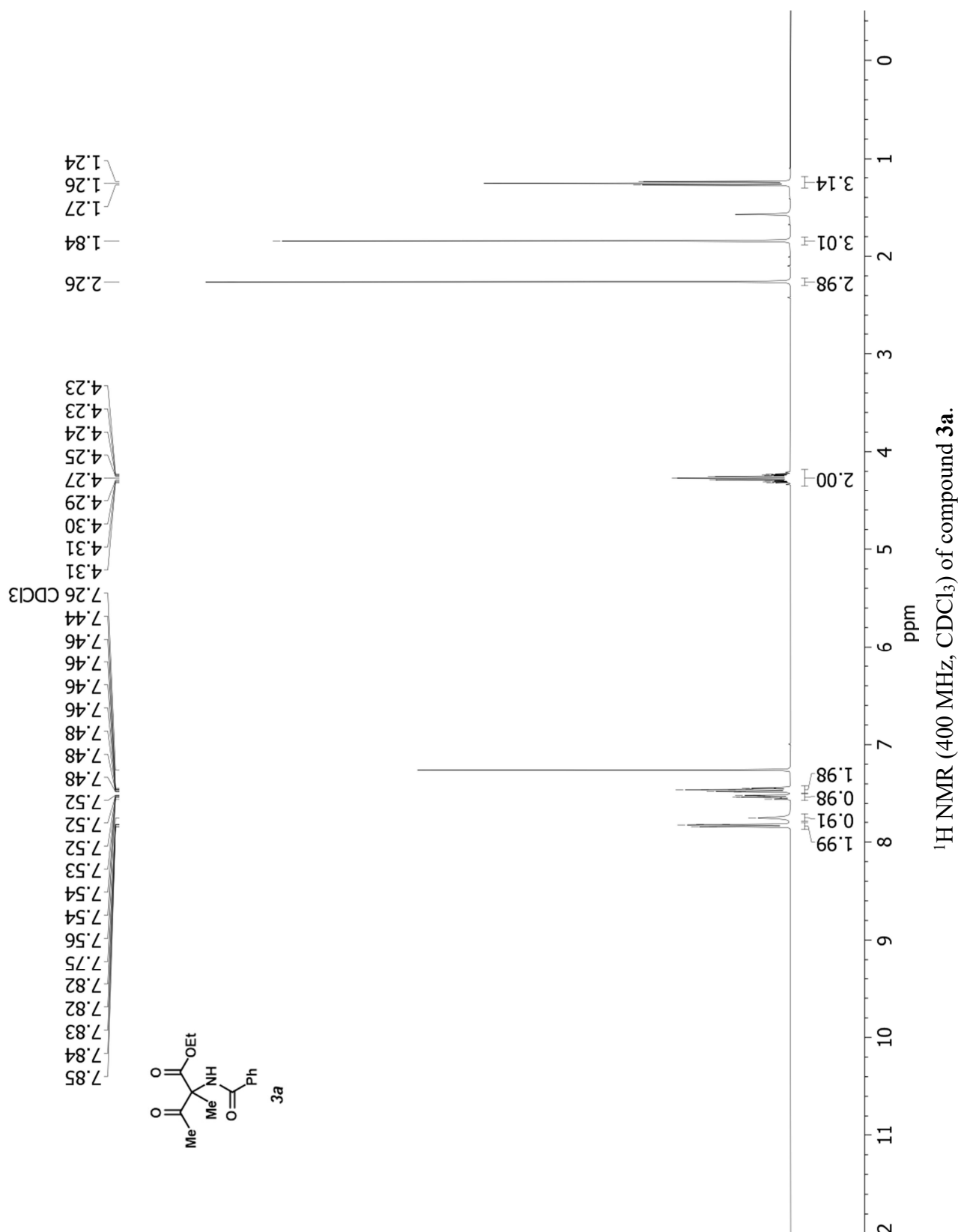
Infrared spectrum (Thin Film, NaCl) of compound **2g**.<sup>13</sup>C NMR (100 MHz, CDCl<sub>3</sub>) of compound **2g**.

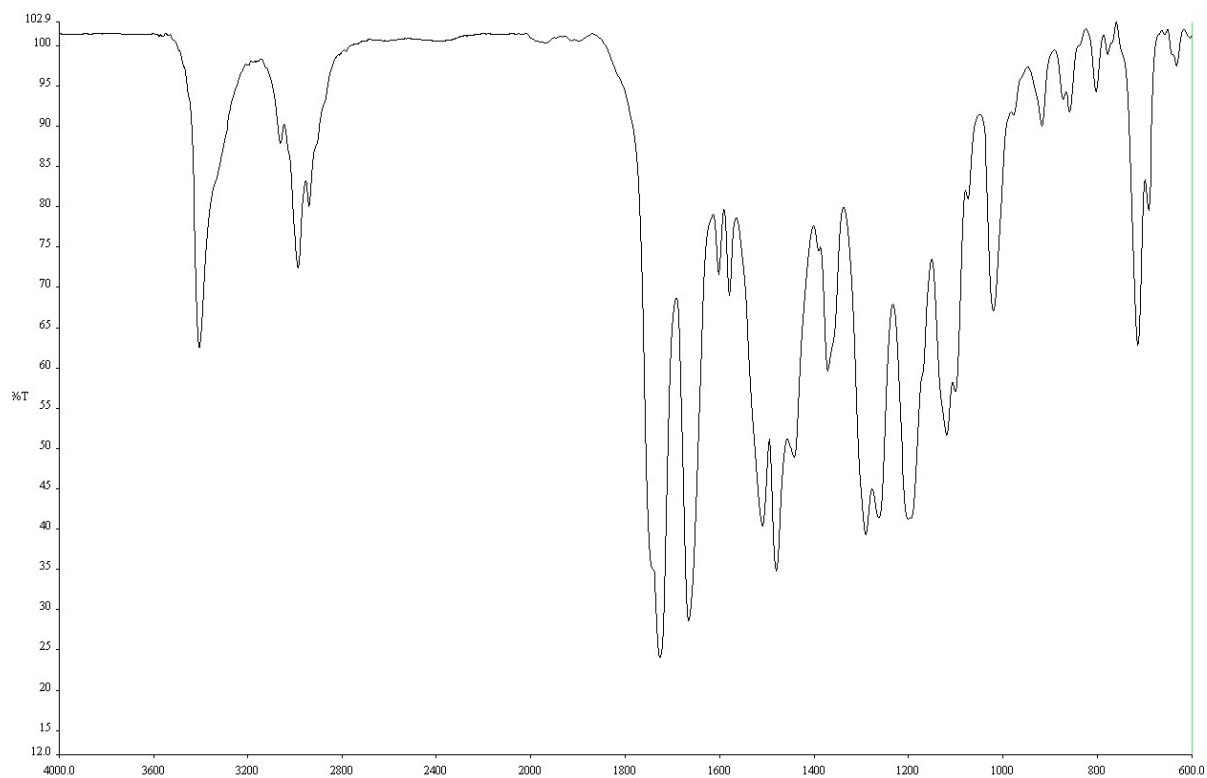
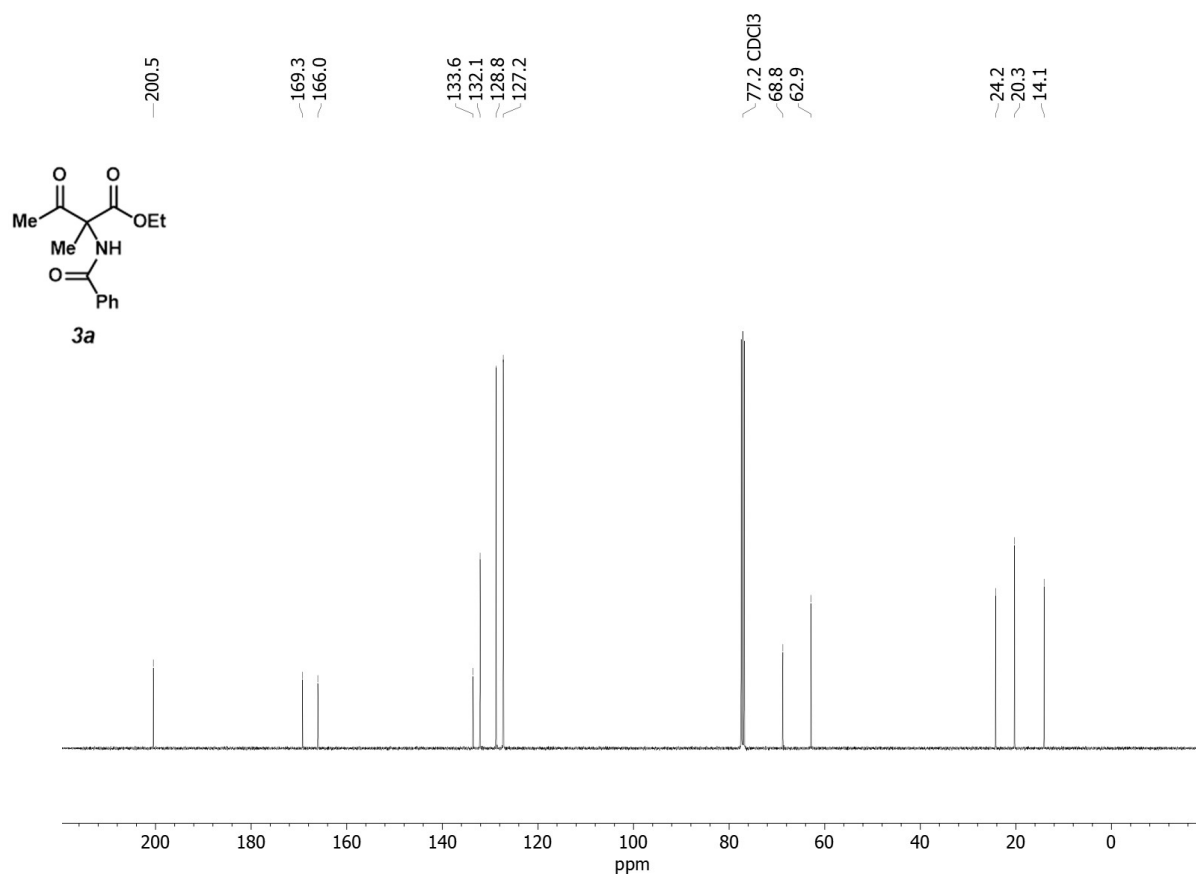


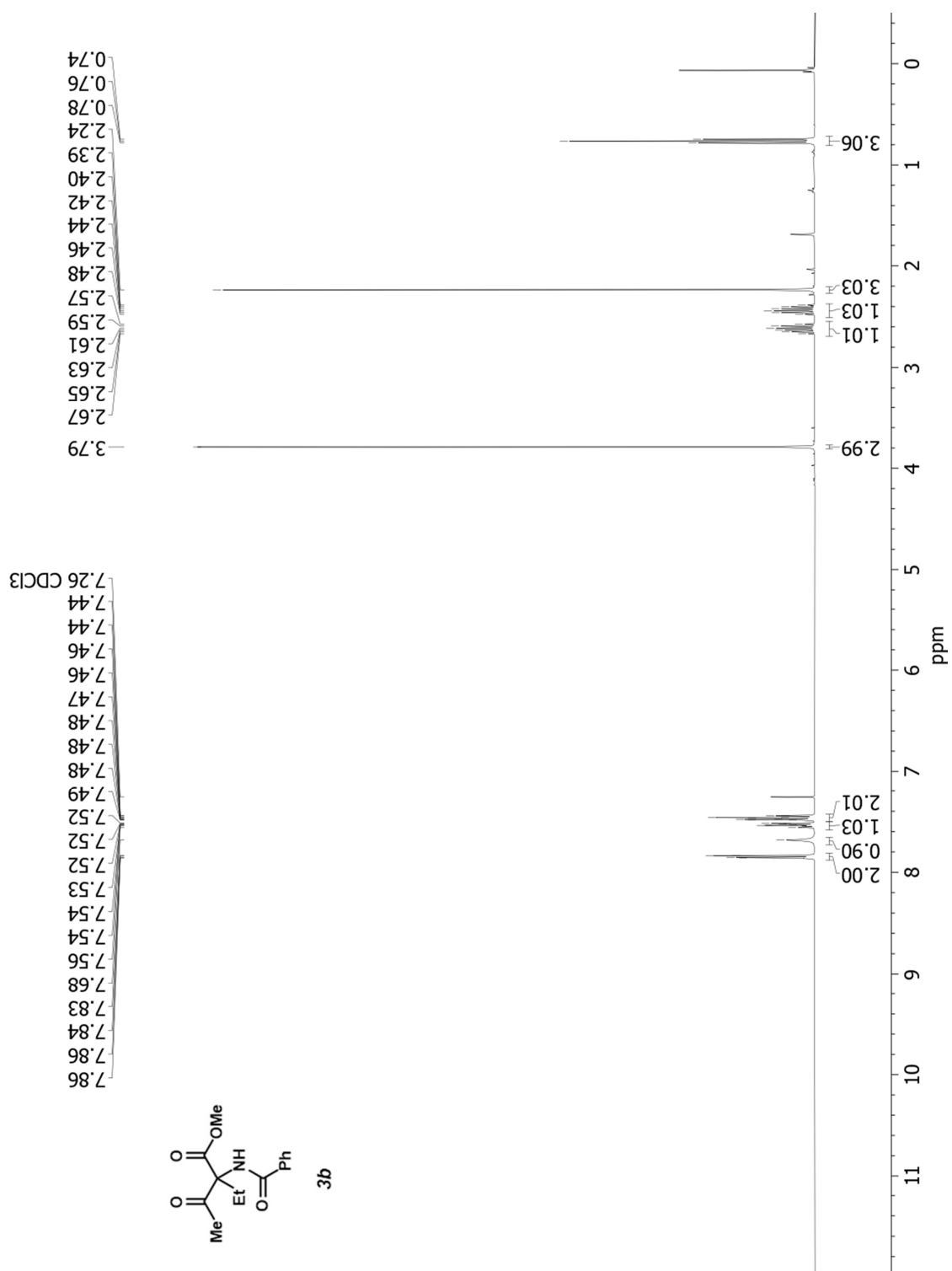
Infrared spectrum (Thin Film, NaCl) of compound **2i**.<sup>13</sup>C NMR (100 MHz, CDCl<sub>3</sub>) of compound **2i**.

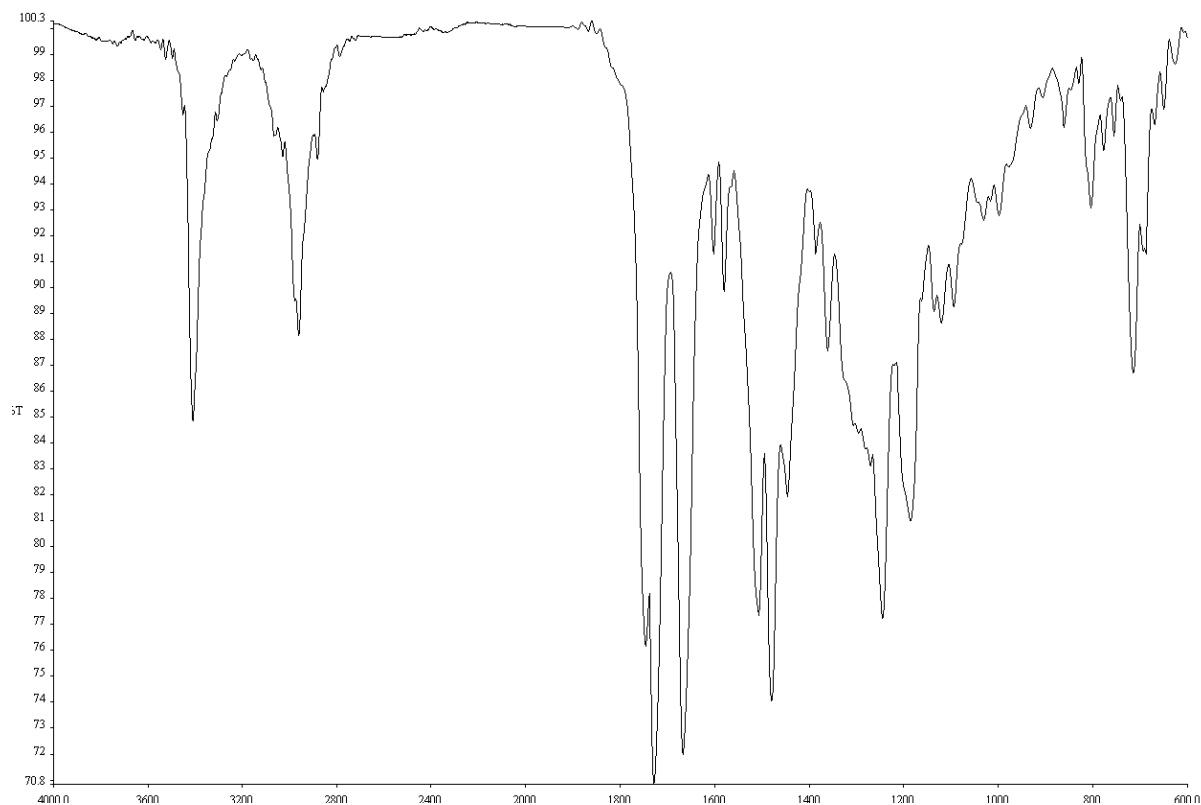
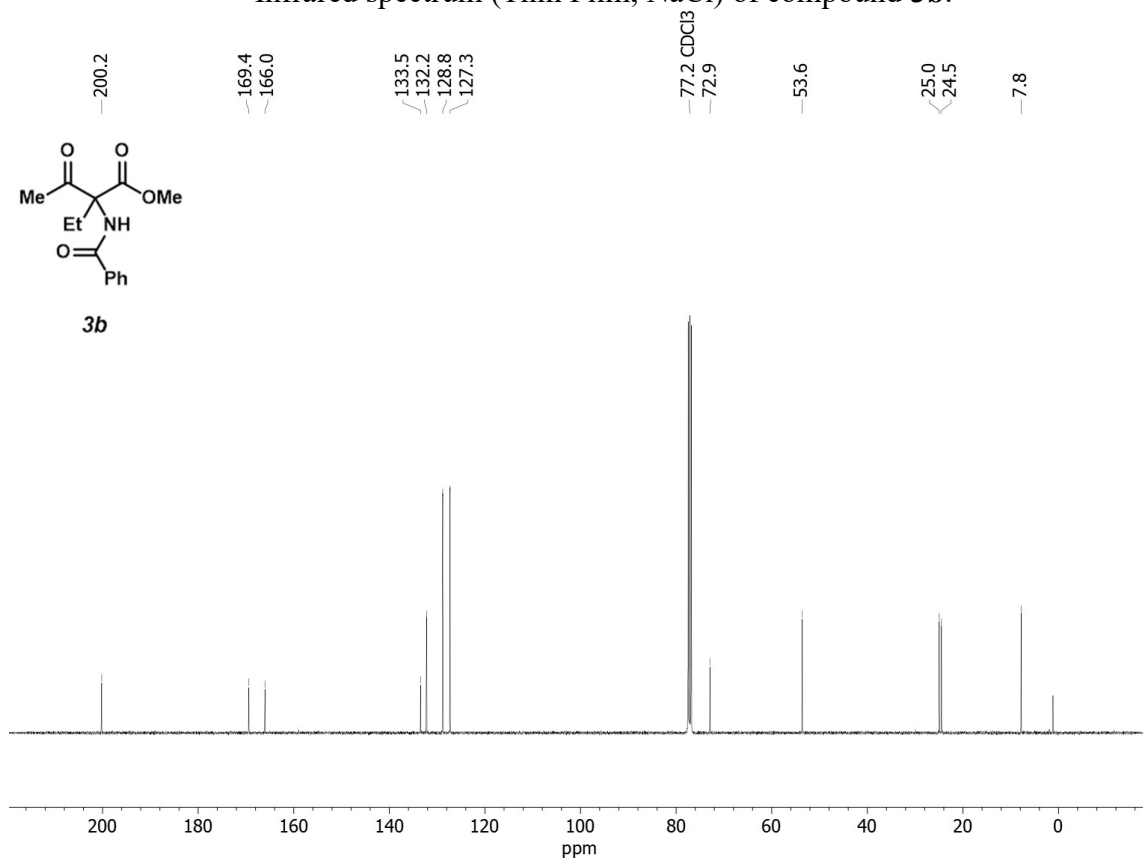


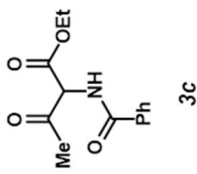
Infrared spectrum (Thin Film, NaCl) of compound **2j**.<sup>13</sup>C NMR (100 MHz, CDCl<sub>3</sub>) of compound **2j**.

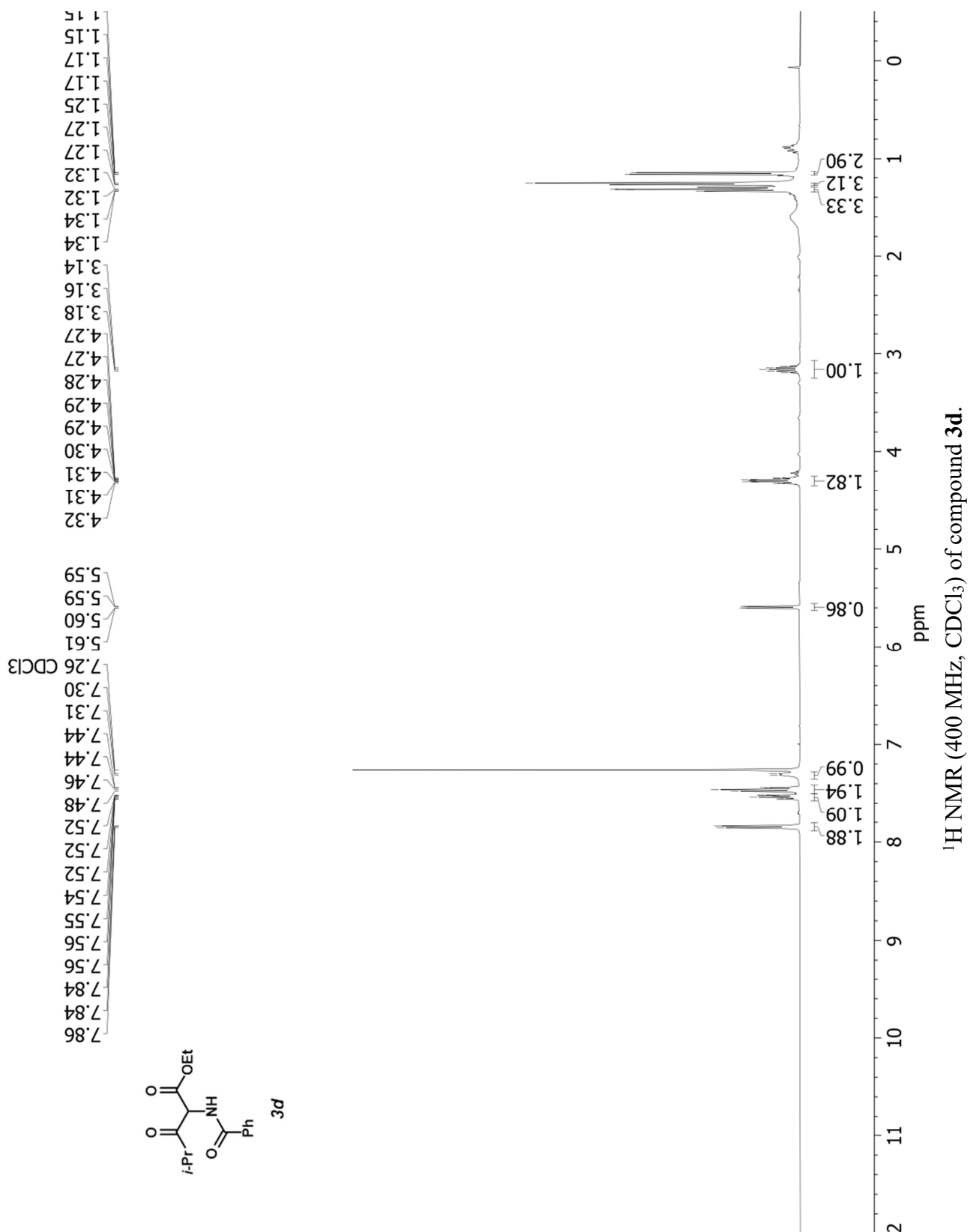


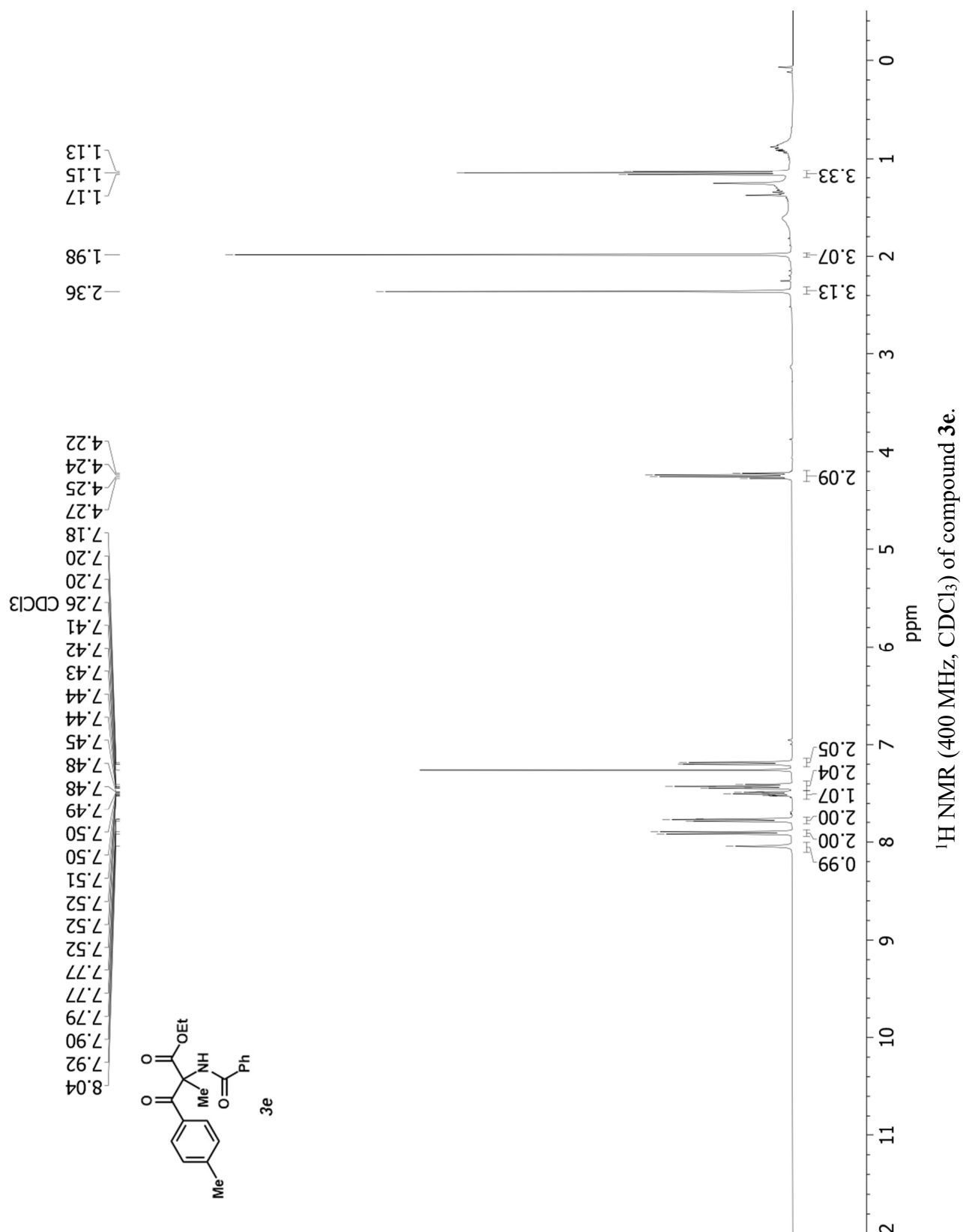
Infrared spectrum (Thin Film, NaCl) of compound **3a**.<sup>13</sup>C NMR (100 MHz, CDCl<sub>3</sub>) of compound **3a**.

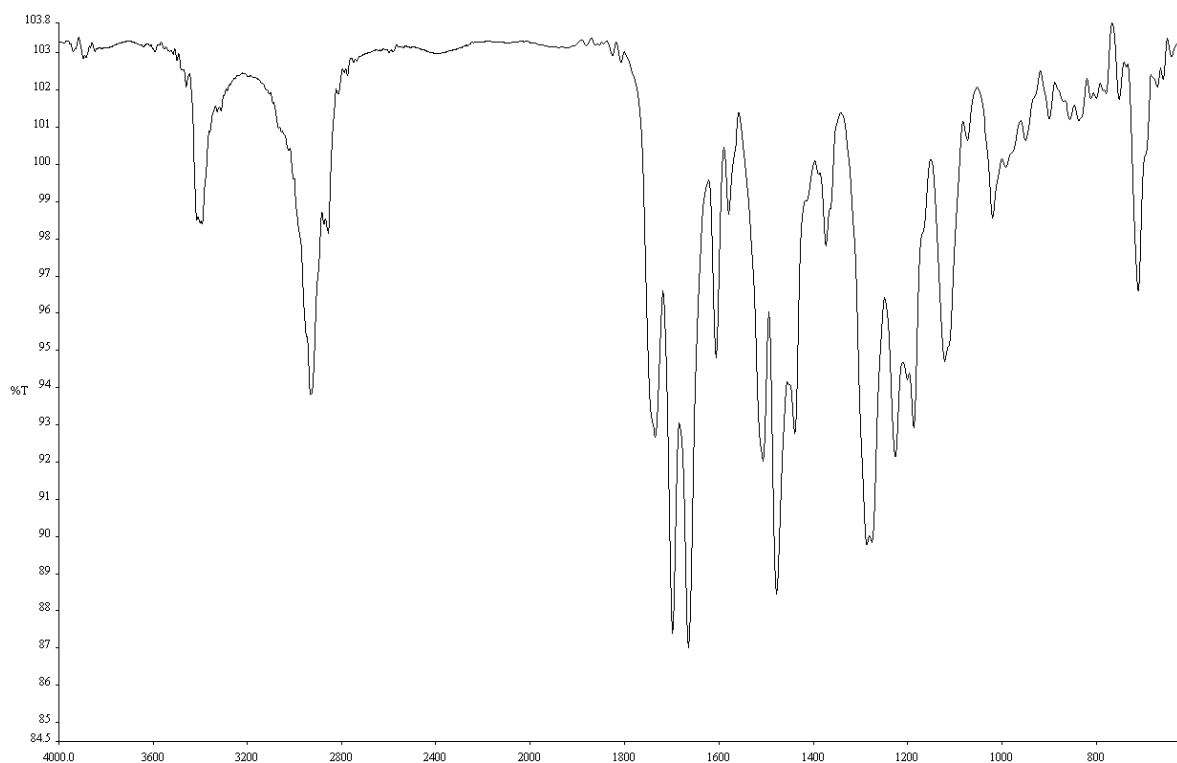
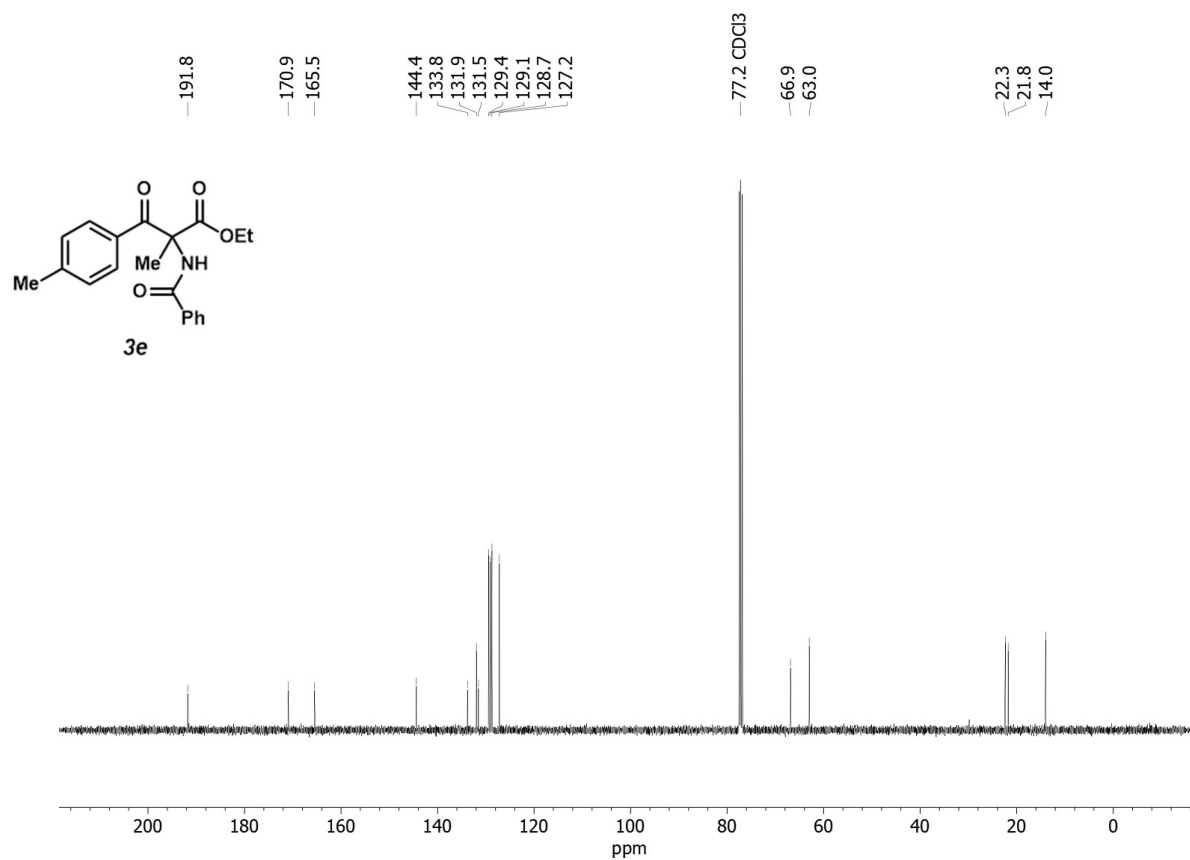


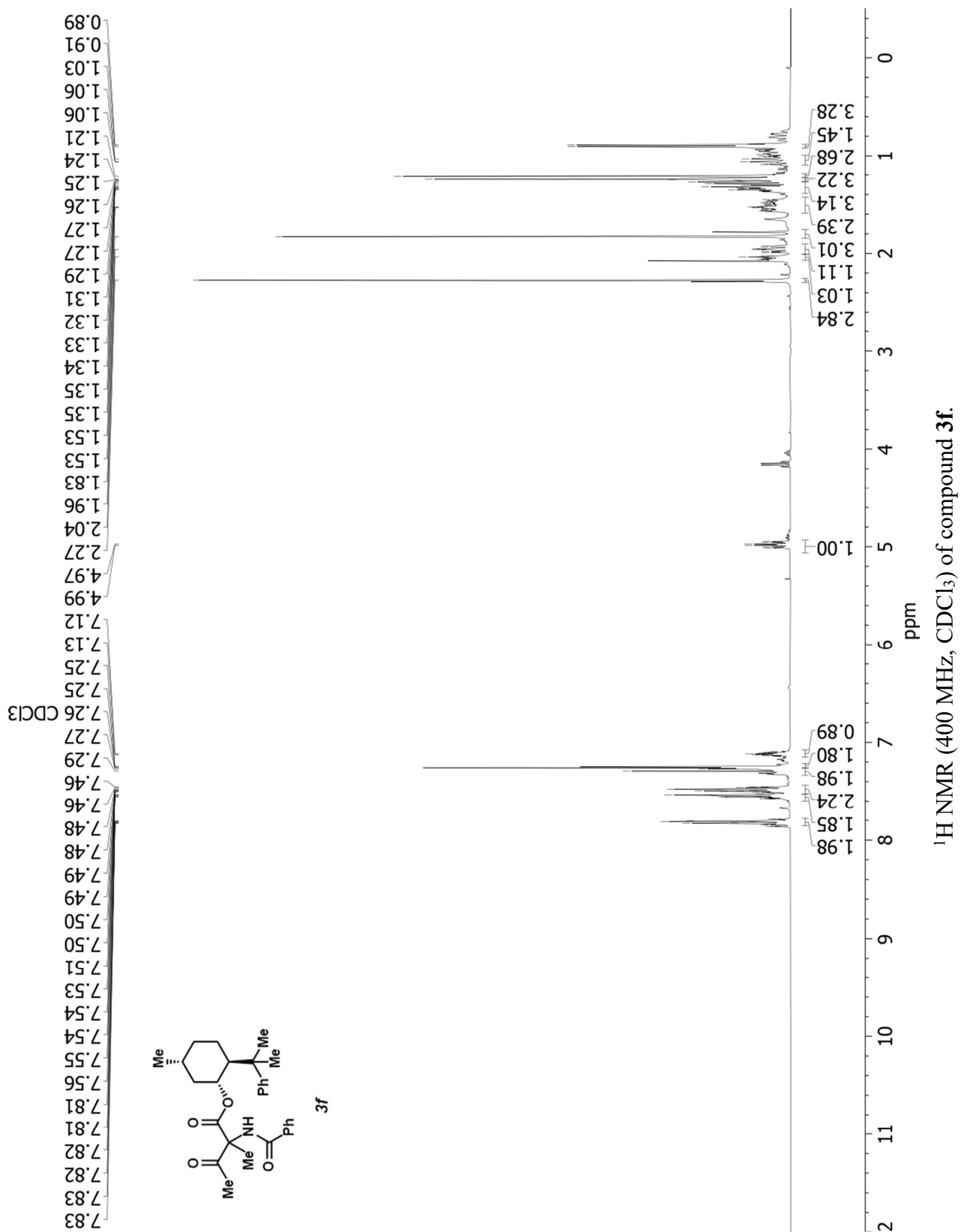
Infrared spectrum (Thin Film, NaCl) of compound **3b**.<sup>13</sup>C NMR (100 MHz, CDCl<sub>3</sub>) of compound **3b**.

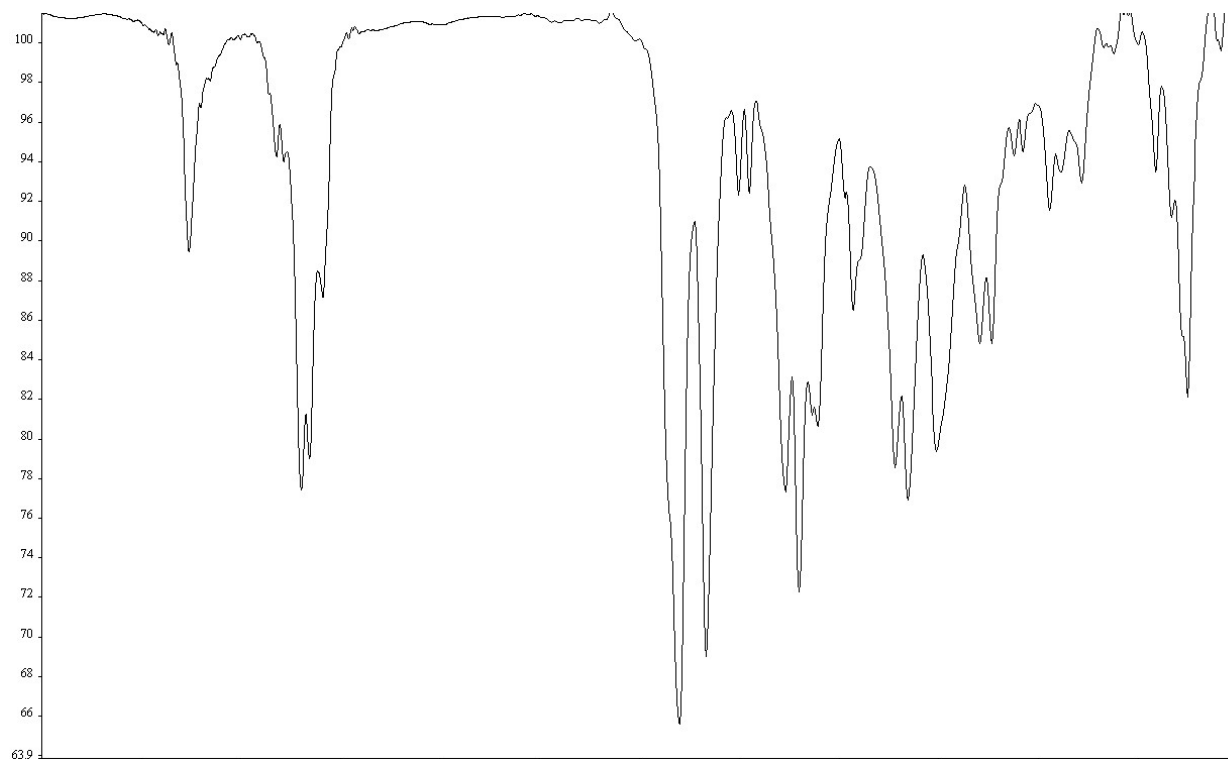
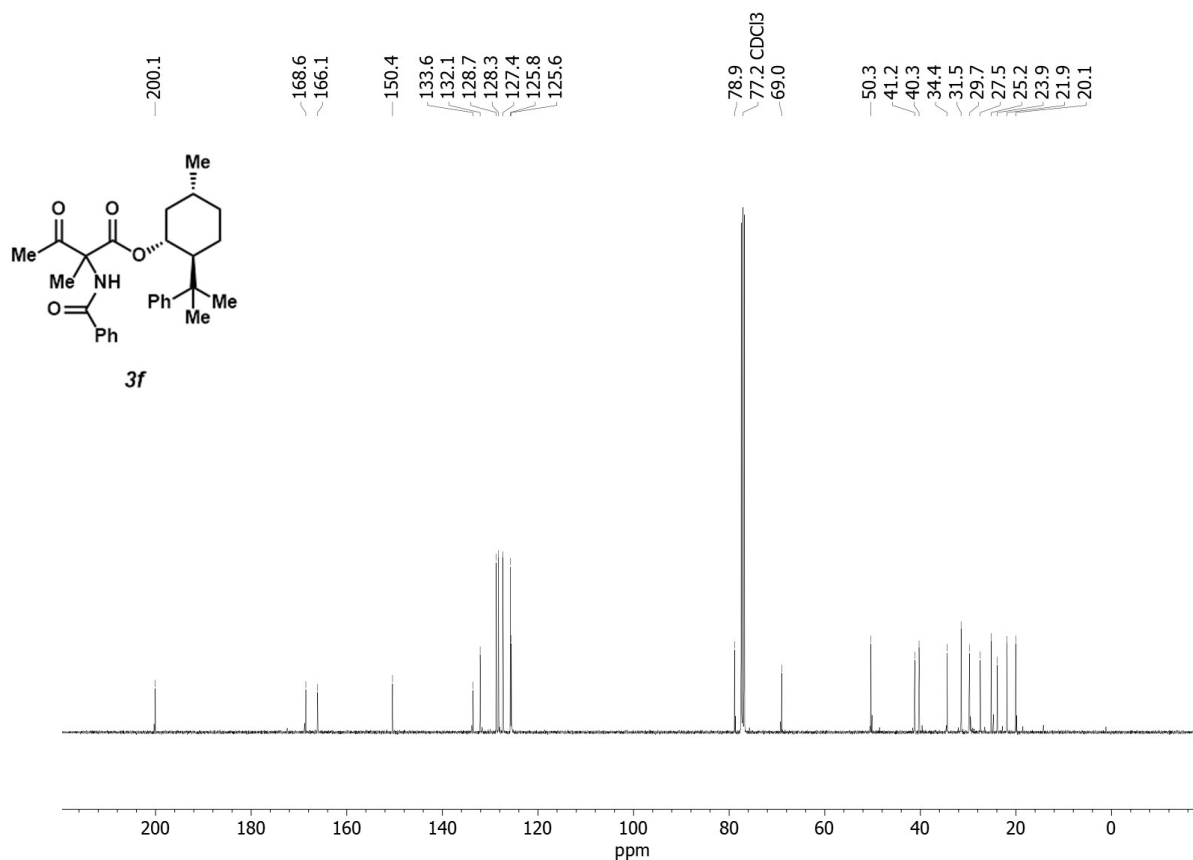


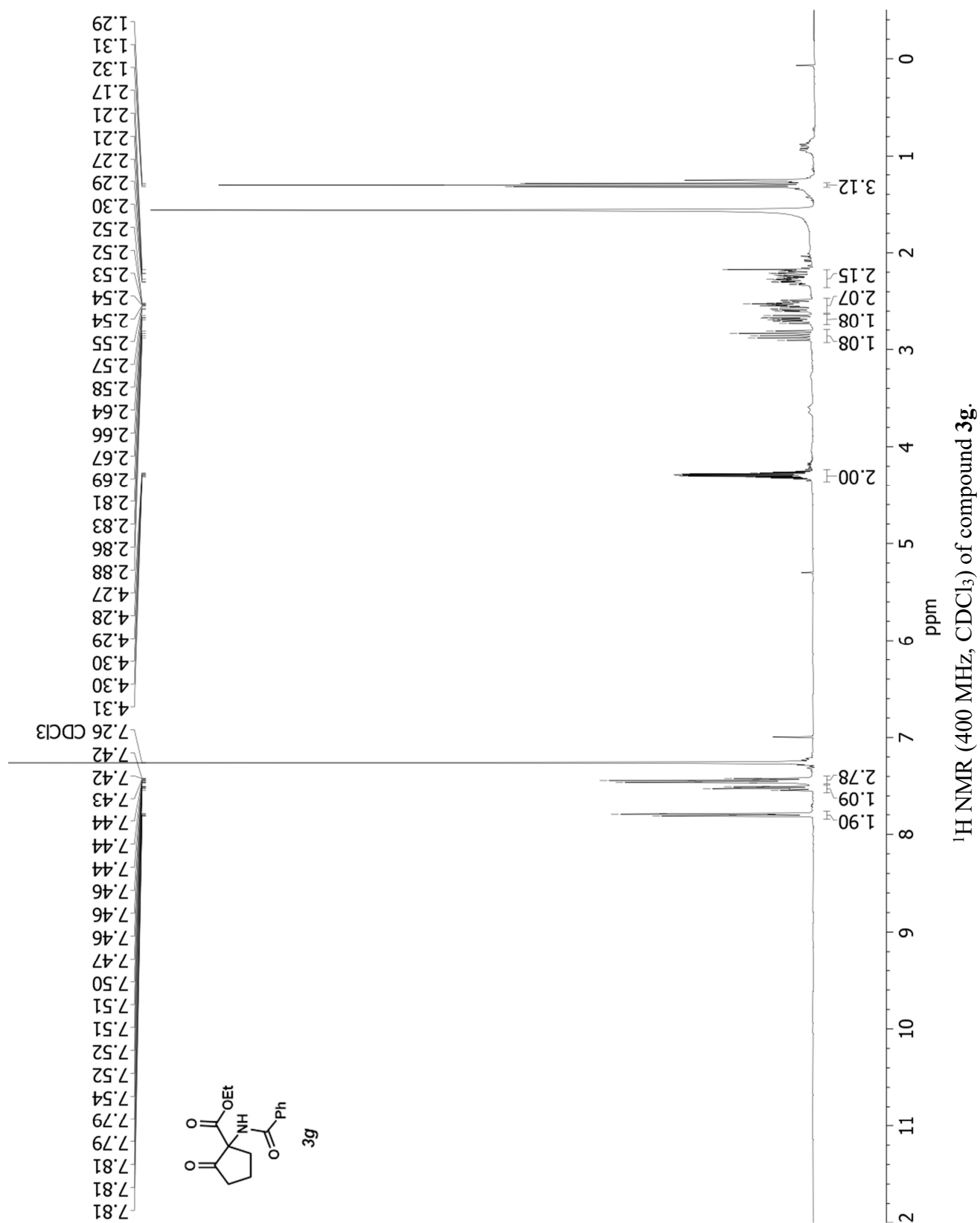


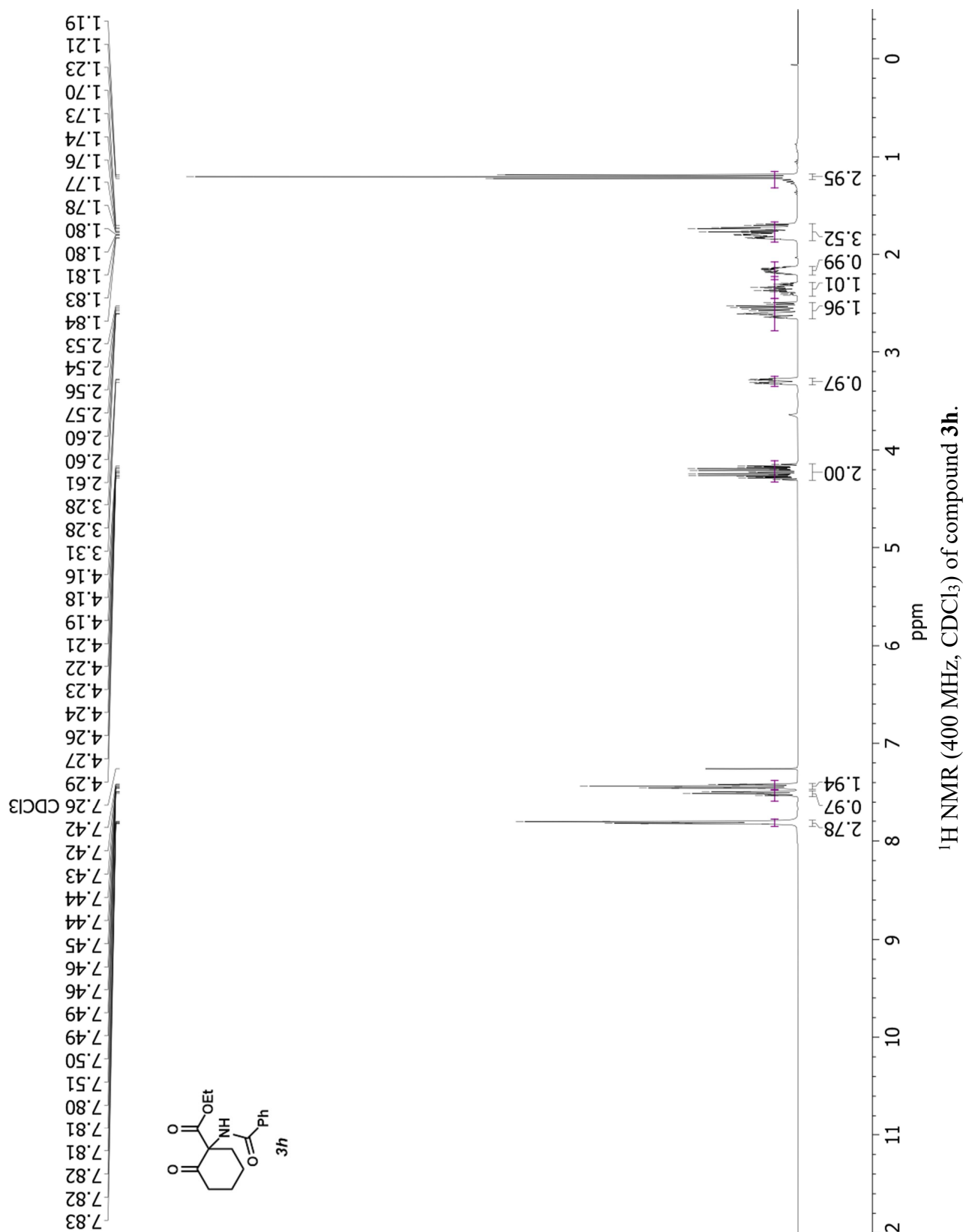


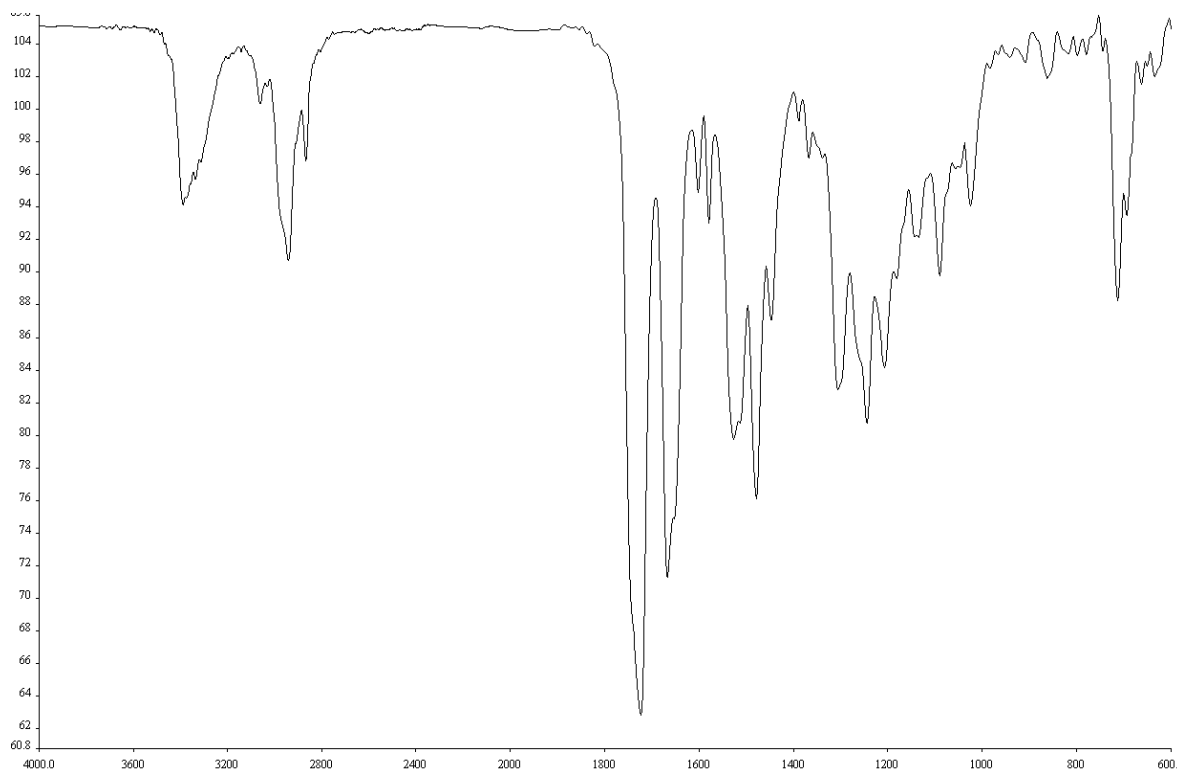
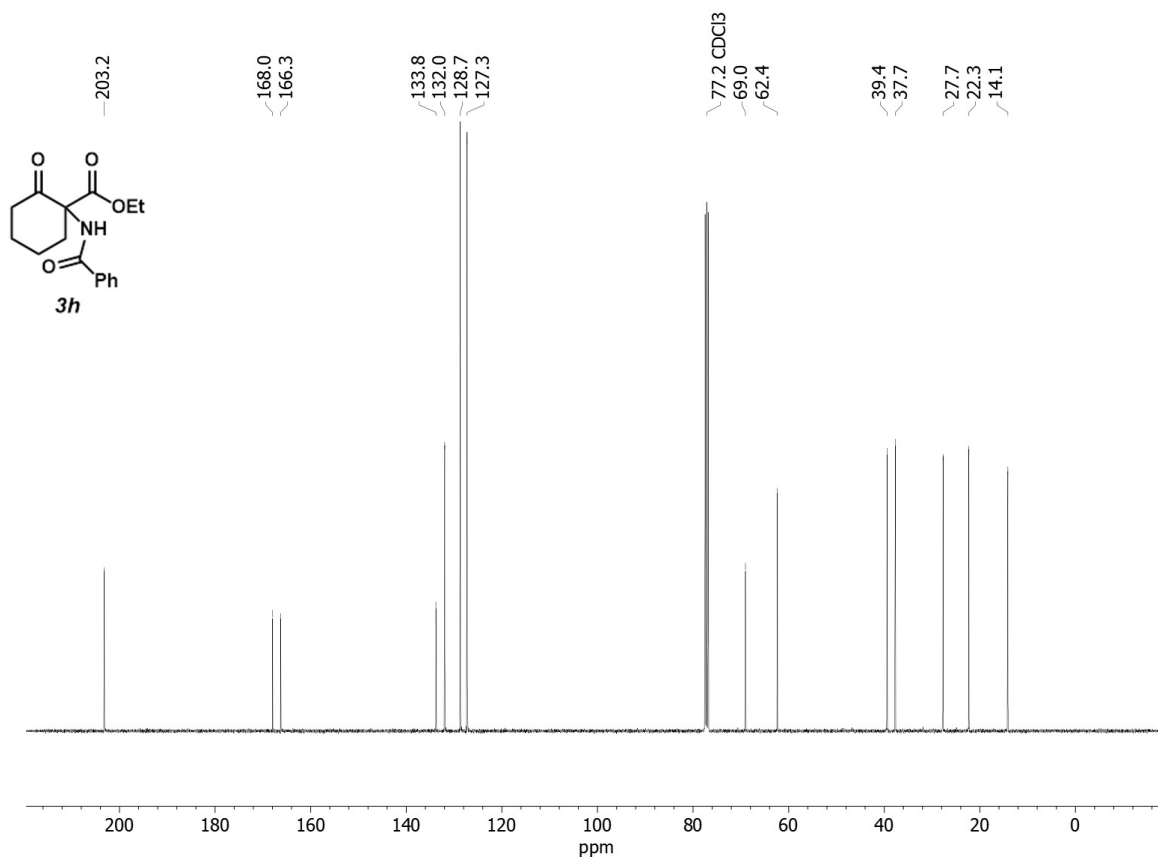
Infrared spectrum (Thin Film, NaCl) of compound **3e**.<sup>13</sup>C NMR (100 MHz, CDCl<sub>3</sub>) of compound **3e**.

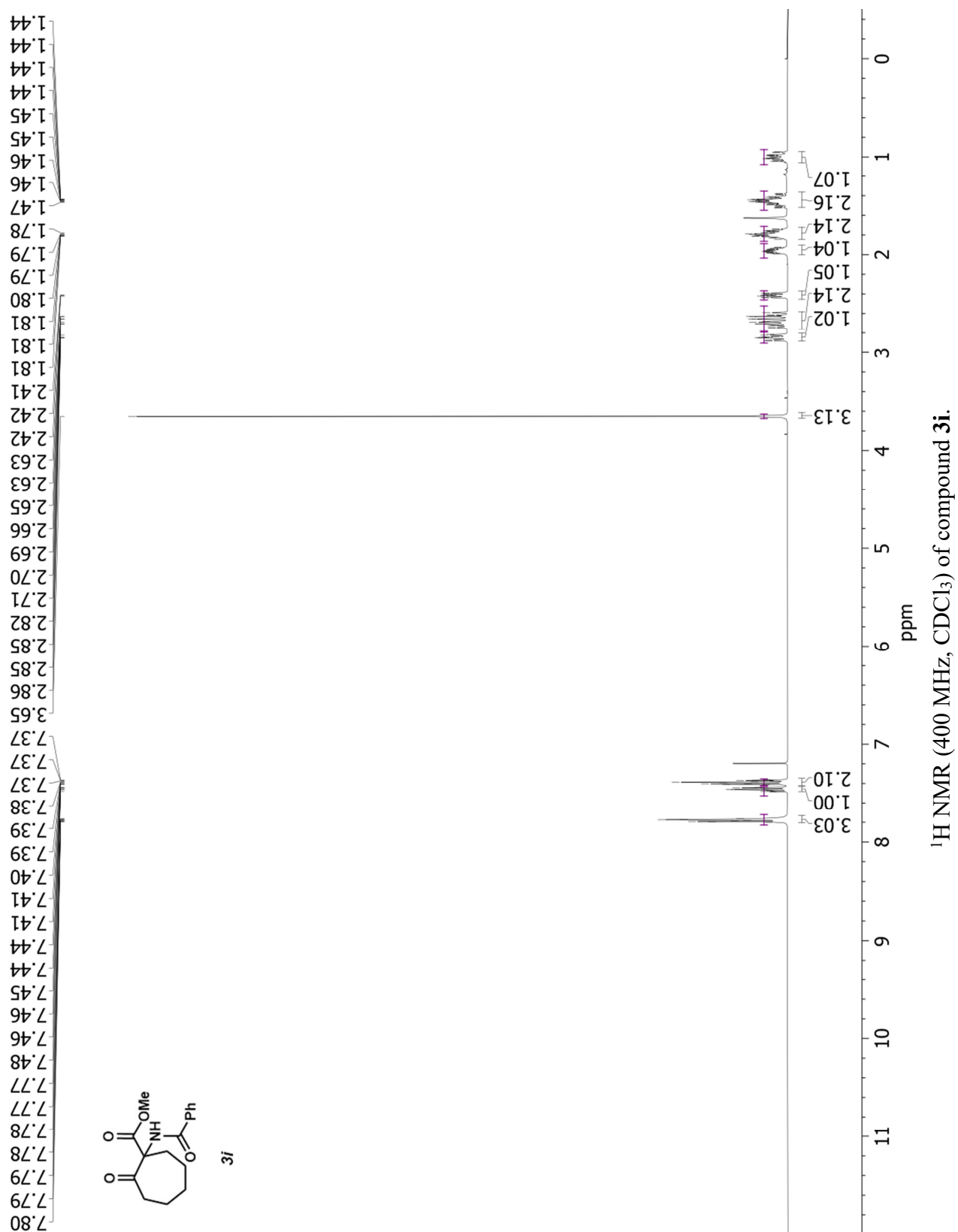


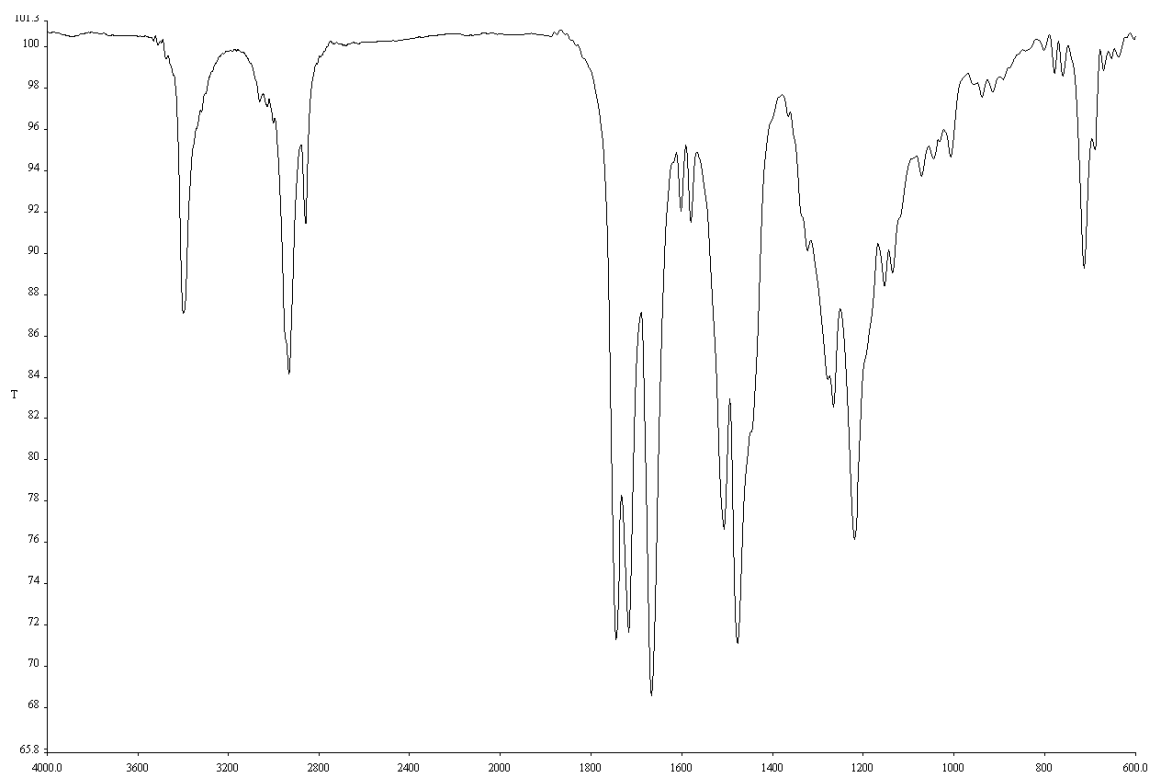
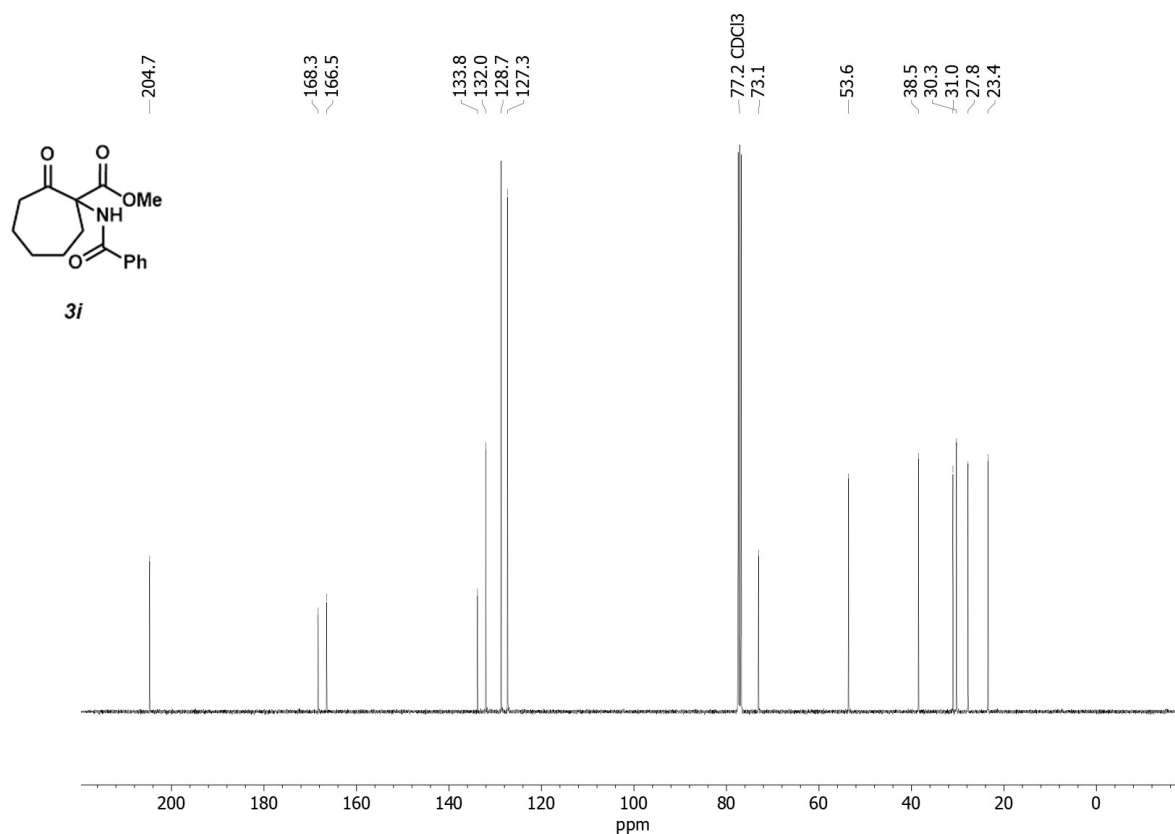
Infrared spectrum (Thin Film, NaCl) of compound **3f**.<sup>13</sup>C NMR (100 MHz, CDCl<sub>3</sub>) of compound **3f**.

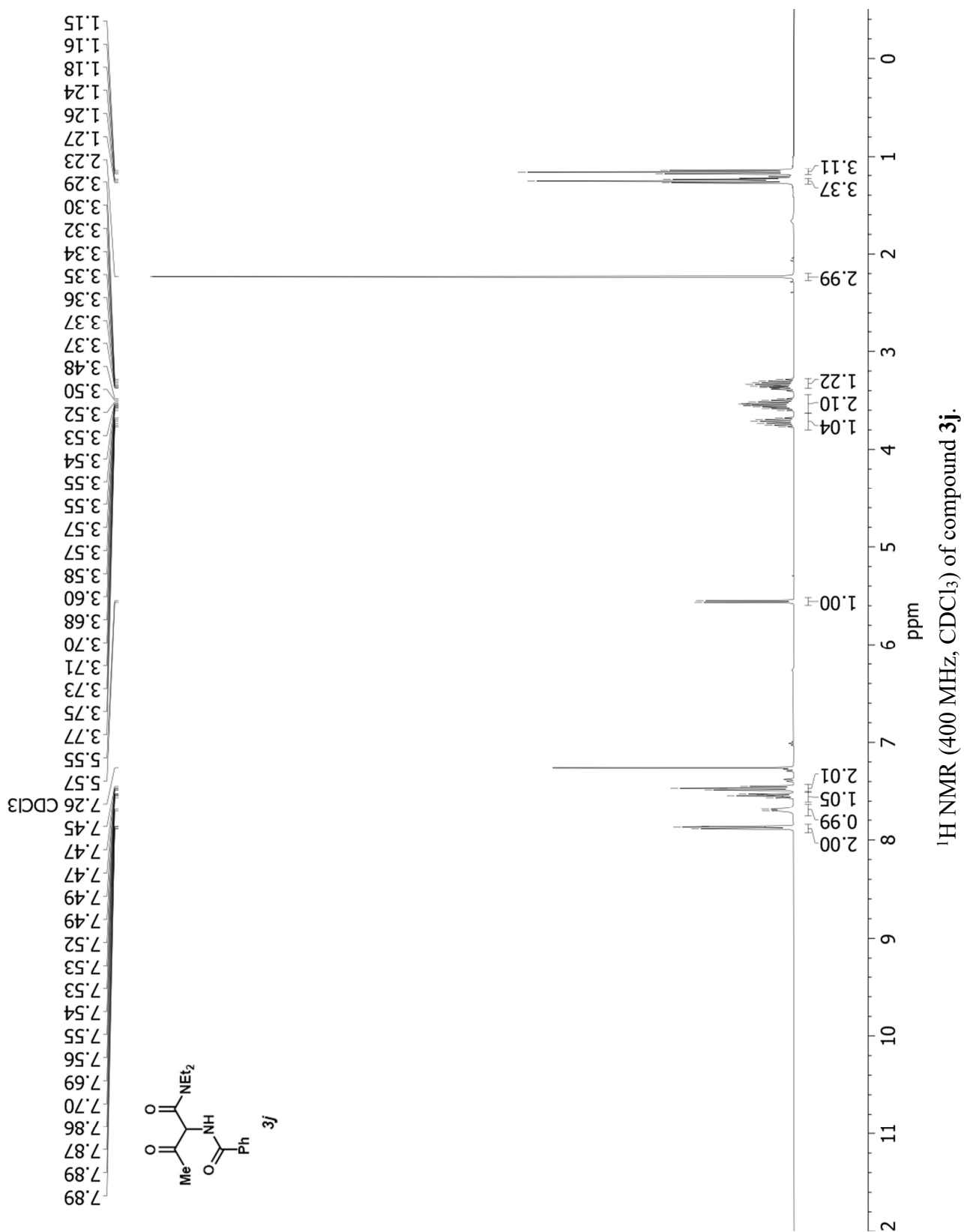


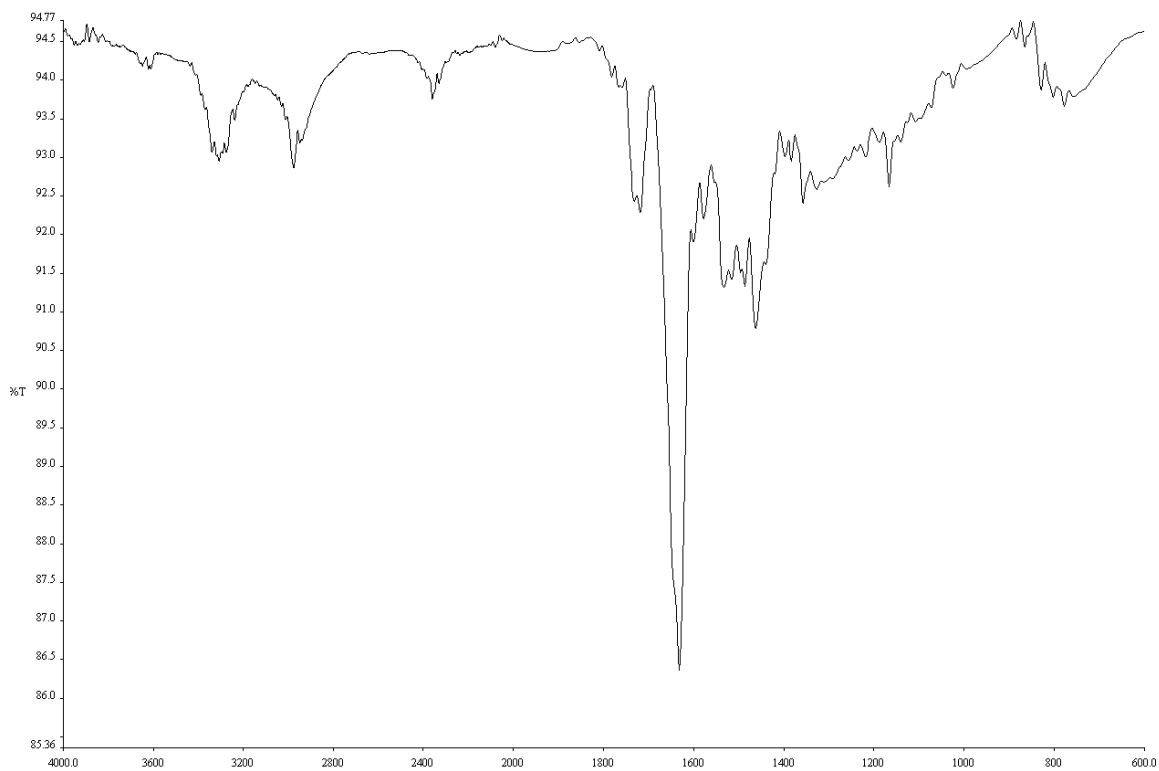
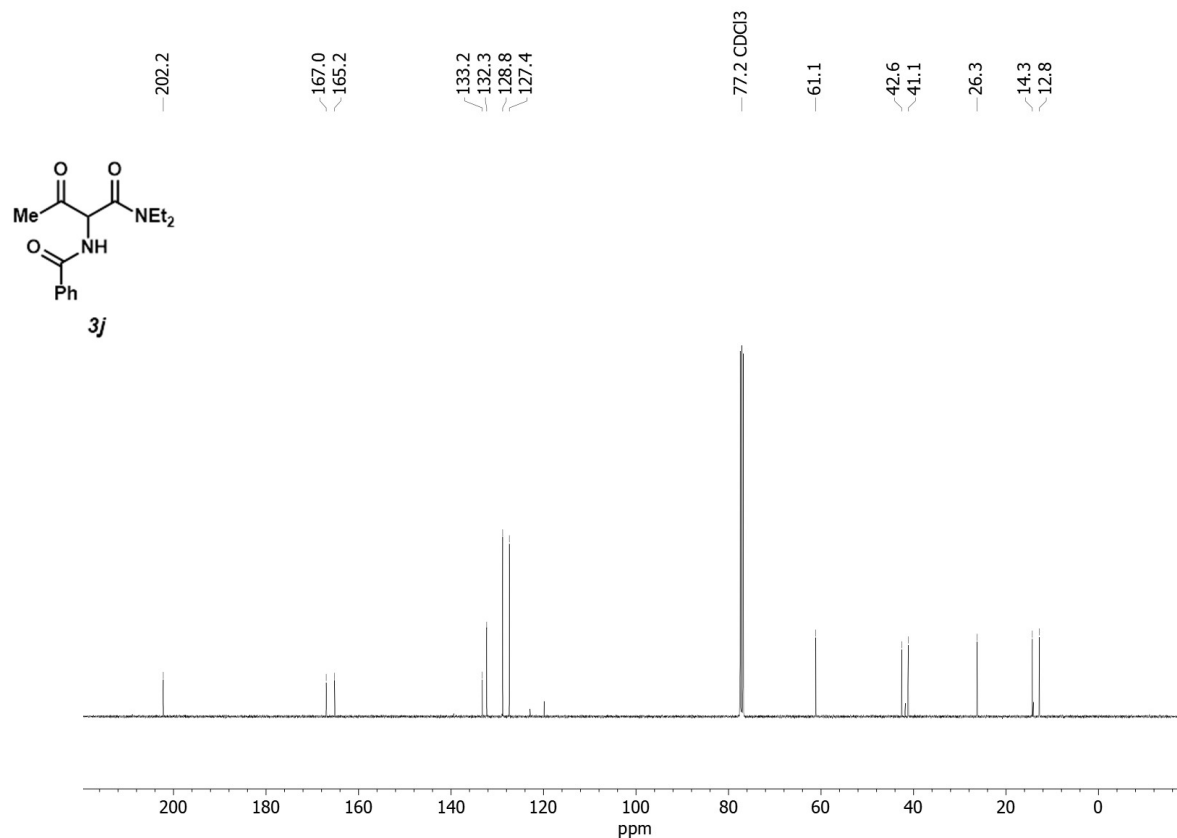


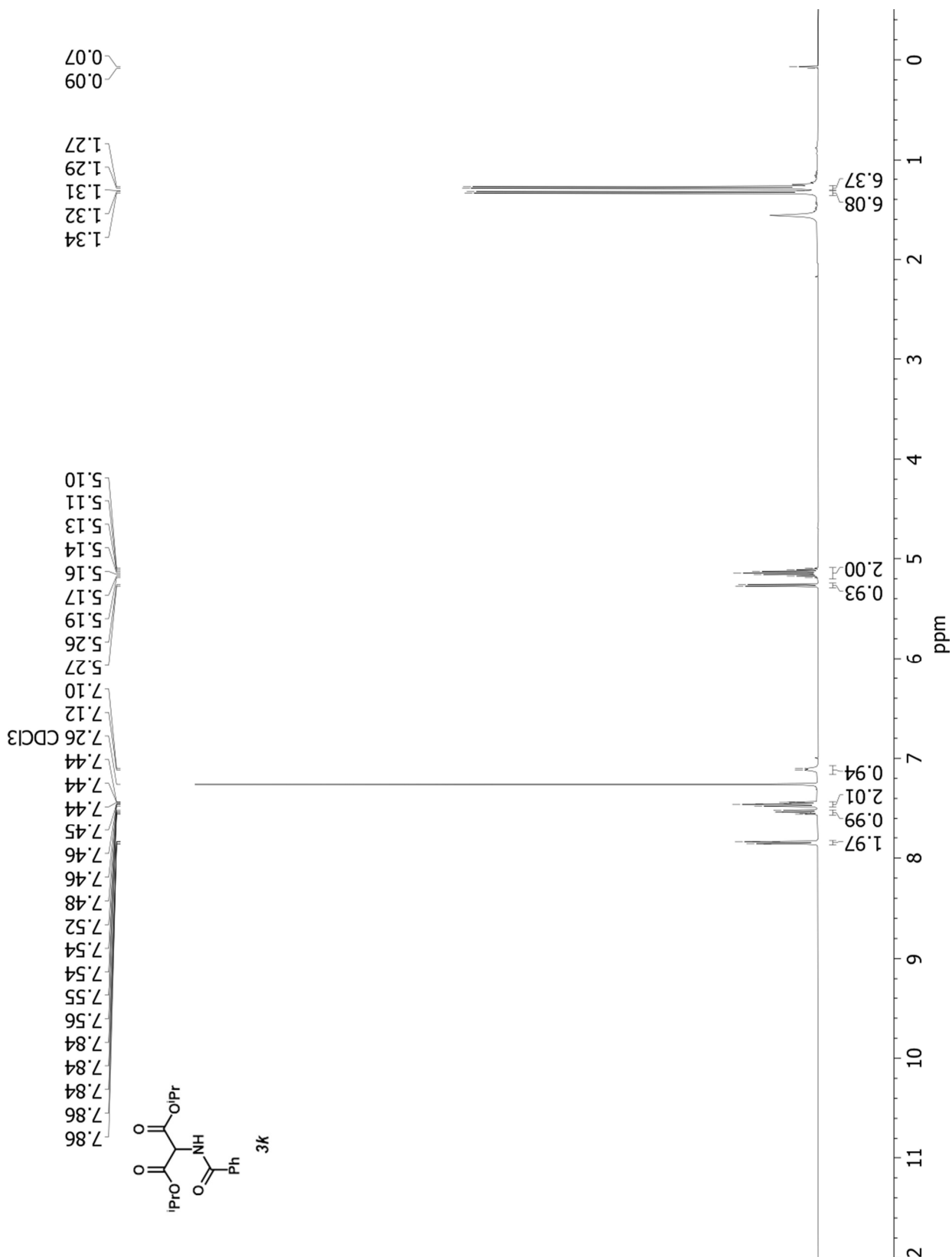
Infrared spectrum (Thin Film, NaCl) of compound **3h**.<sup>13</sup>C NMR (100 MHz, CDCl<sub>3</sub>) of compound **3h**.

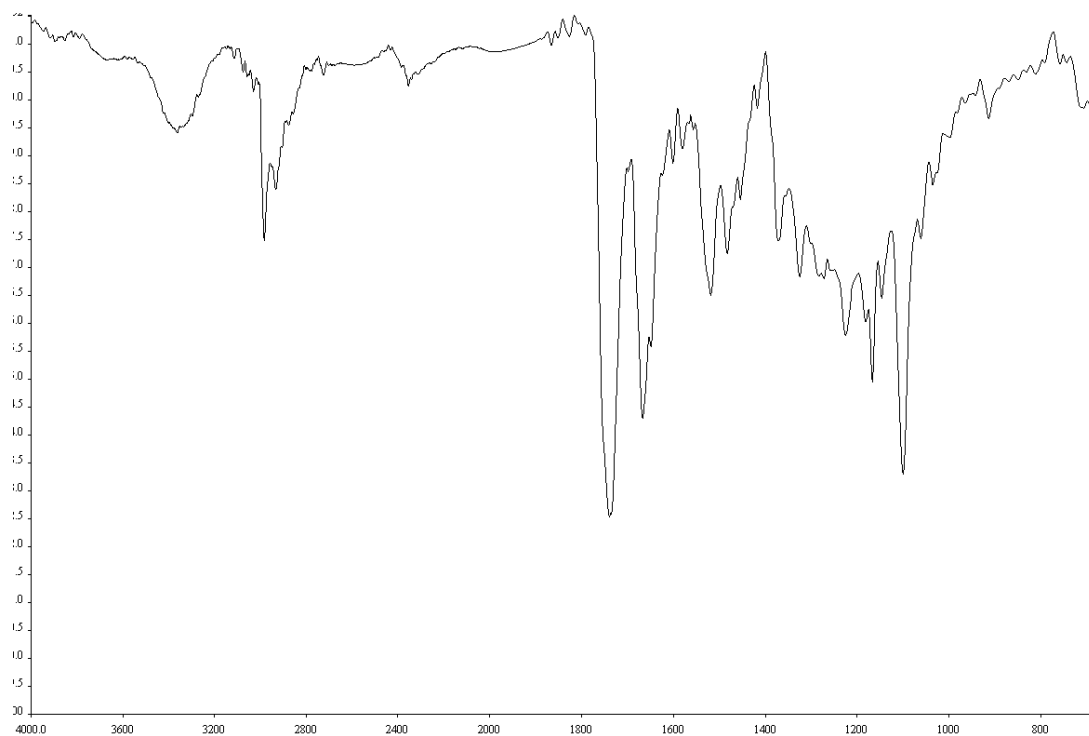
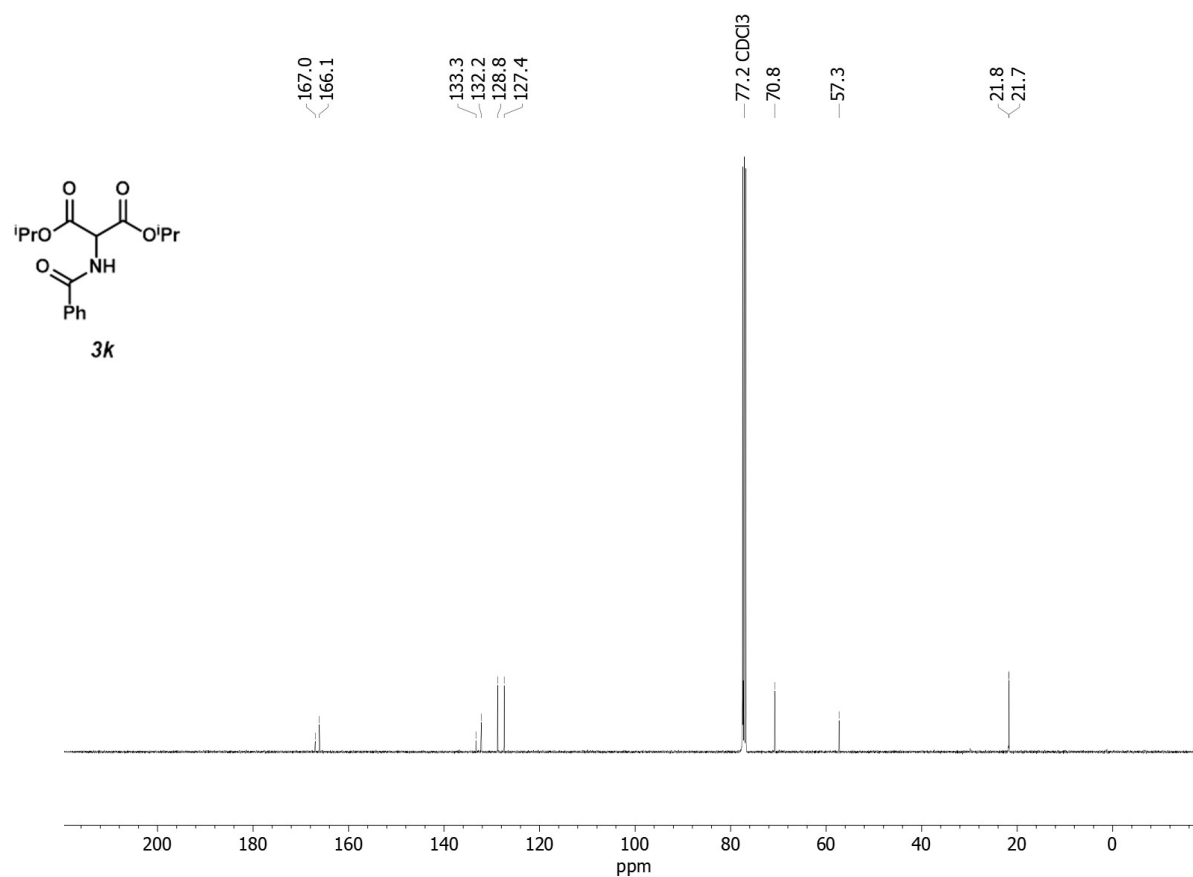


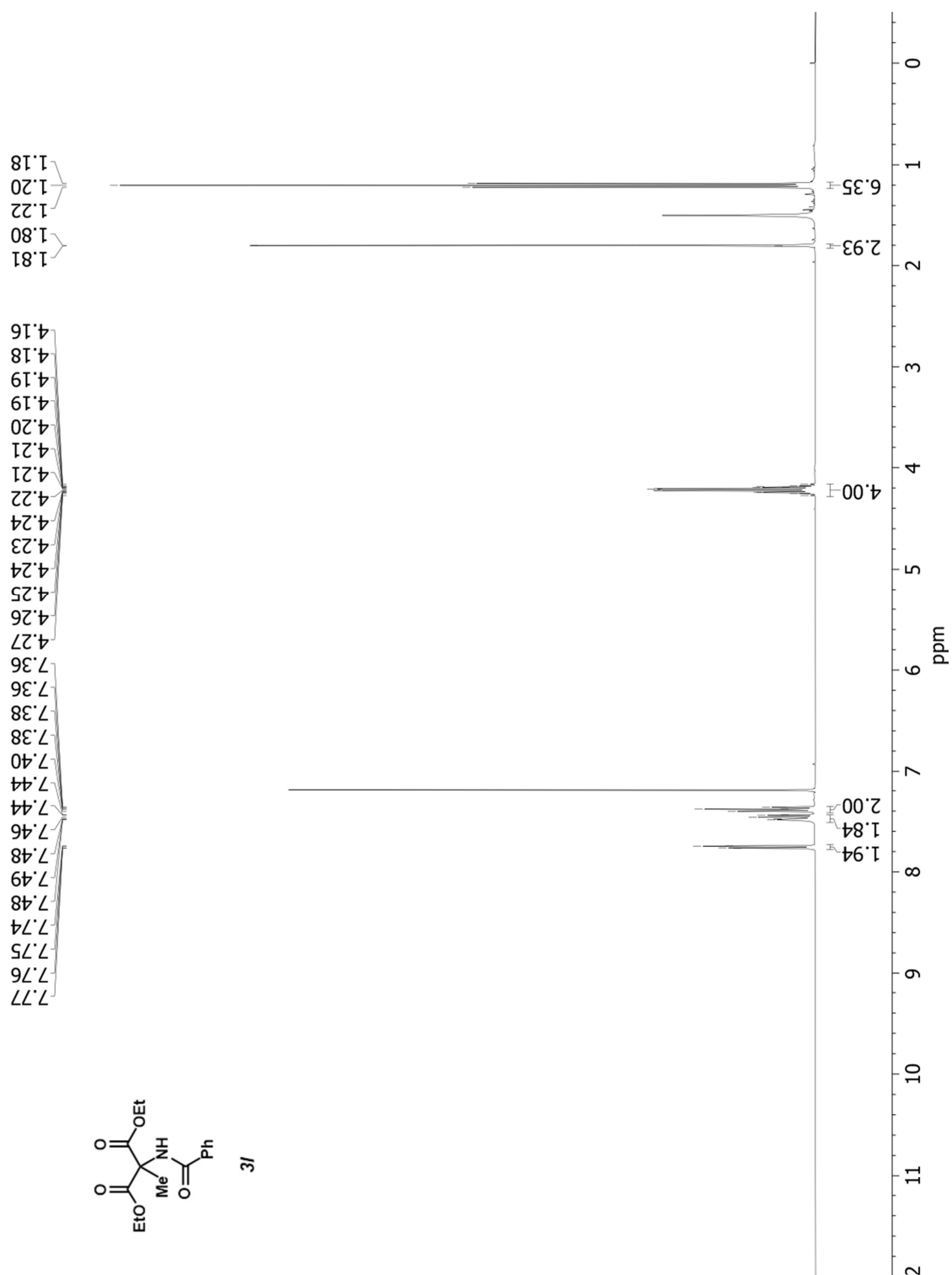
Infrared spectrum (Thin Film, NaCl) of compound **3i**.<sup>13</sup>C NMR (100 MHz, CDCl<sub>3</sub>) of compound **3i**.

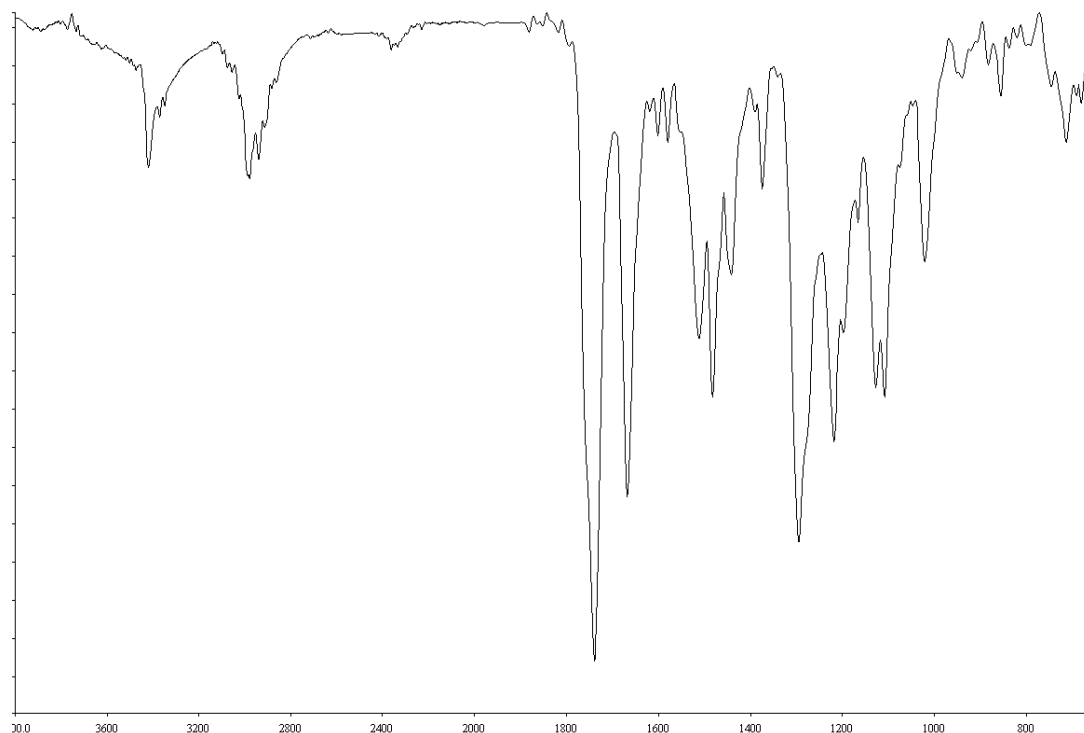
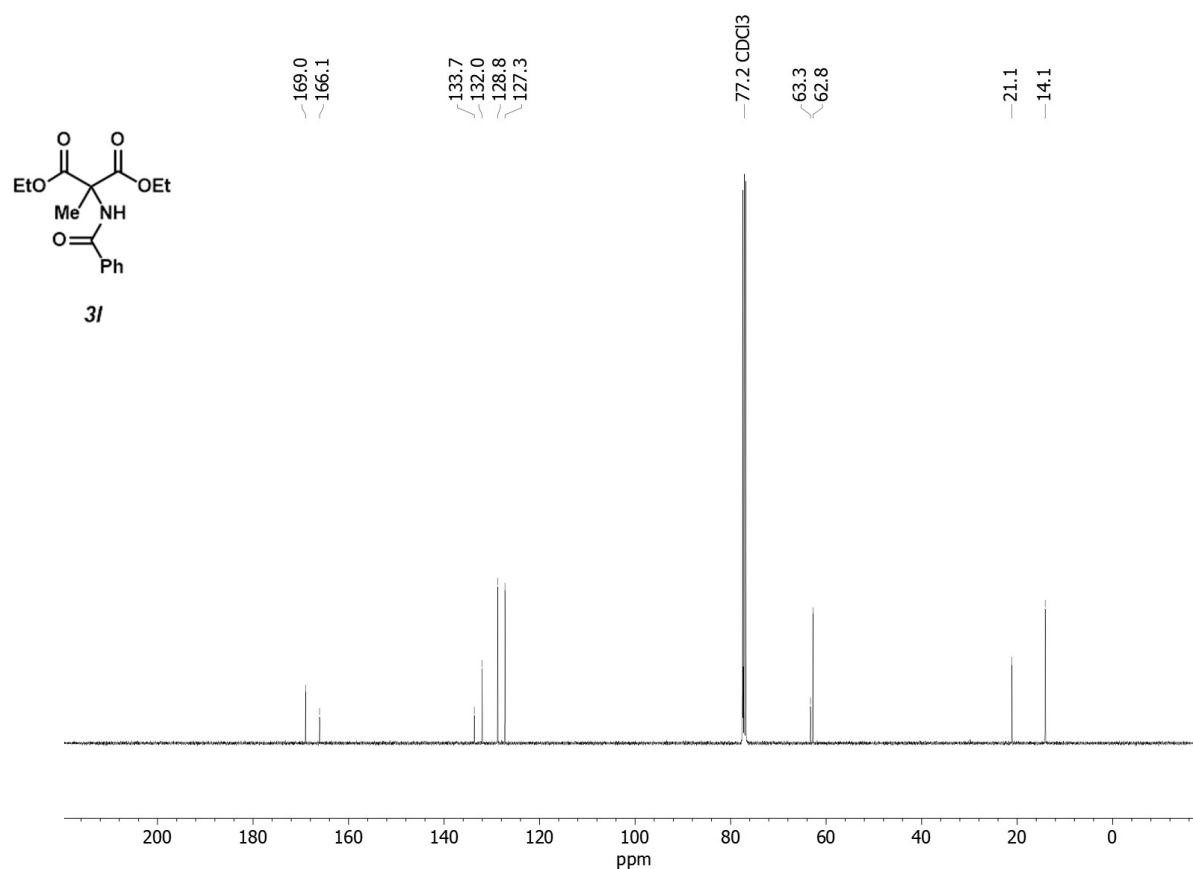


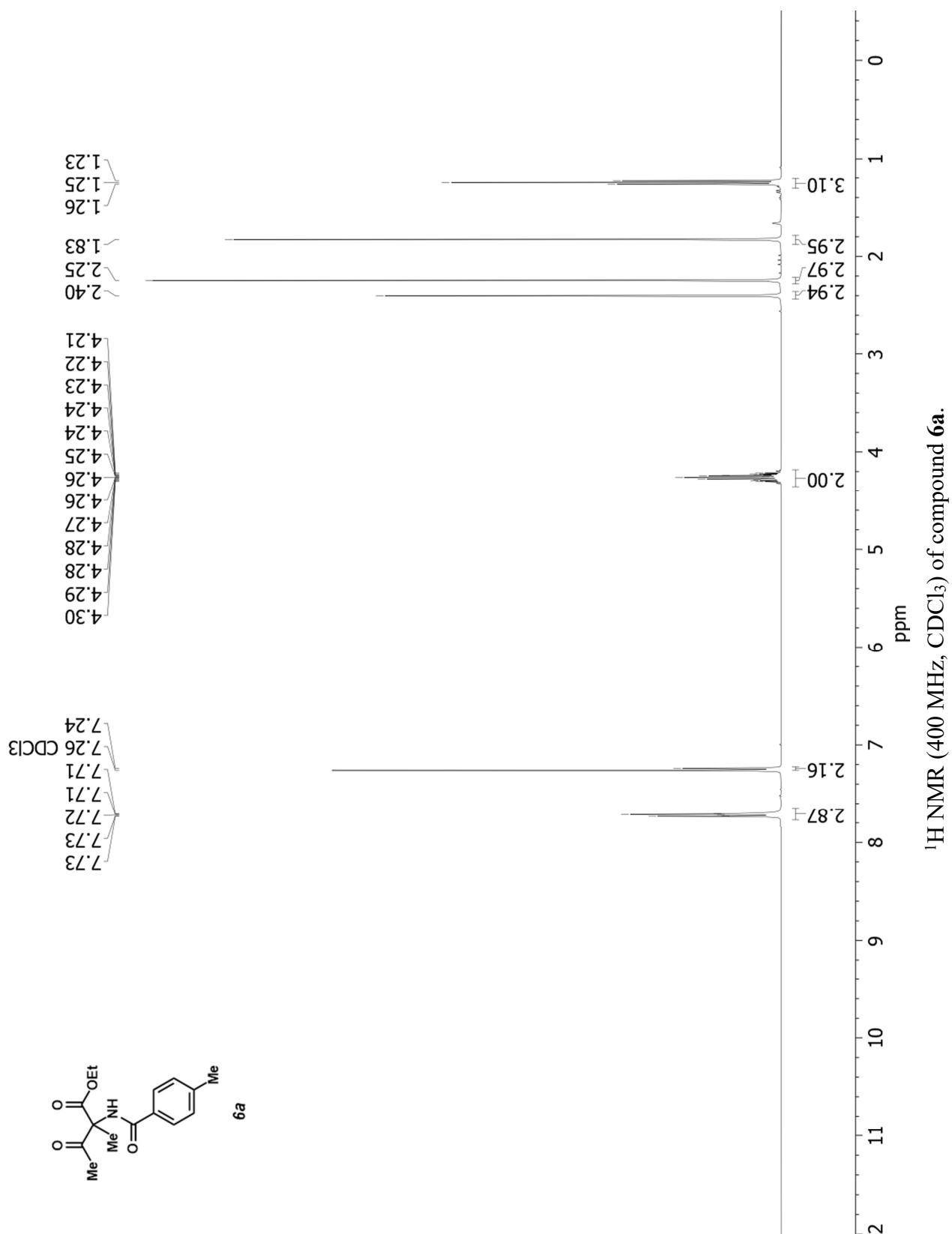
Infrared spectrum (Thin Film, NaCl) of compound **3j**.<sup>13</sup>C NMR (100 MHz, CDCl<sub>3</sub>) of compound **3j**.

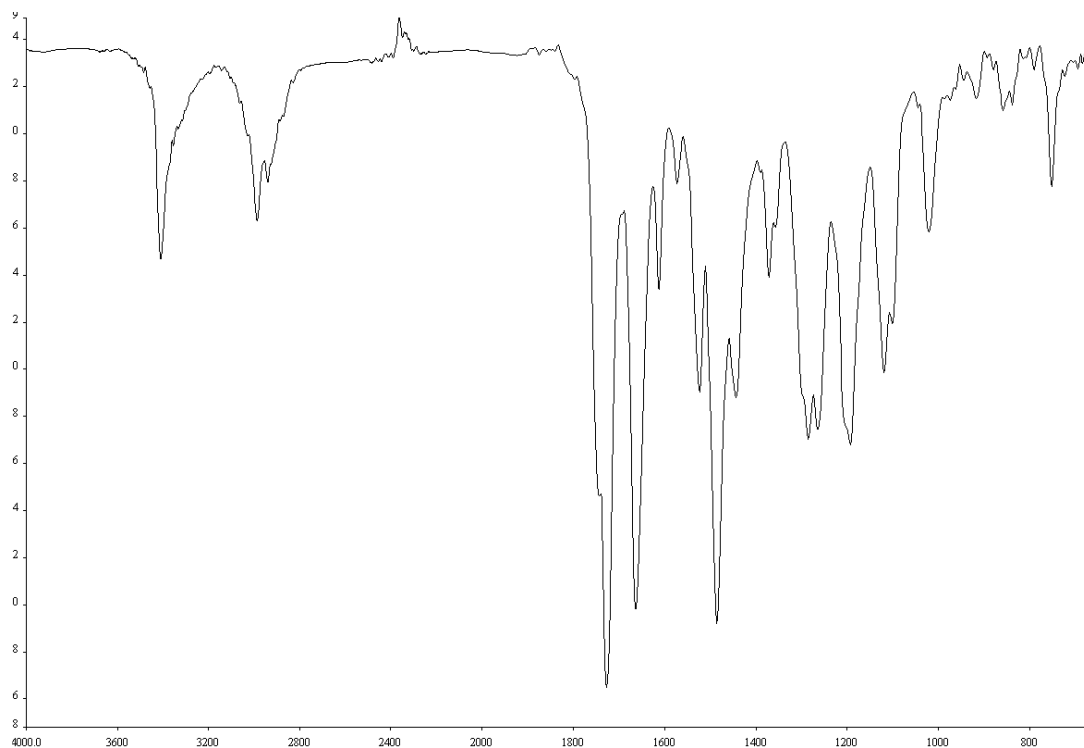
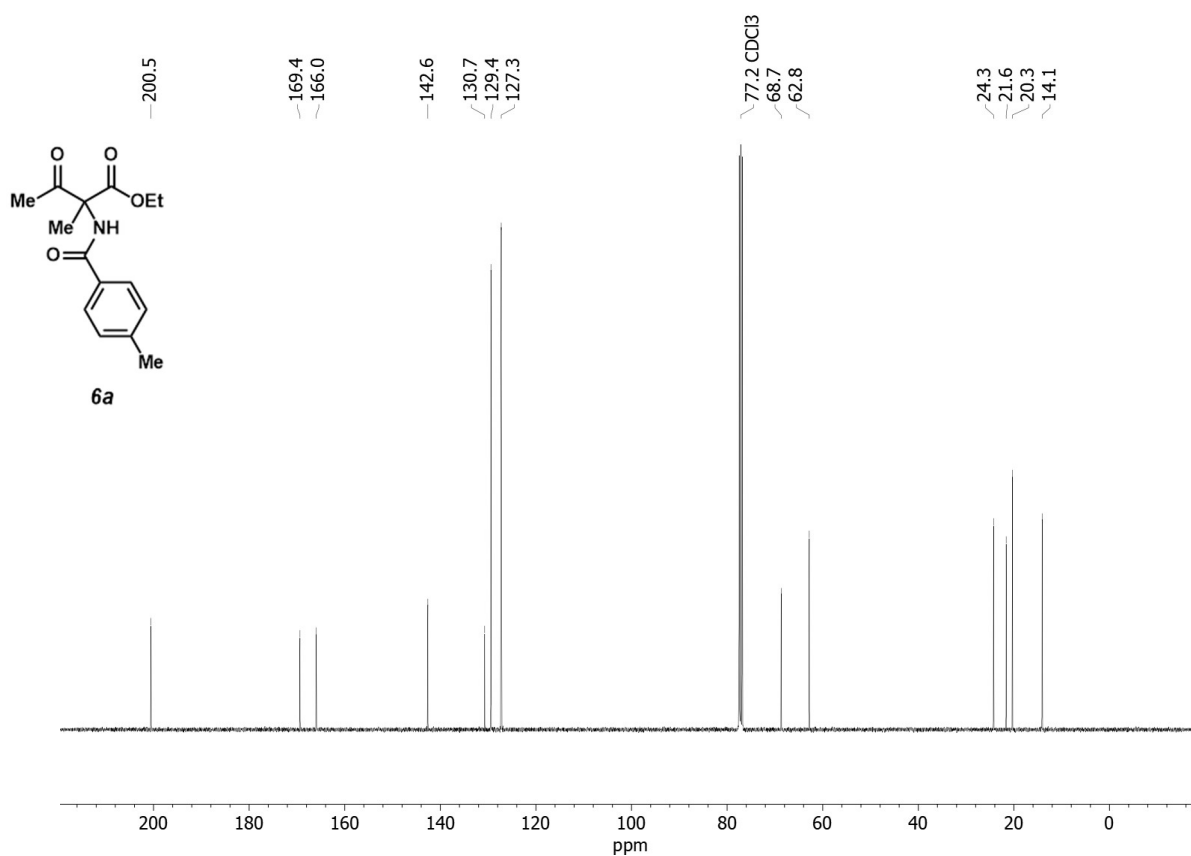
<sup>1</sup>H NMR (400 MHz, CDCl<sub>3</sub>) of compound **3k**.

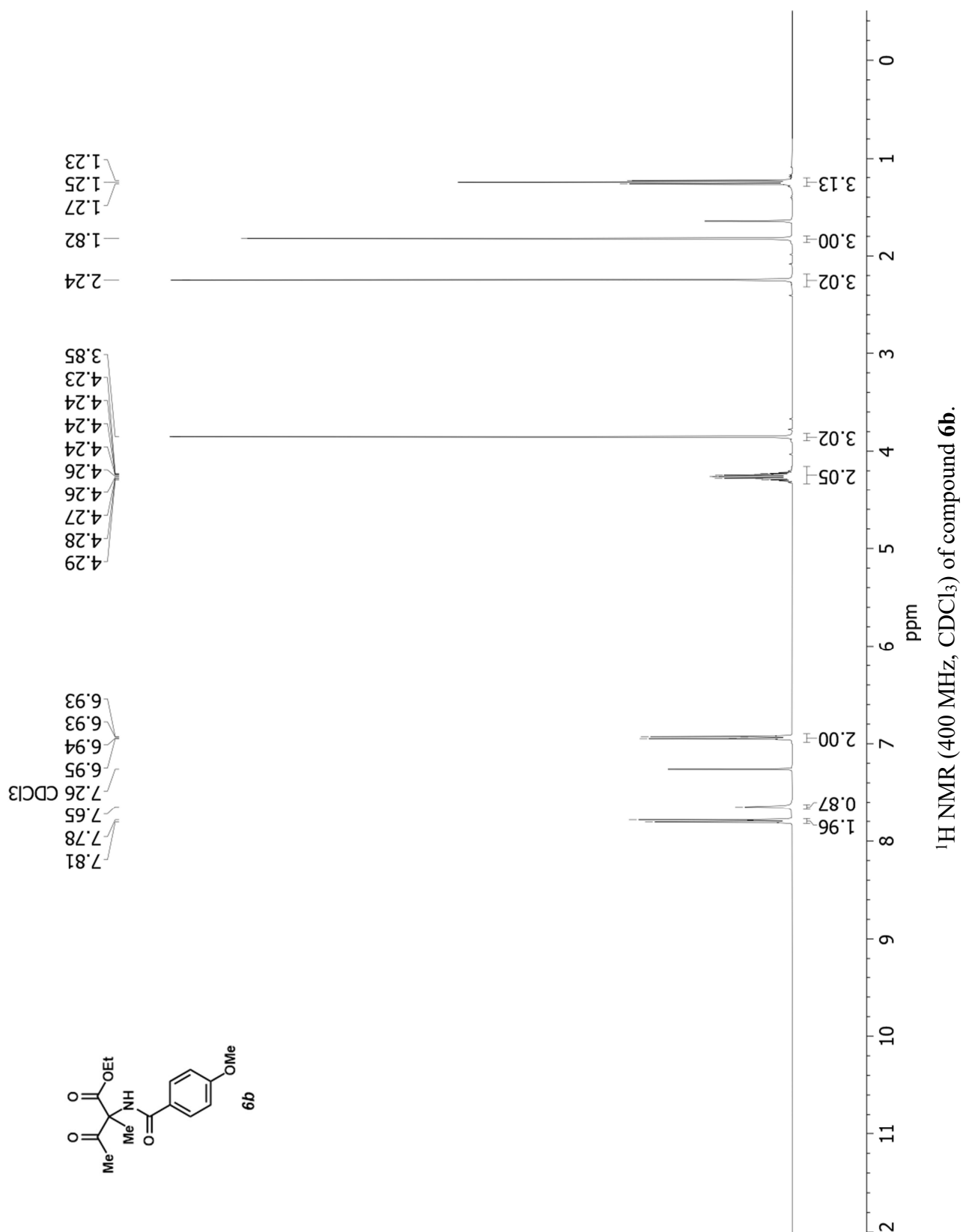
Infrared spectrum (Thin Film, NaCl) of compound **3k**.<sup>13</sup>C NMR (100 MHz, CDCl<sub>3</sub>) of compound **3k**.

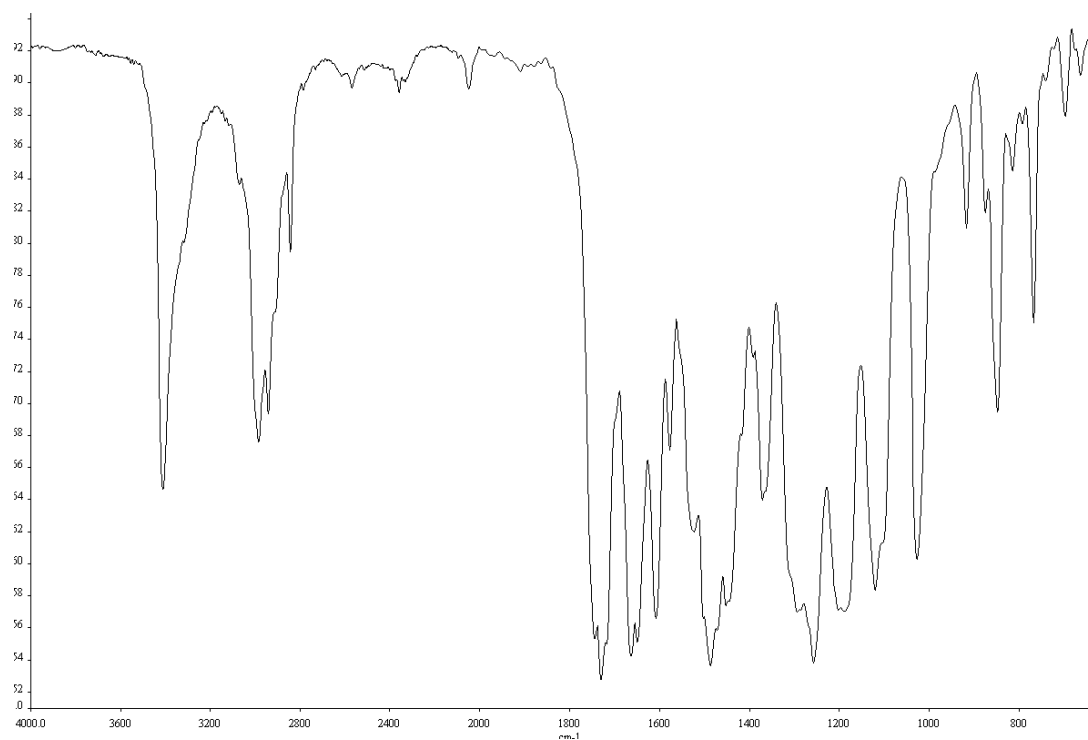
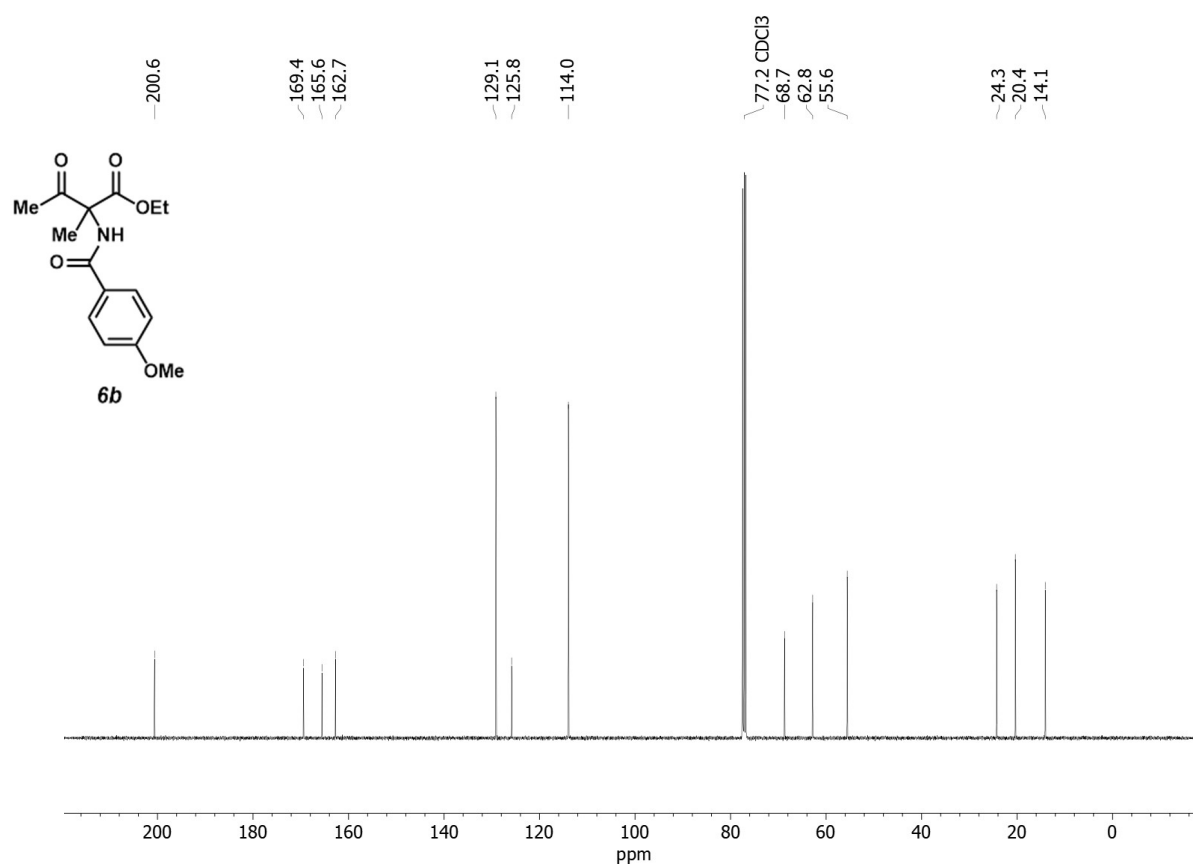


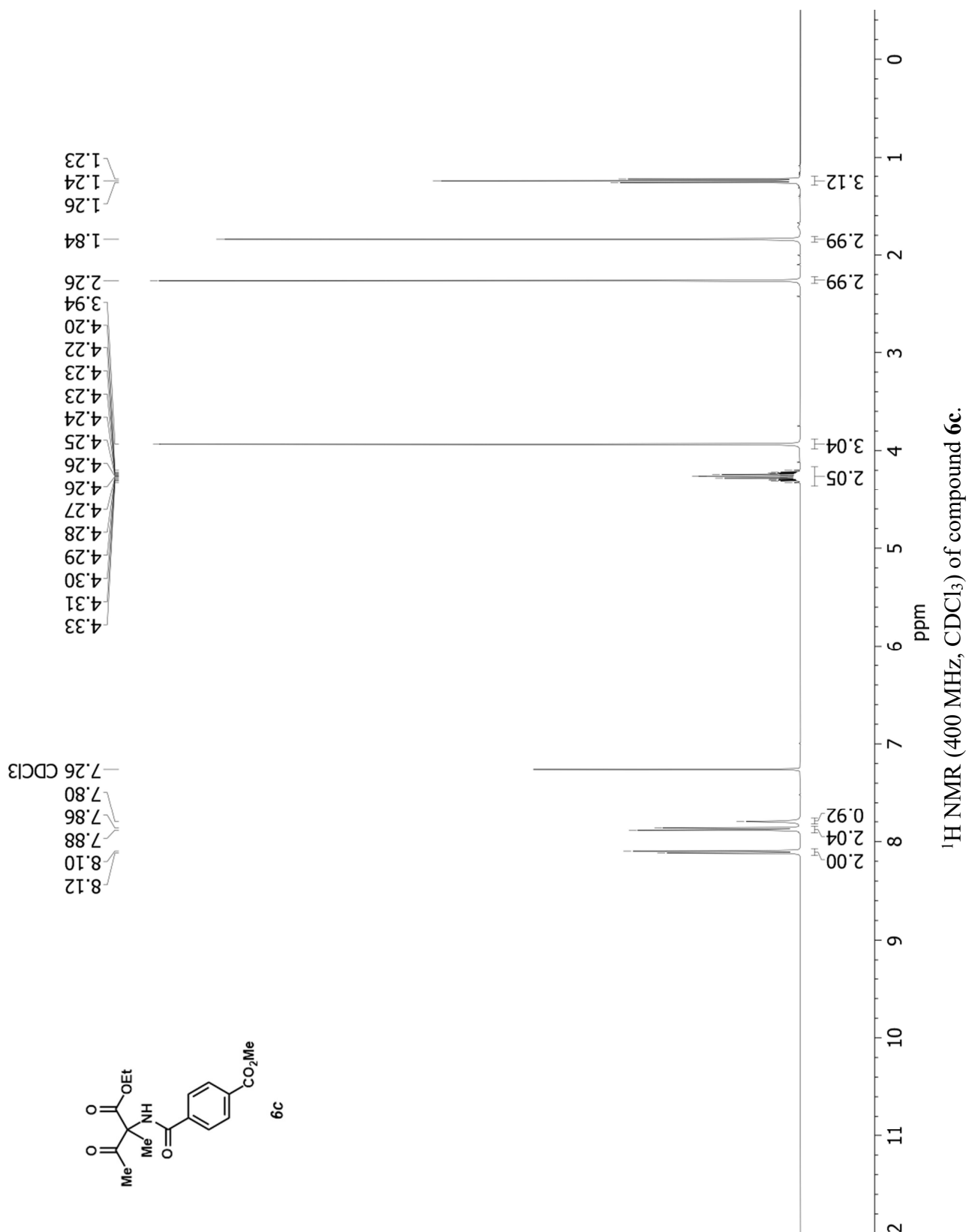
Infrared spectrum (Thin Film, NaCl) of compound **3I**.<sup>13</sup>C NMR (100 MHz, CDCl<sub>3</sub>) of compound **3I**.

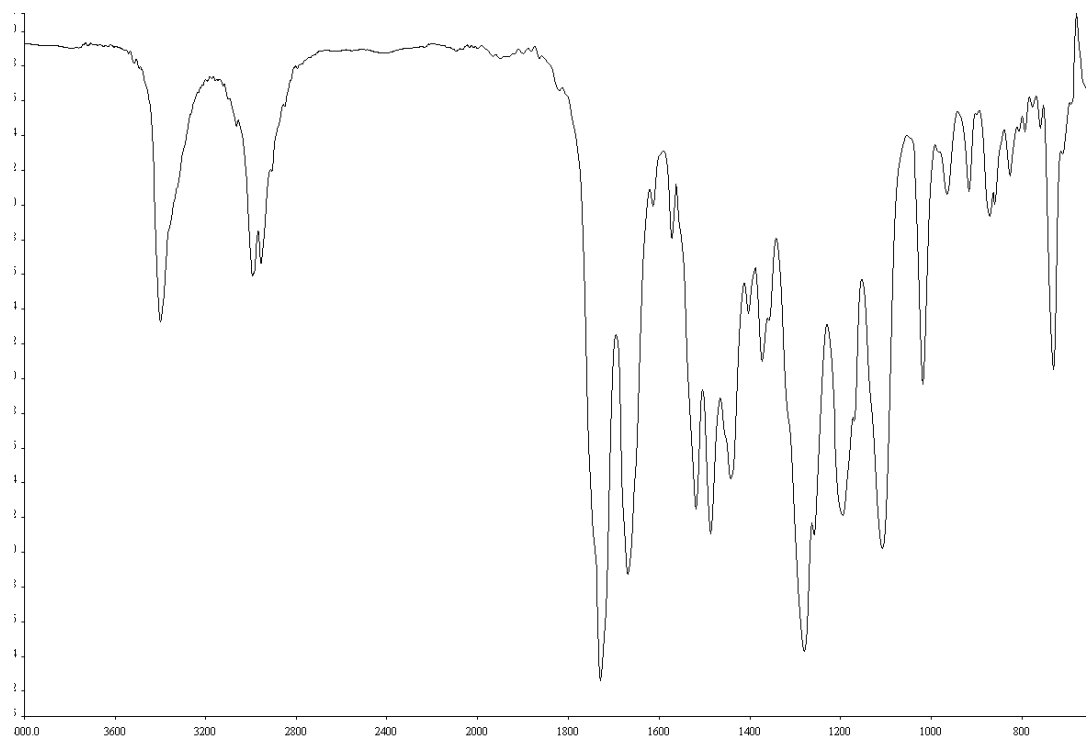
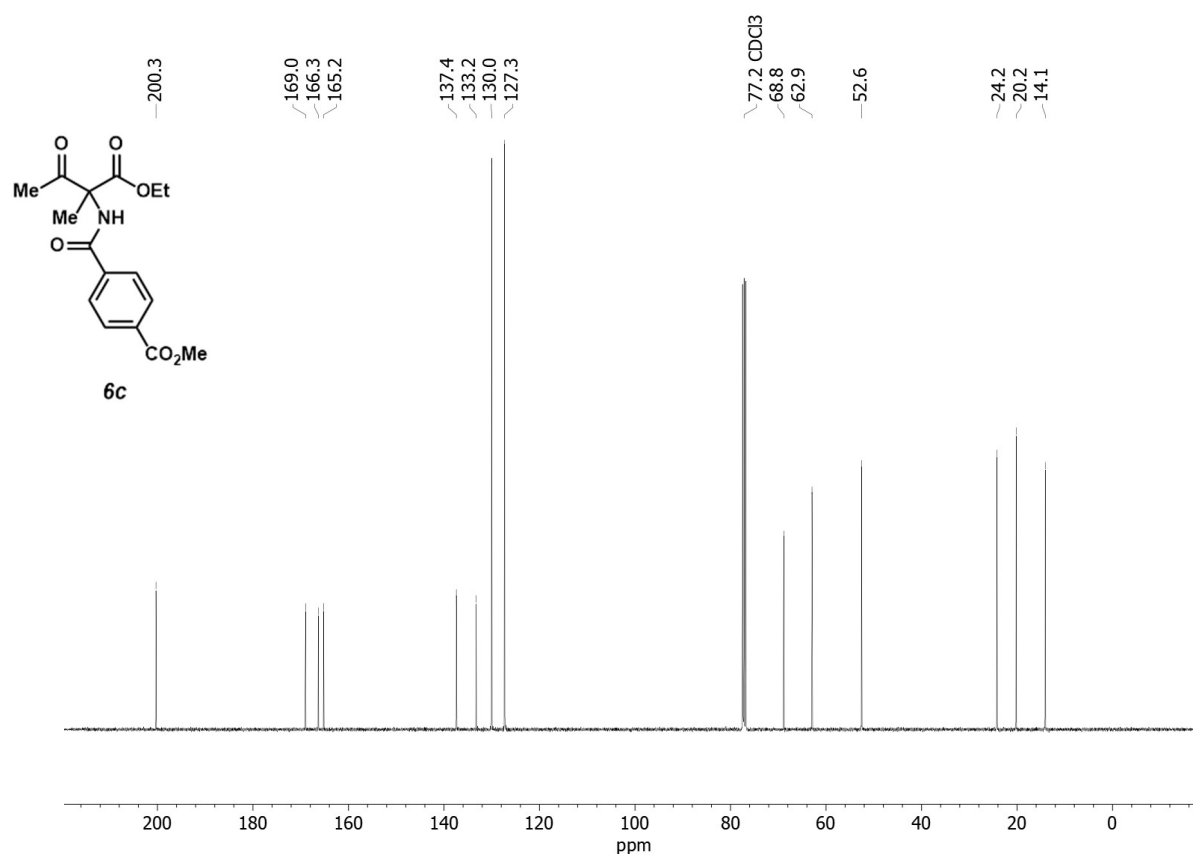


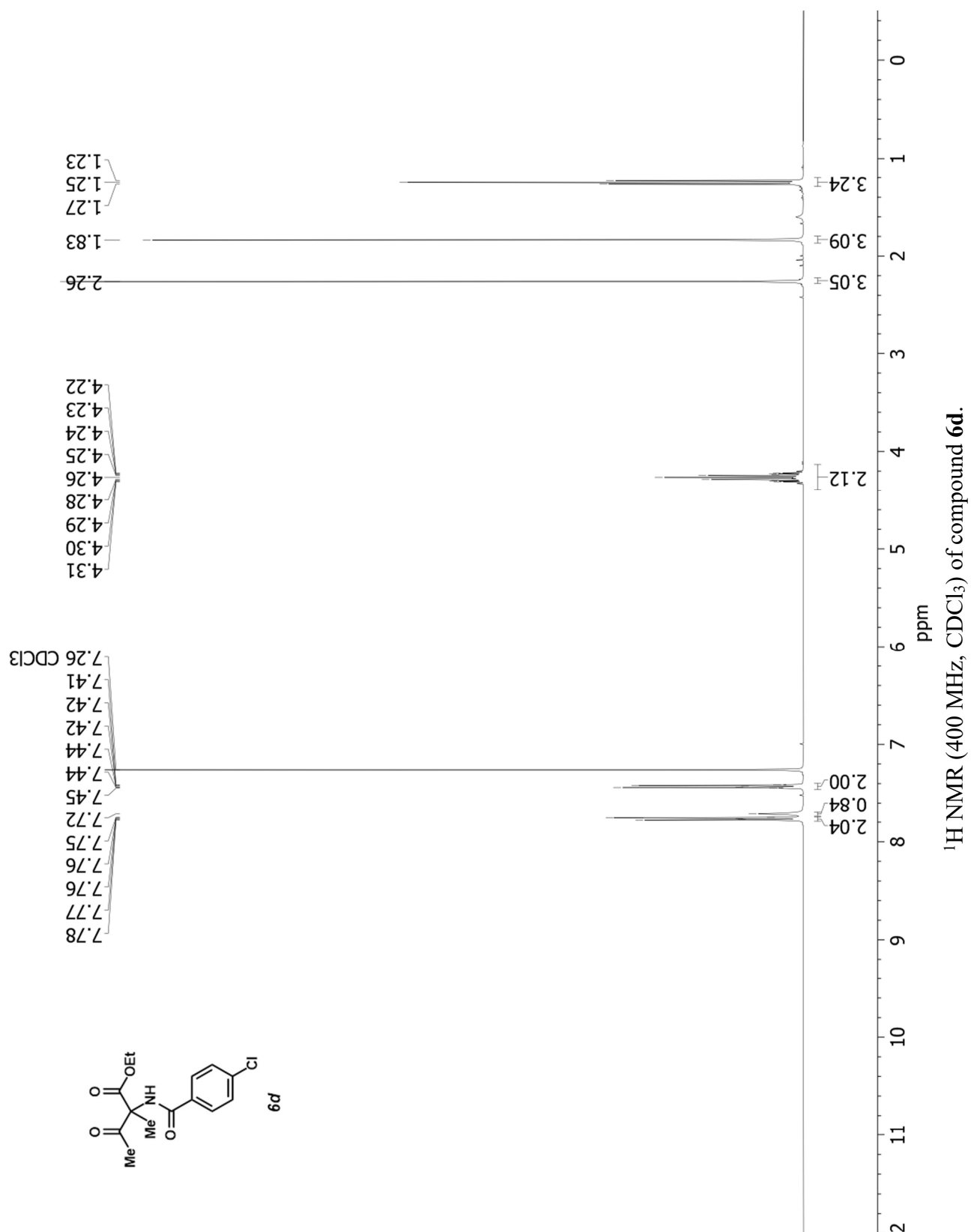
Infrared spectrum (Thin Film, NaCl) of compound **6a**.<sup>13</sup>C NMR (100 MHz, CDCl<sub>3</sub>) of compound **6a**.

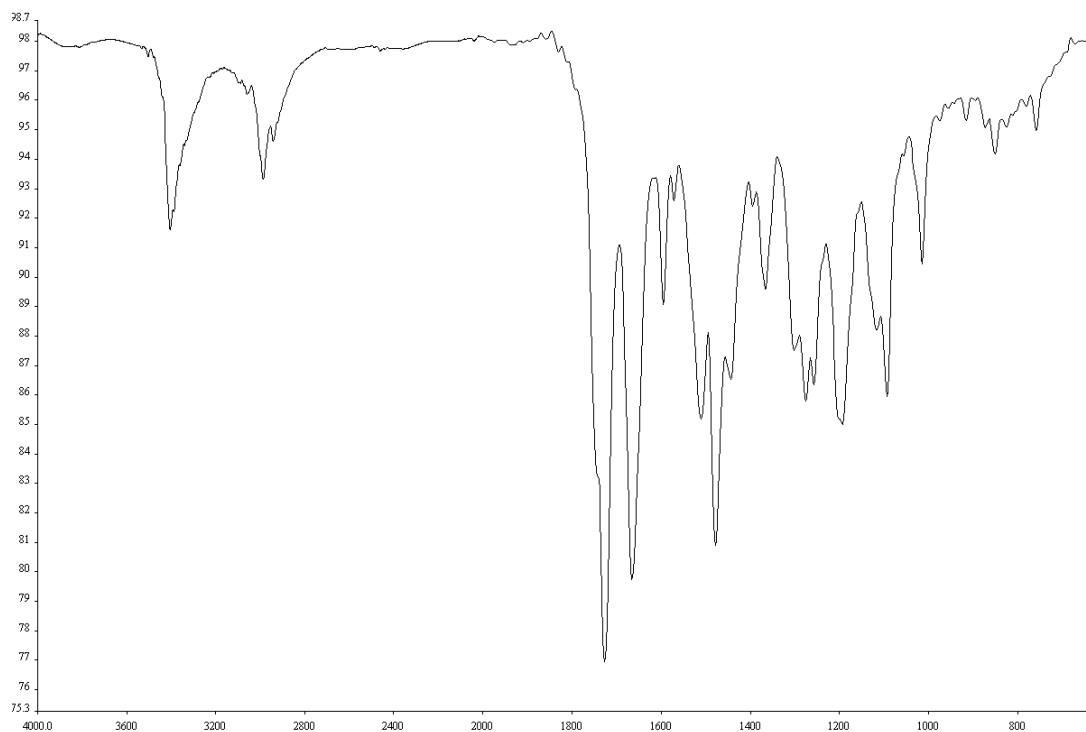
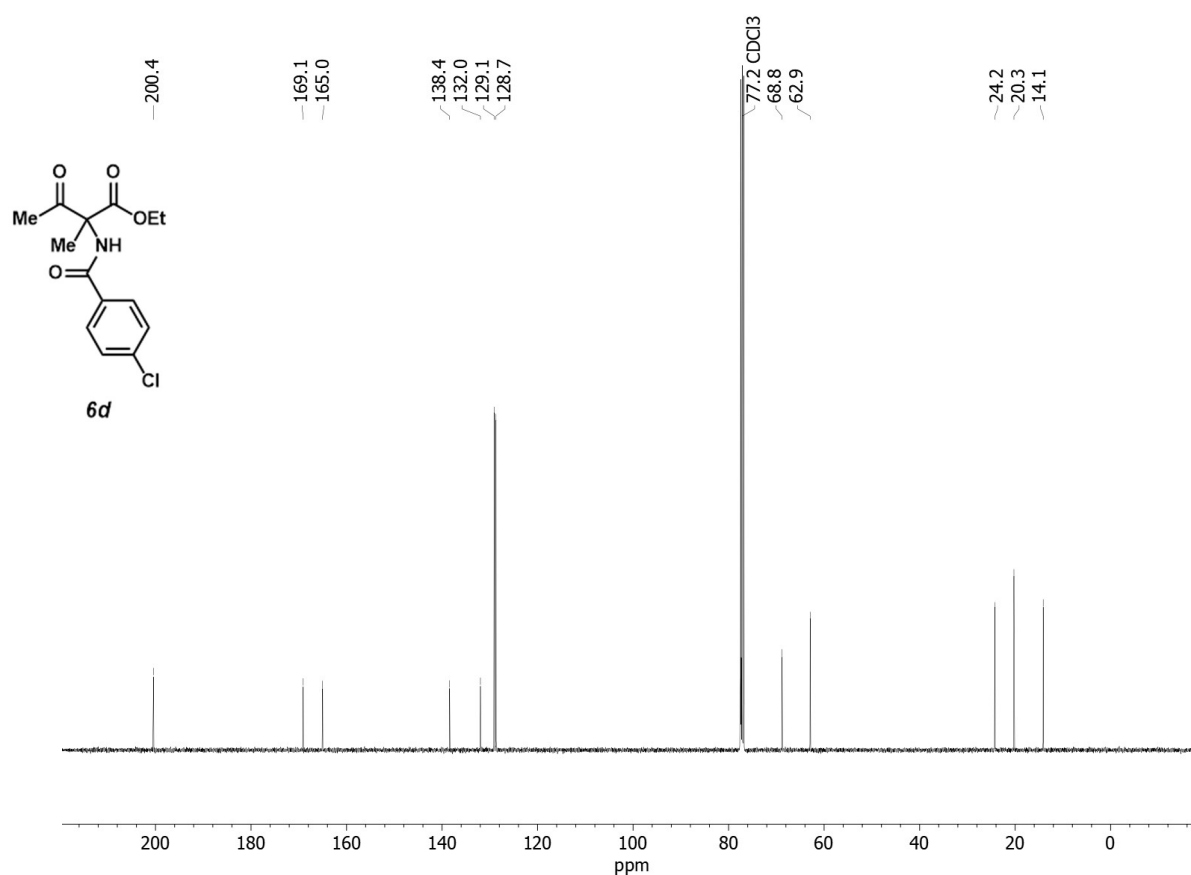


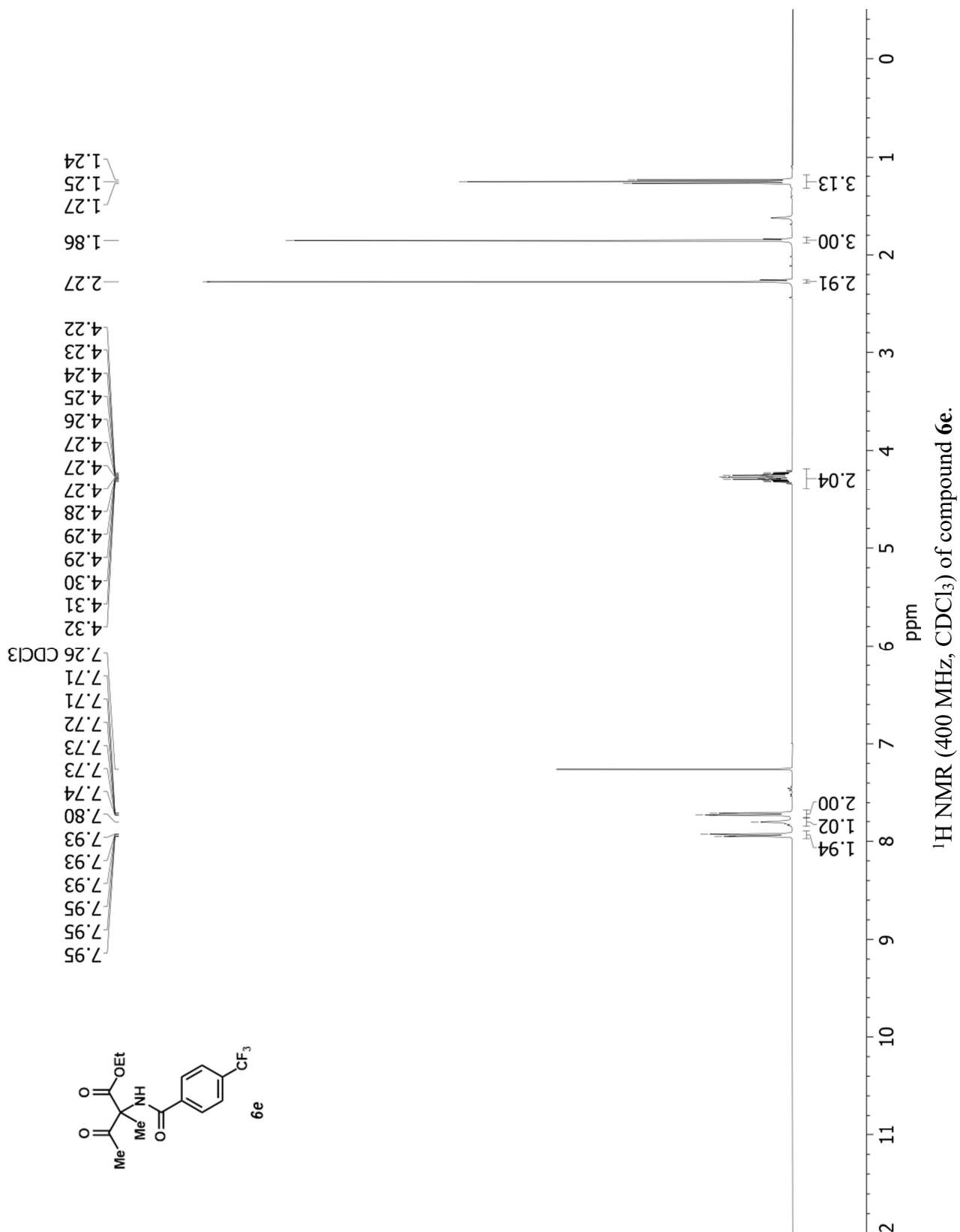
Infrared spectrum (Thin Film, NaCl) of compound **6b**.<sup>13</sup>C NMR (100 MHz, CDCl<sub>3</sub>) of compound **6b**.

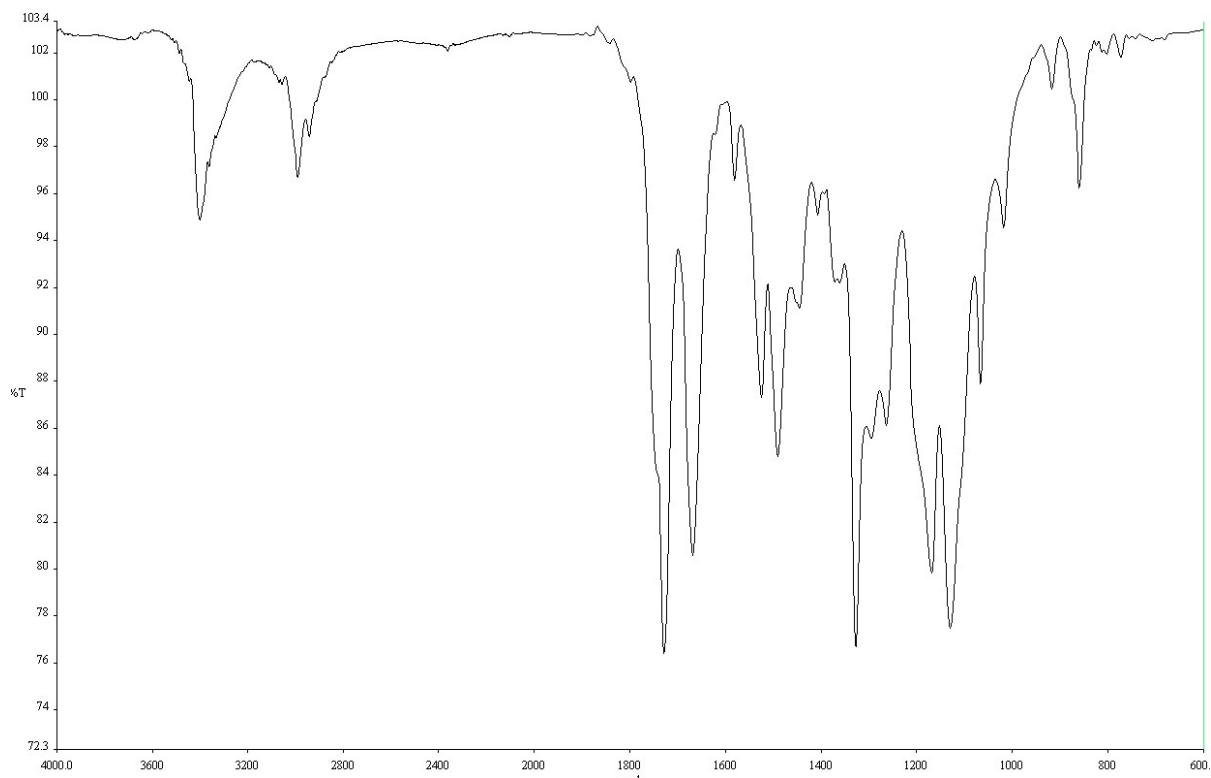
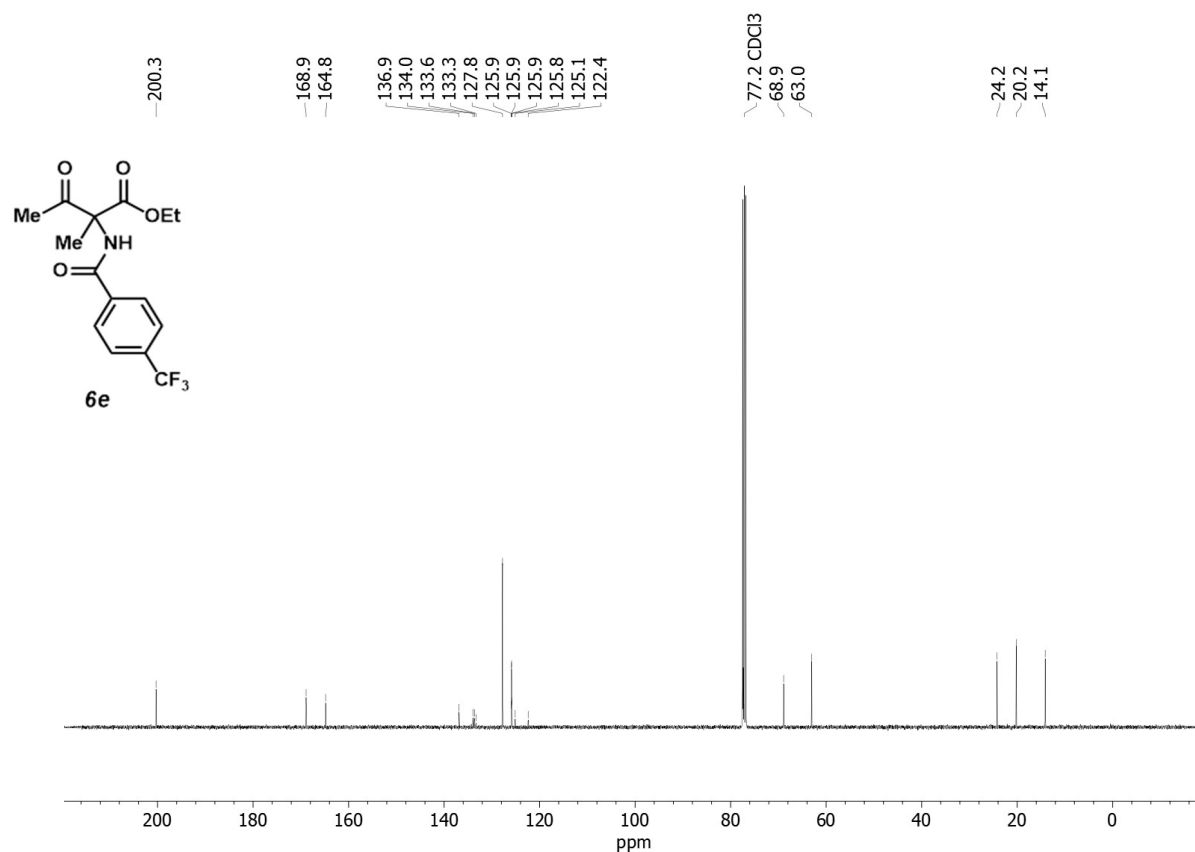


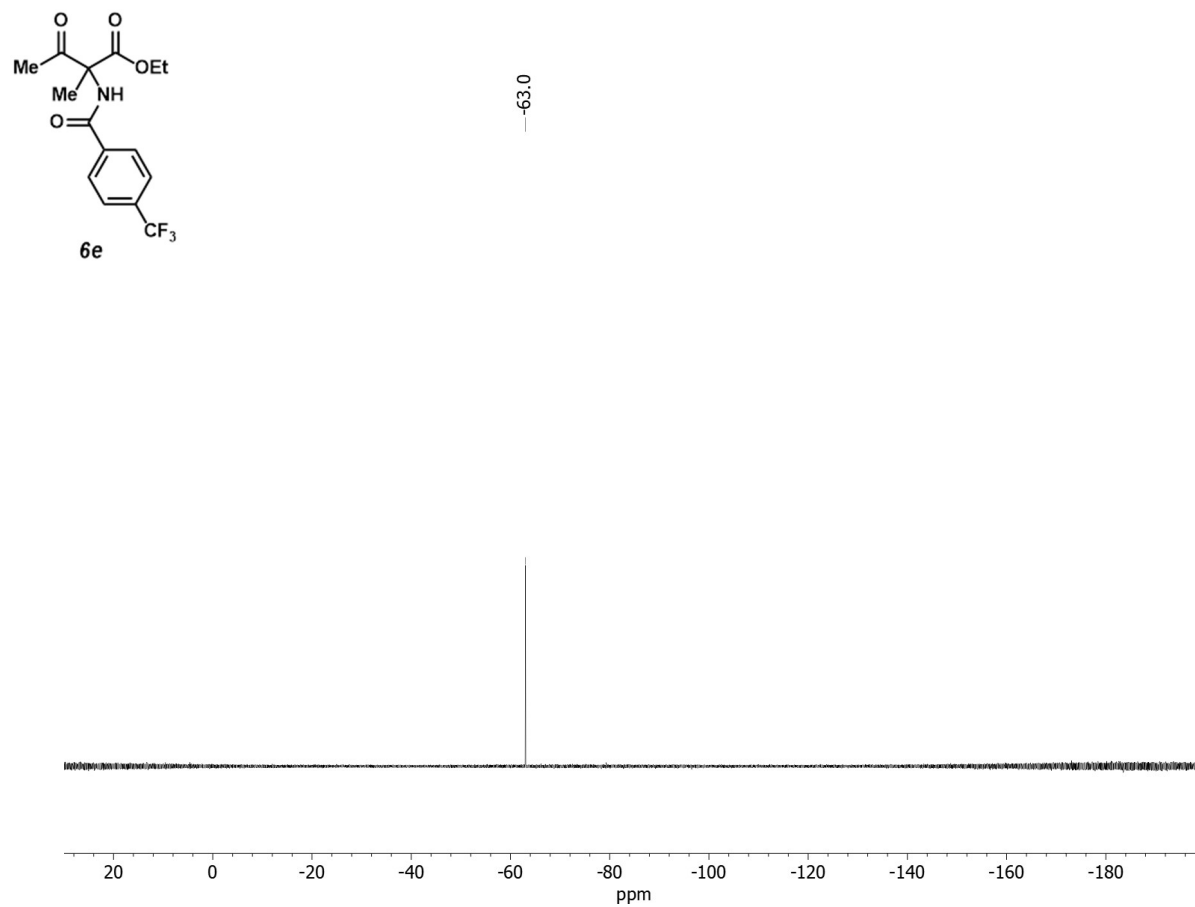
Infrared spectrum (Thin Film, NaCl) of compound **6c**.<sup>13</sup>C NMR (100 MHz, CDCl<sub>3</sub>) of compound **6c**.

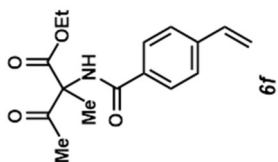


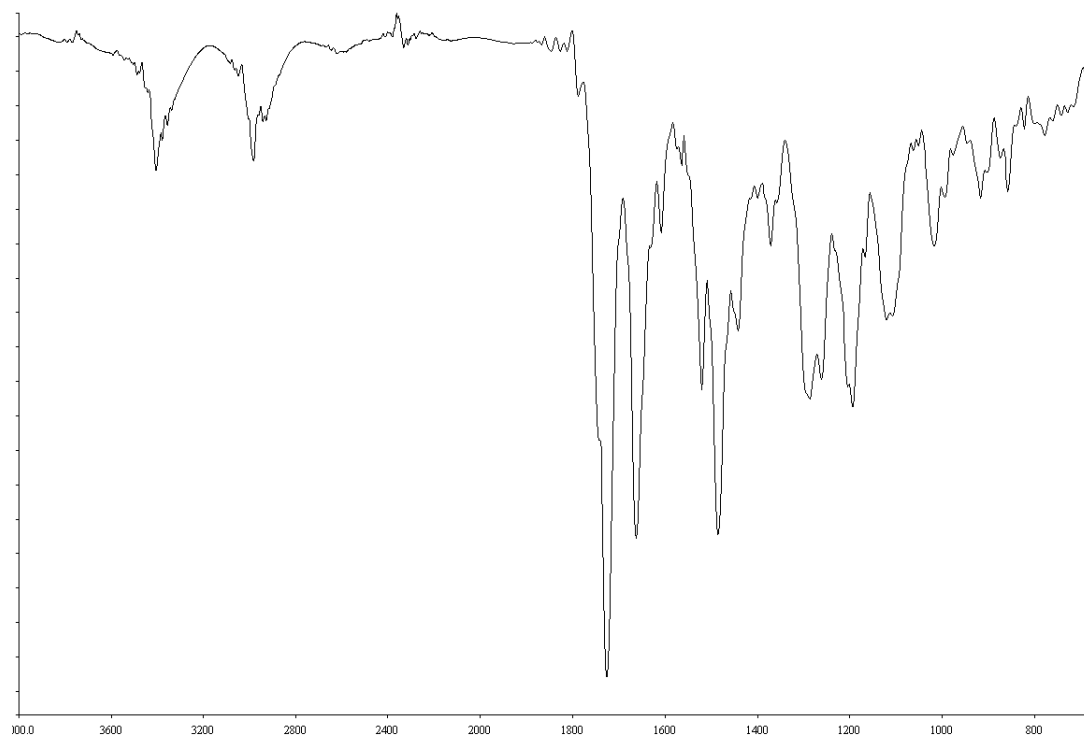
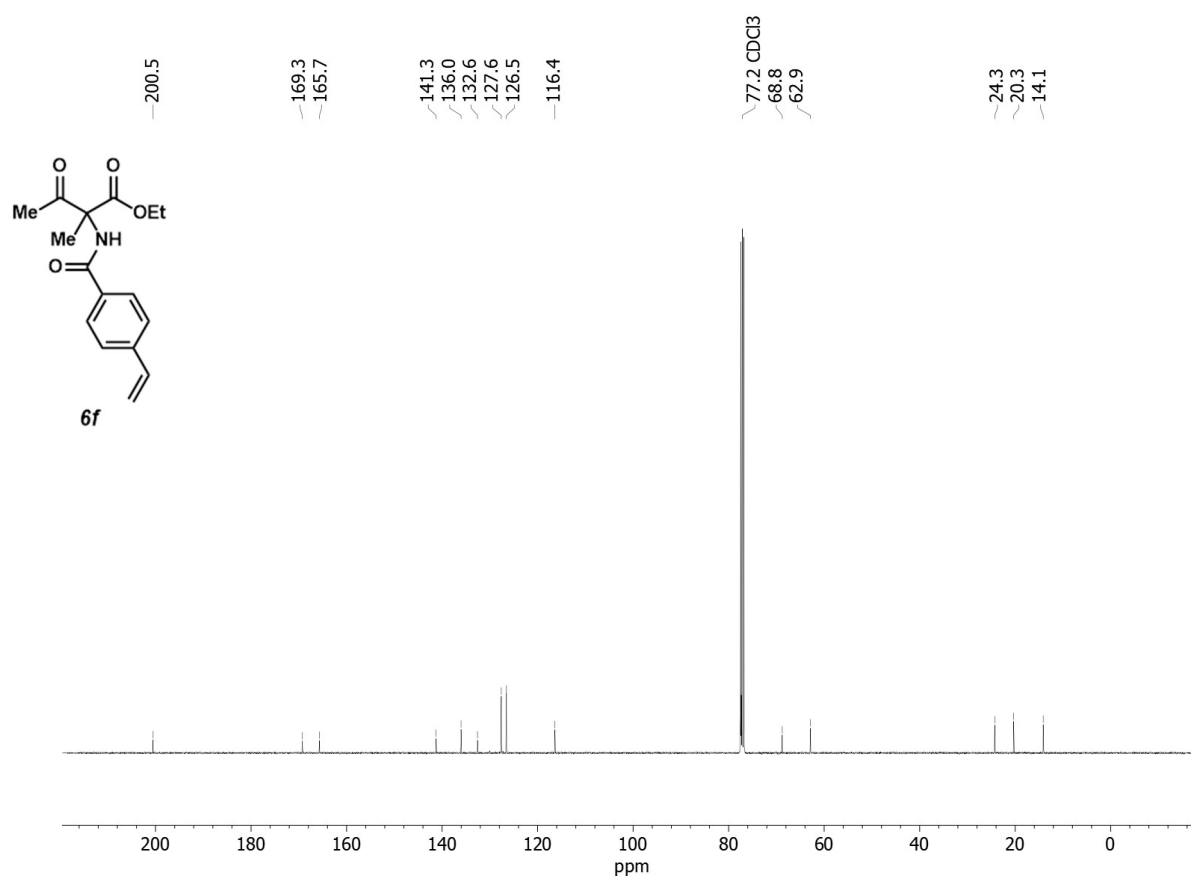
Infrared spectrum (Thin Film, NaCl) of compound **6d**.<sup>13</sup>C NMR (100 MHz, CDCl<sub>3</sub>) of compound **6d**.

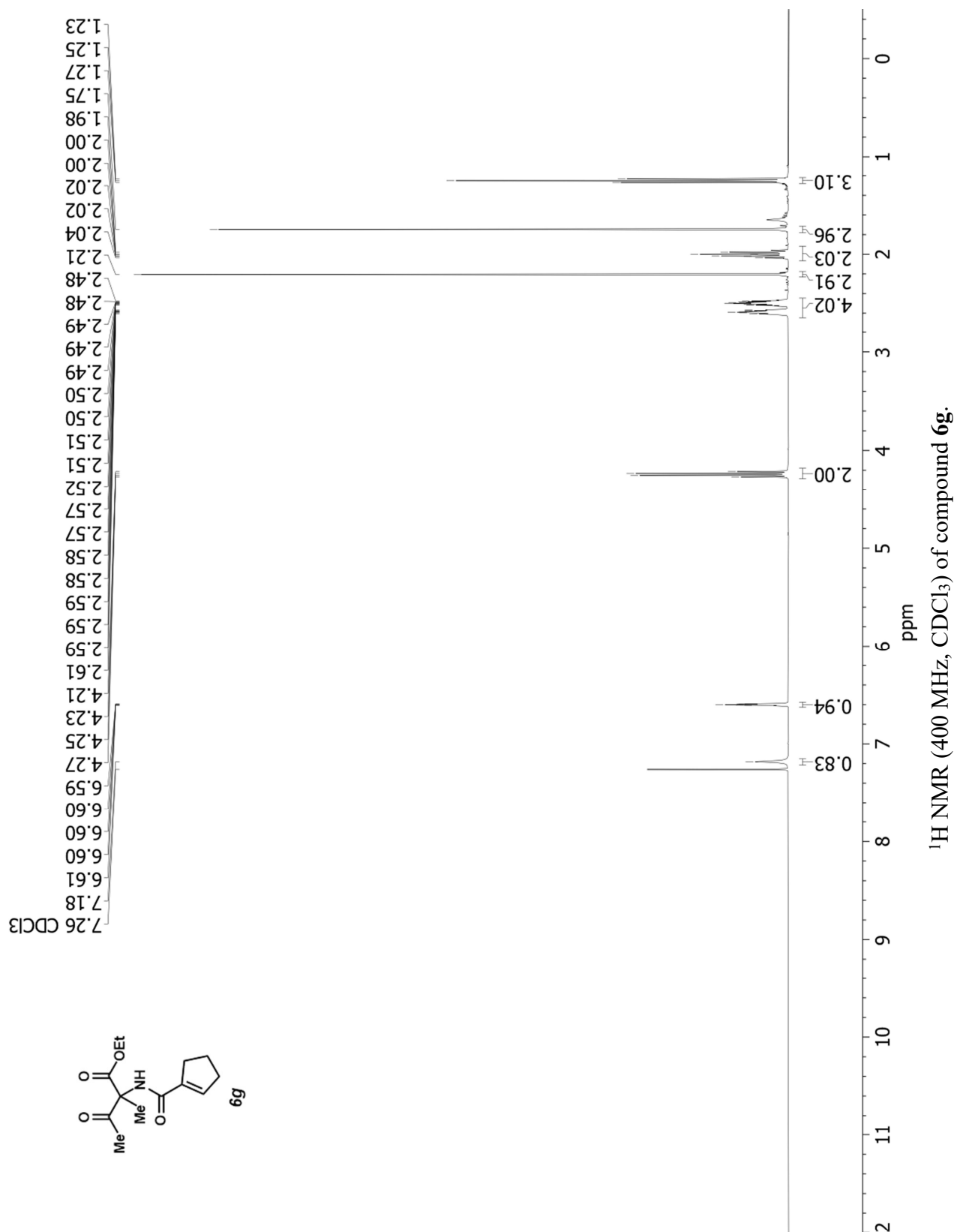


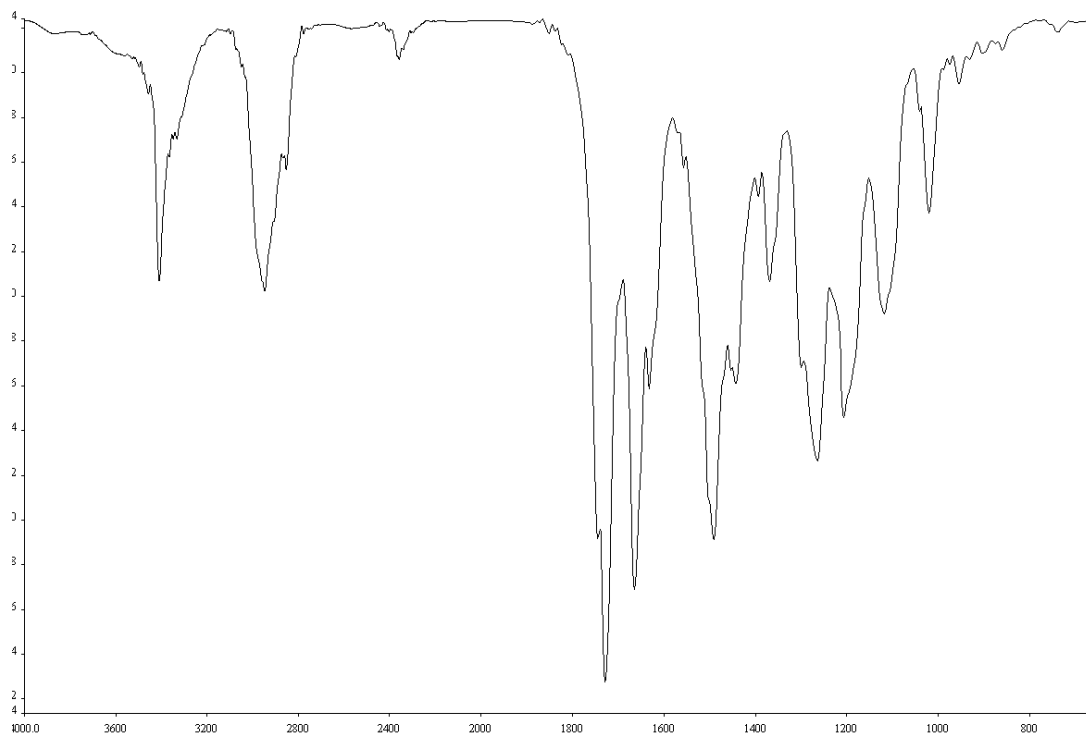
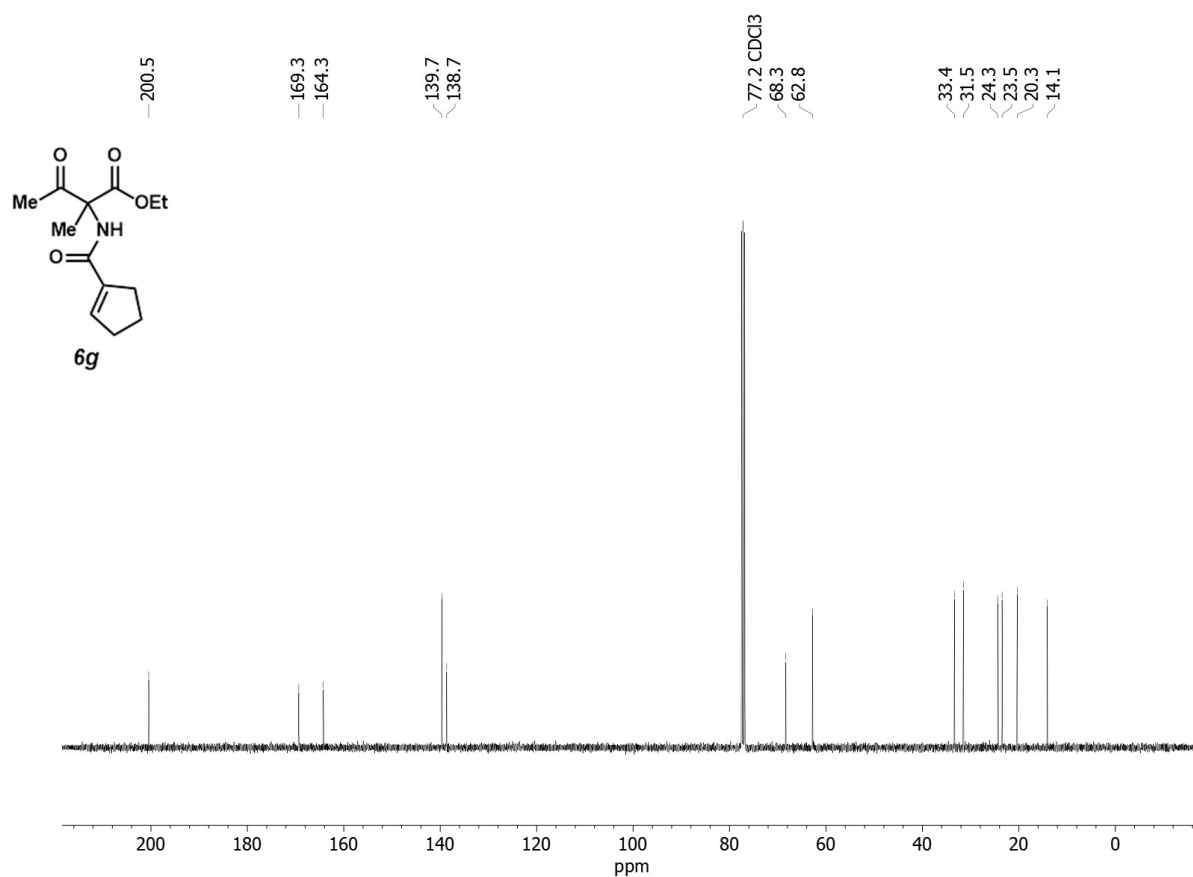
Infrared spectrum (Thin Film, NaCl) of compound **6e**.<sup>13</sup>C NMR (100 MHz, CDCl<sub>3</sub>) of compound **6e**.

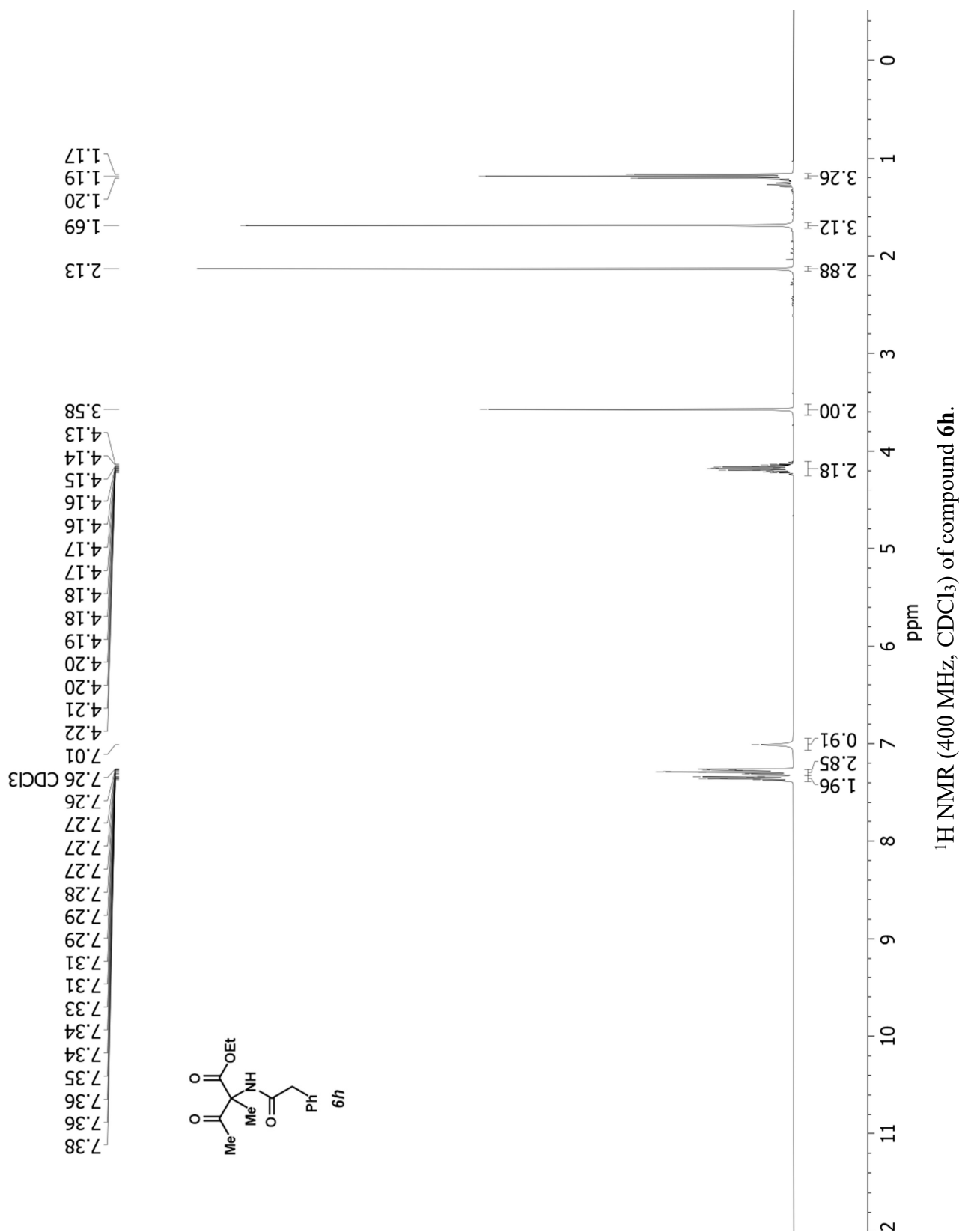
 $^{19}\text{F}$  NMR (282 MHz,  $\text{CDCl}_3$ ) of compound **6e**.

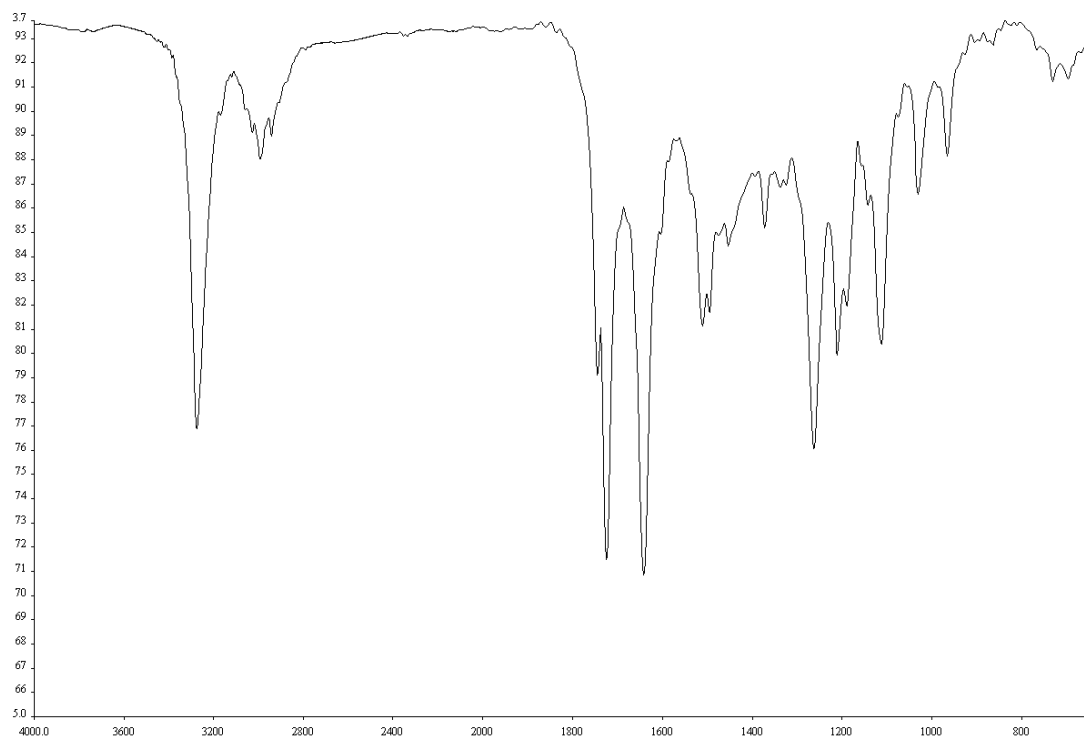
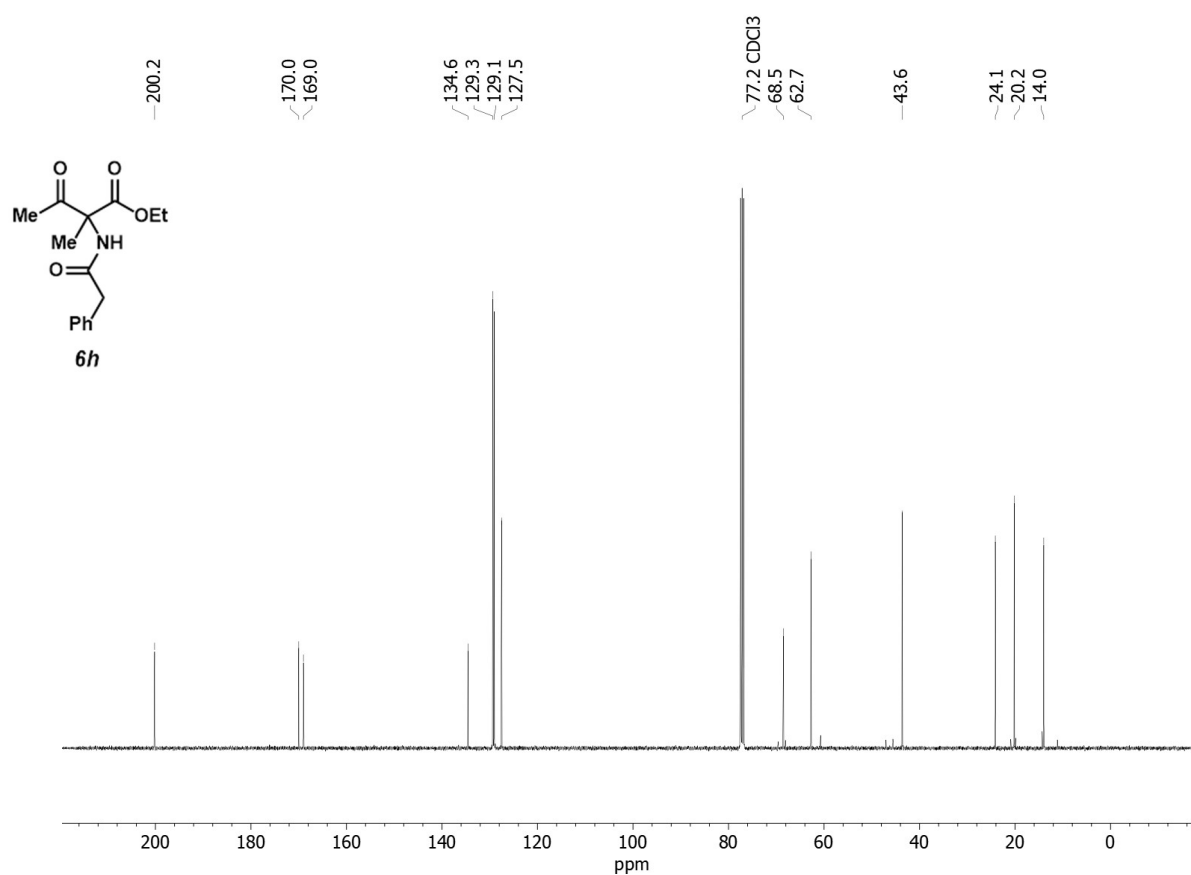


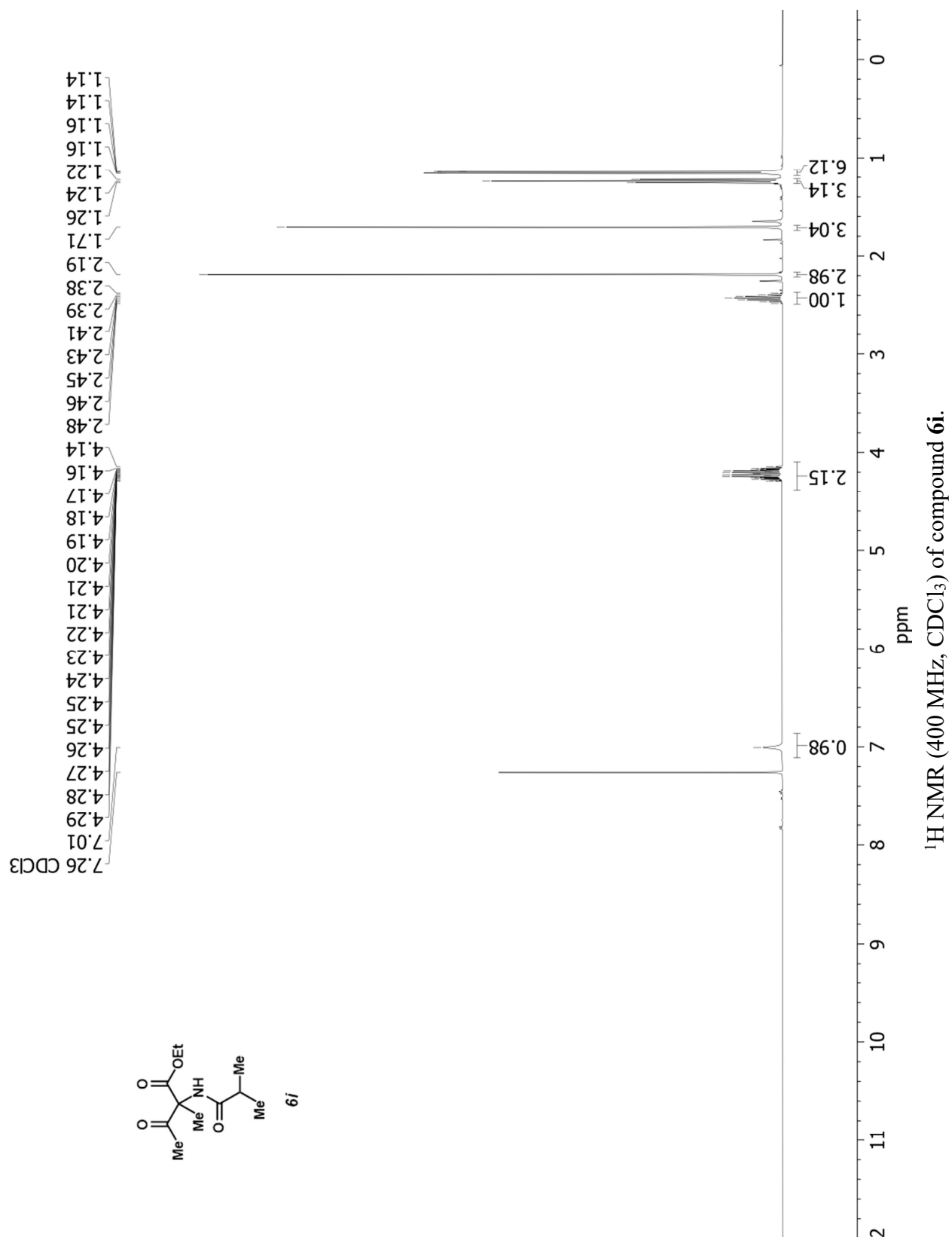
Infrared spectrum (Thin Film, NaCl) of compound **6f**.<sup>13</sup>C NMR (100 MHz, CDCl<sub>3</sub>) of compound **6f**.

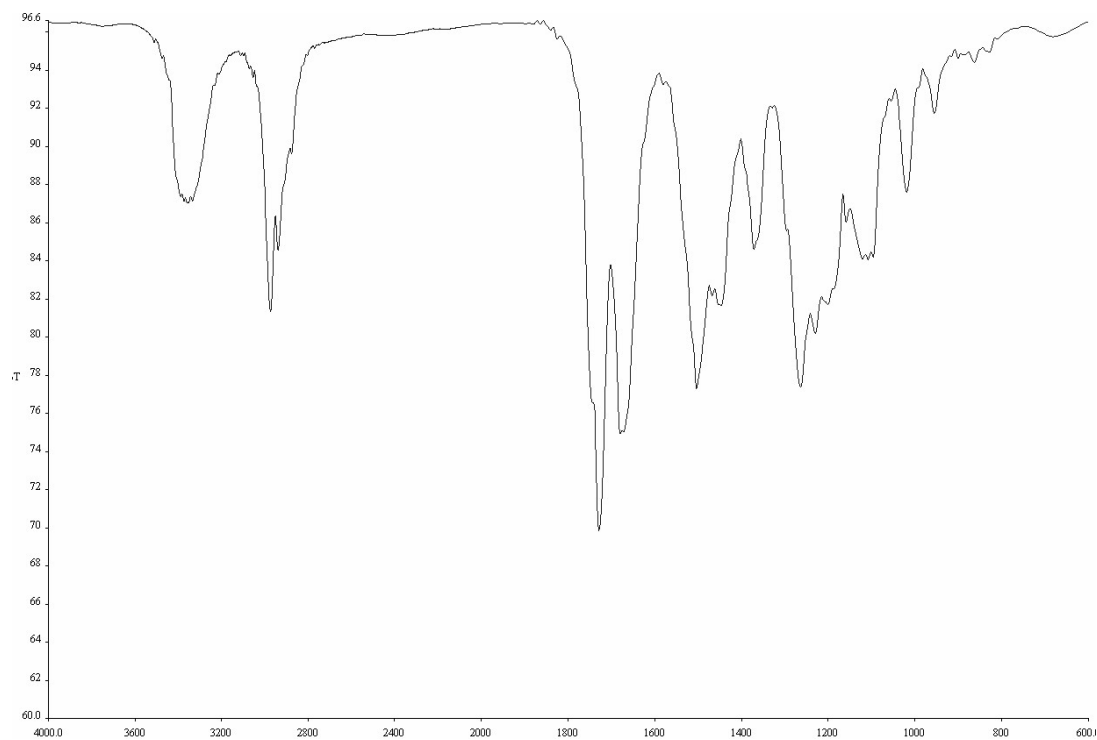
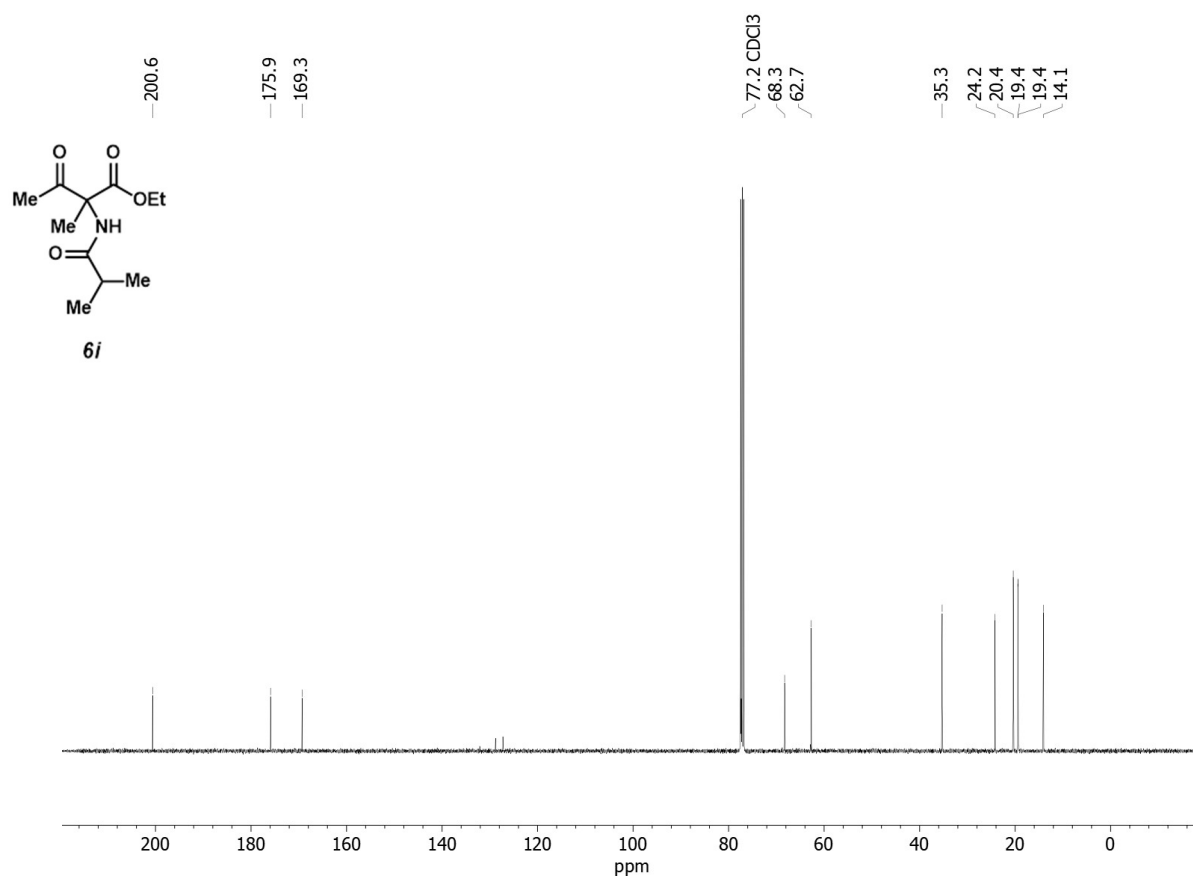


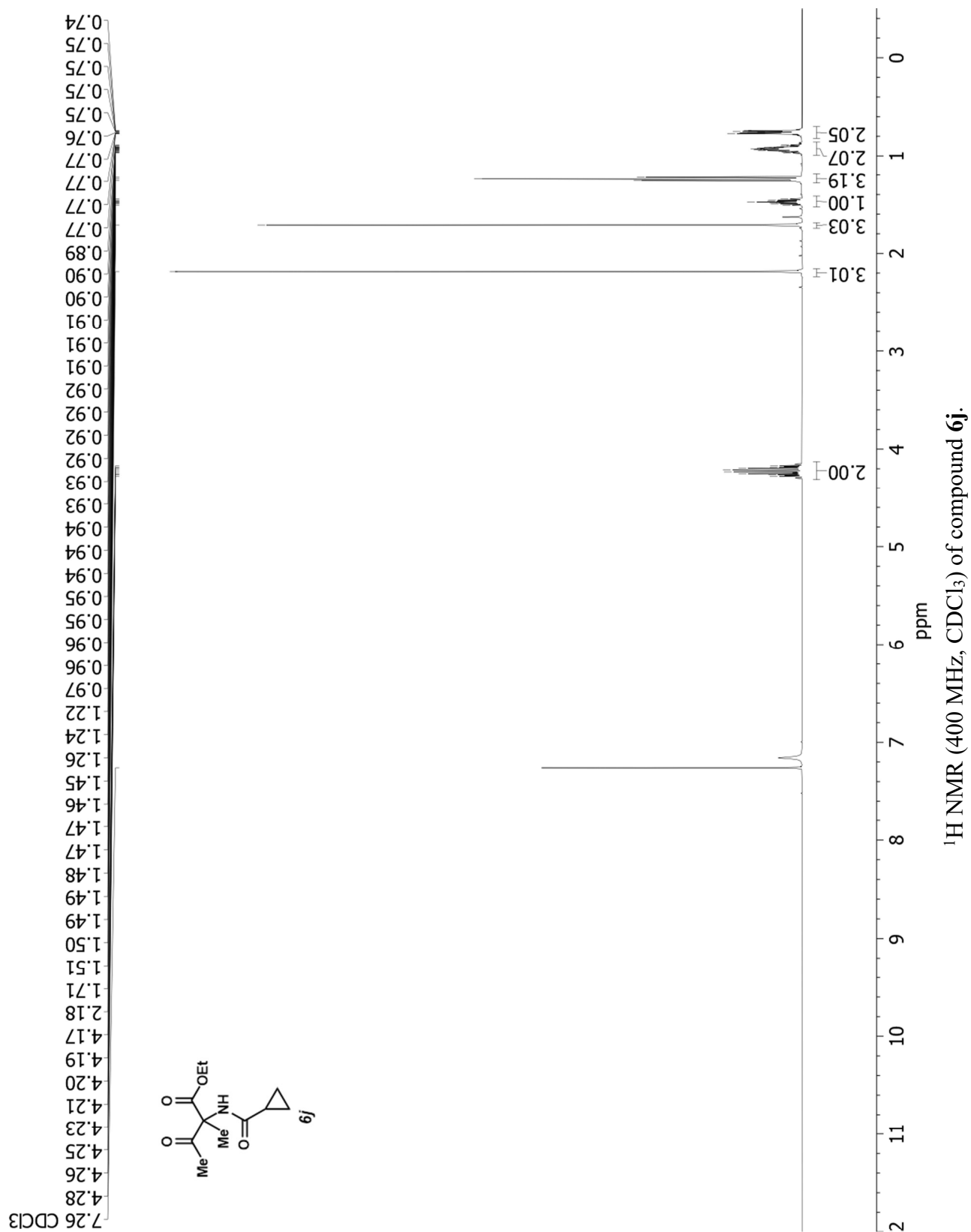
Infrared spectrum (Thin Film, NaCl) of compound **6g**.<sup>13</sup>C NMR (100 MHz, CDCl<sub>3</sub>) of compound **6g**.

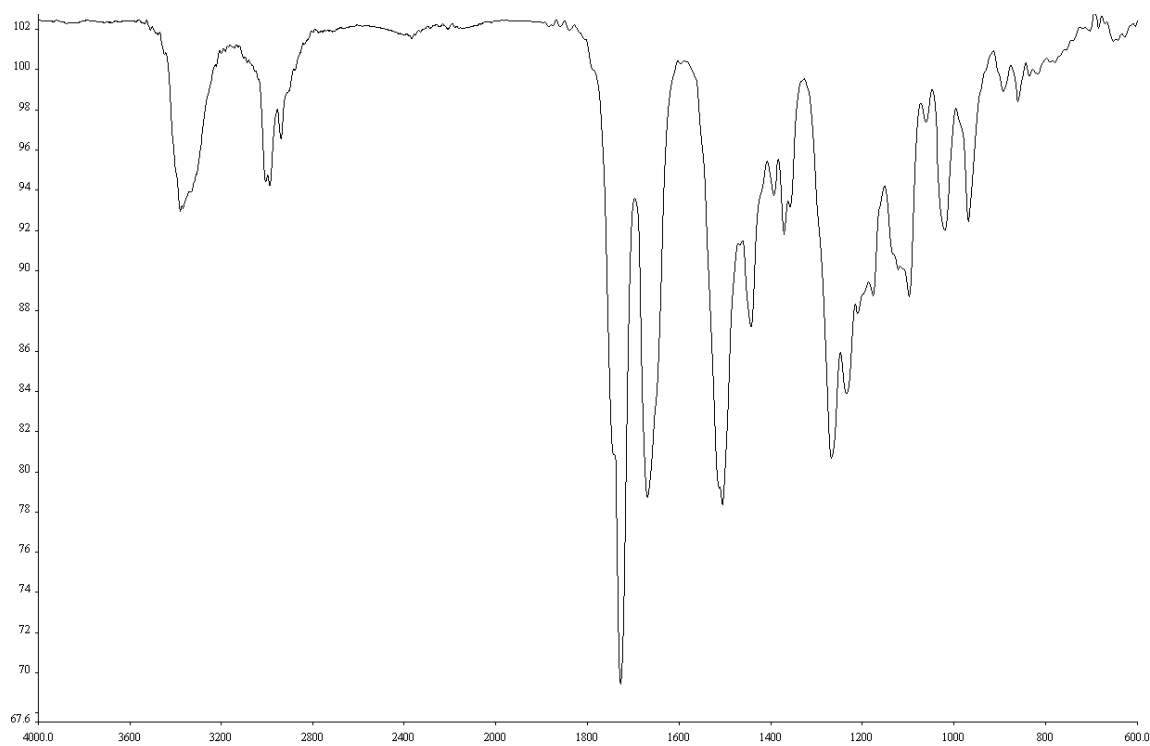
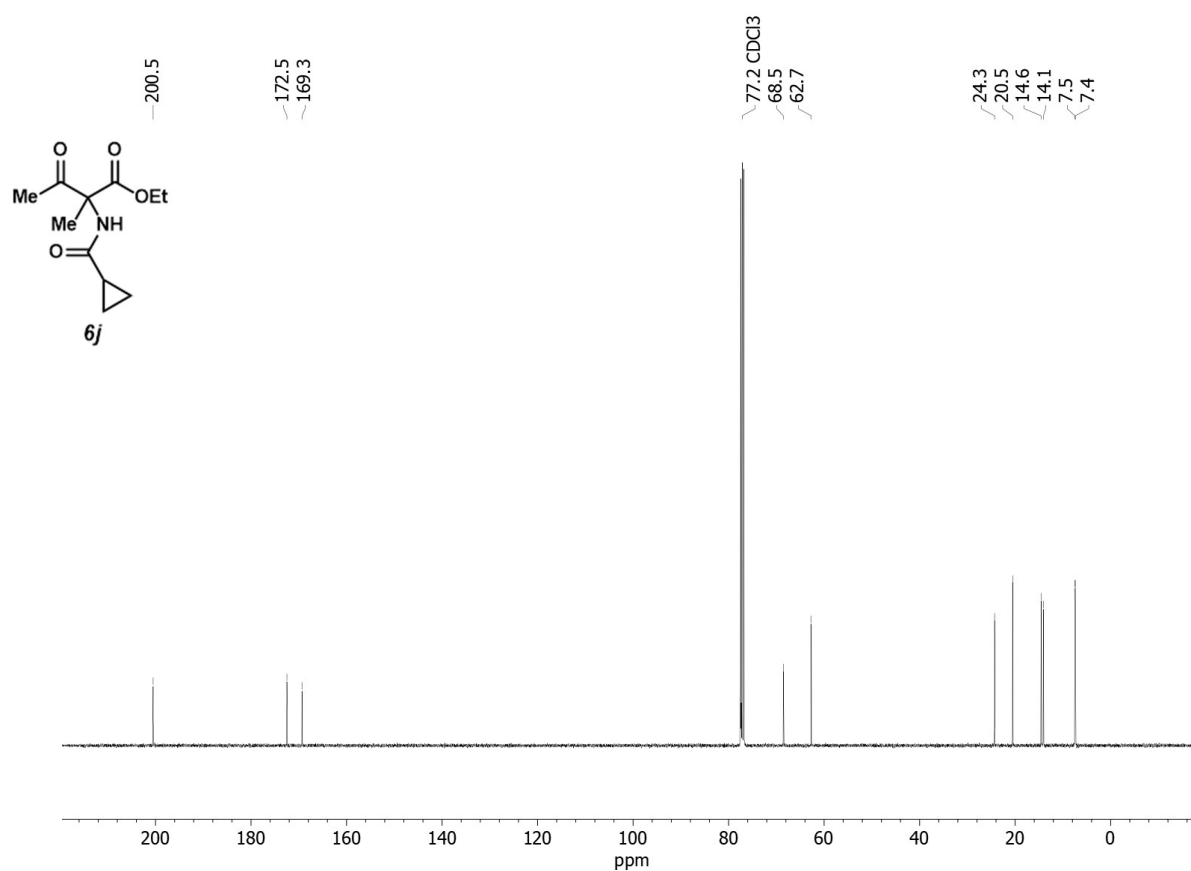


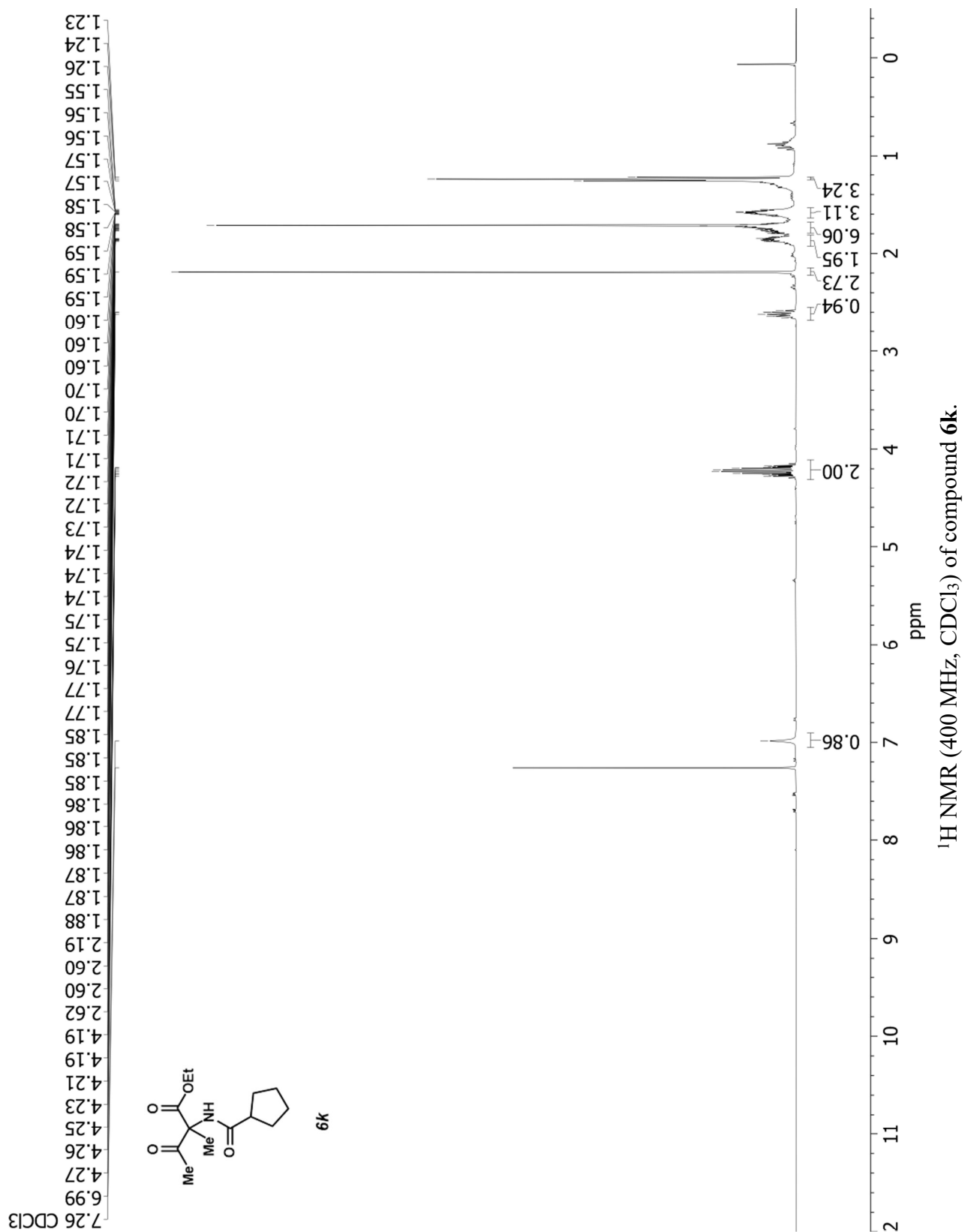
Infrared spectrum (Thin Film, NaCl) of compound **6h**.<sup>13</sup>C NMR (100 MHz, CDCl<sub>3</sub>) of compound **6h**.

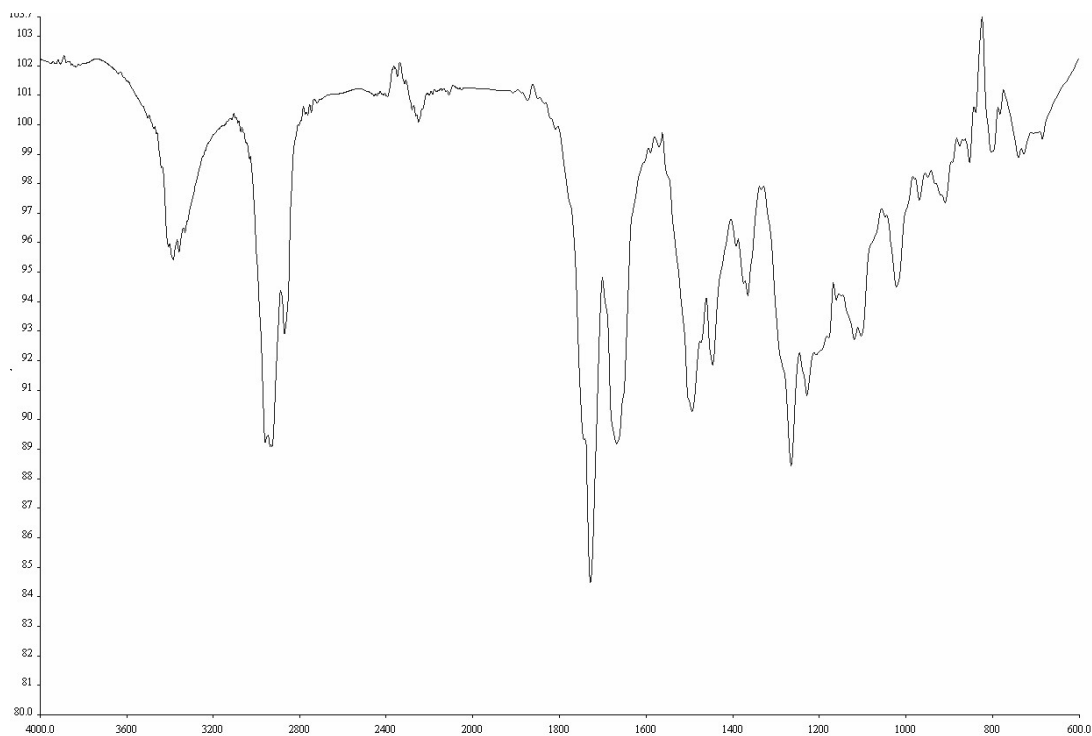
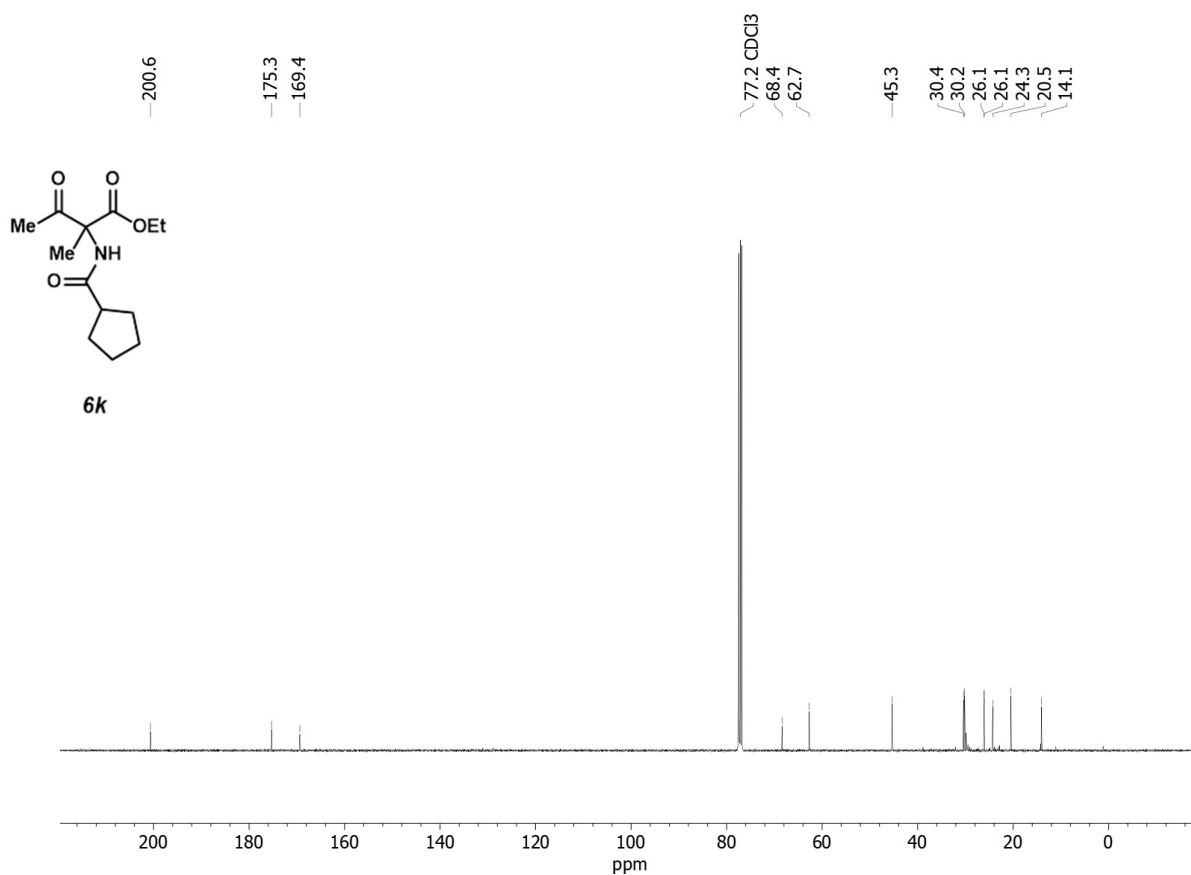


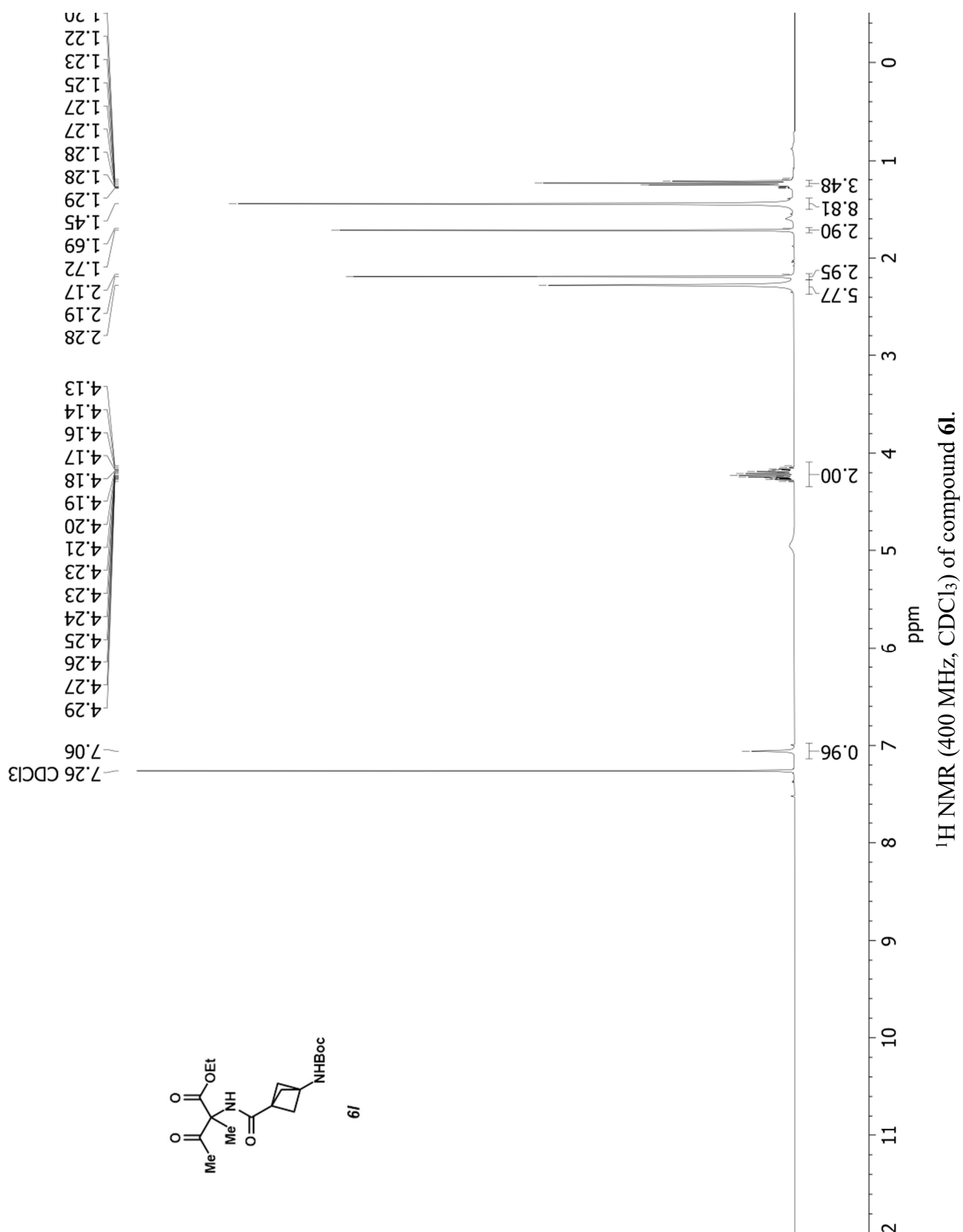
Infrared spectrum (Thin Film, NaCl) of compound **6i**.<sup>13</sup>C NMR (100 MHz, CDCl<sub>3</sub>) of compound **6i**.

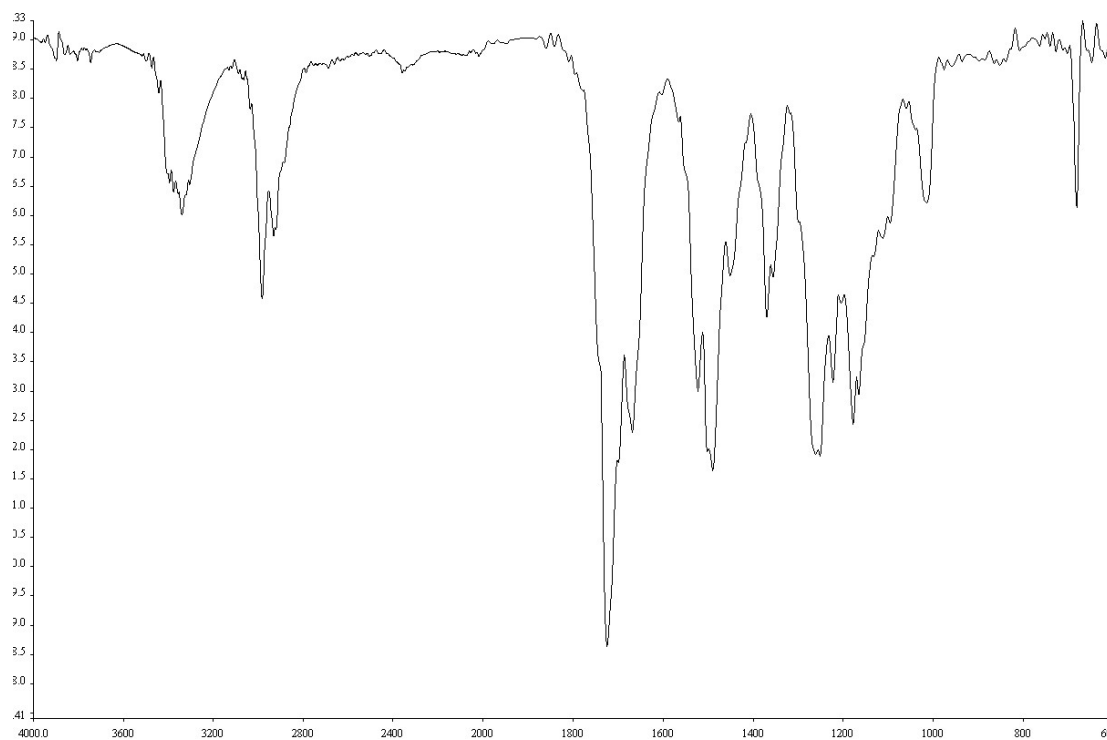
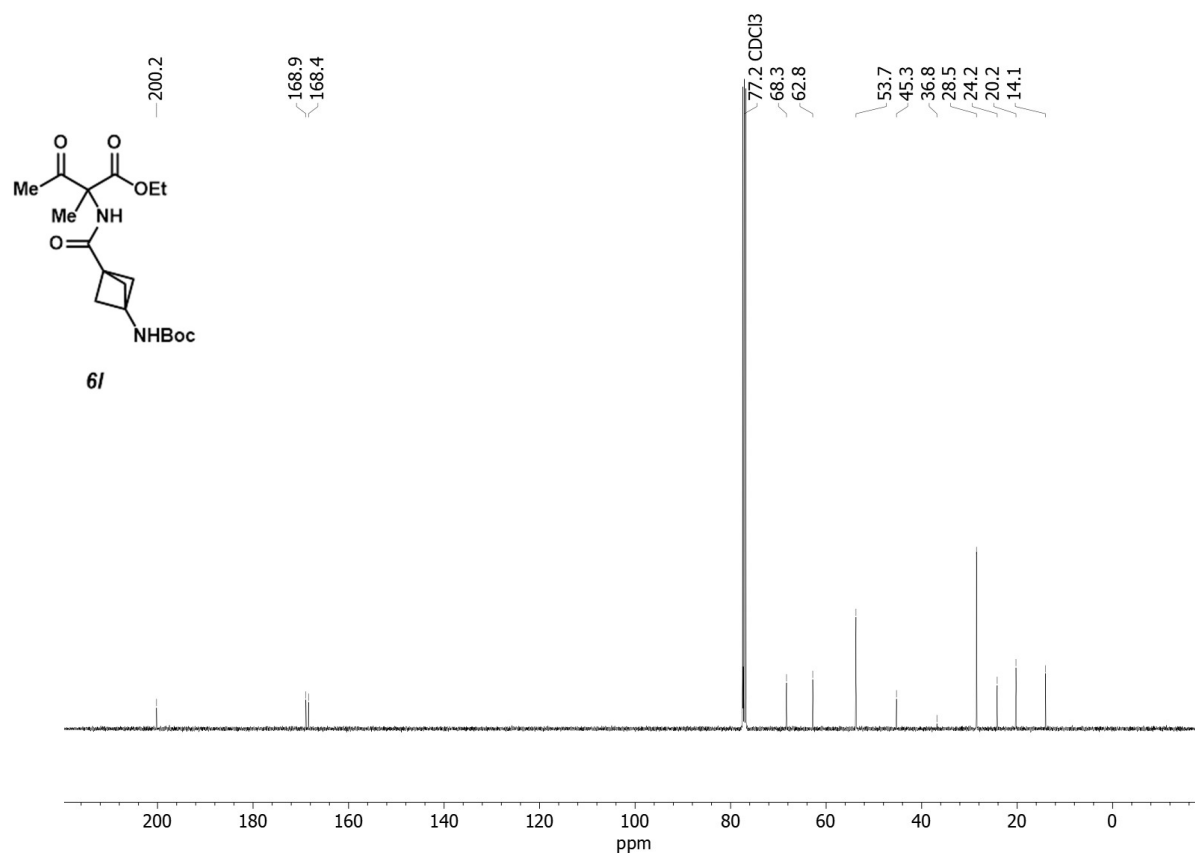


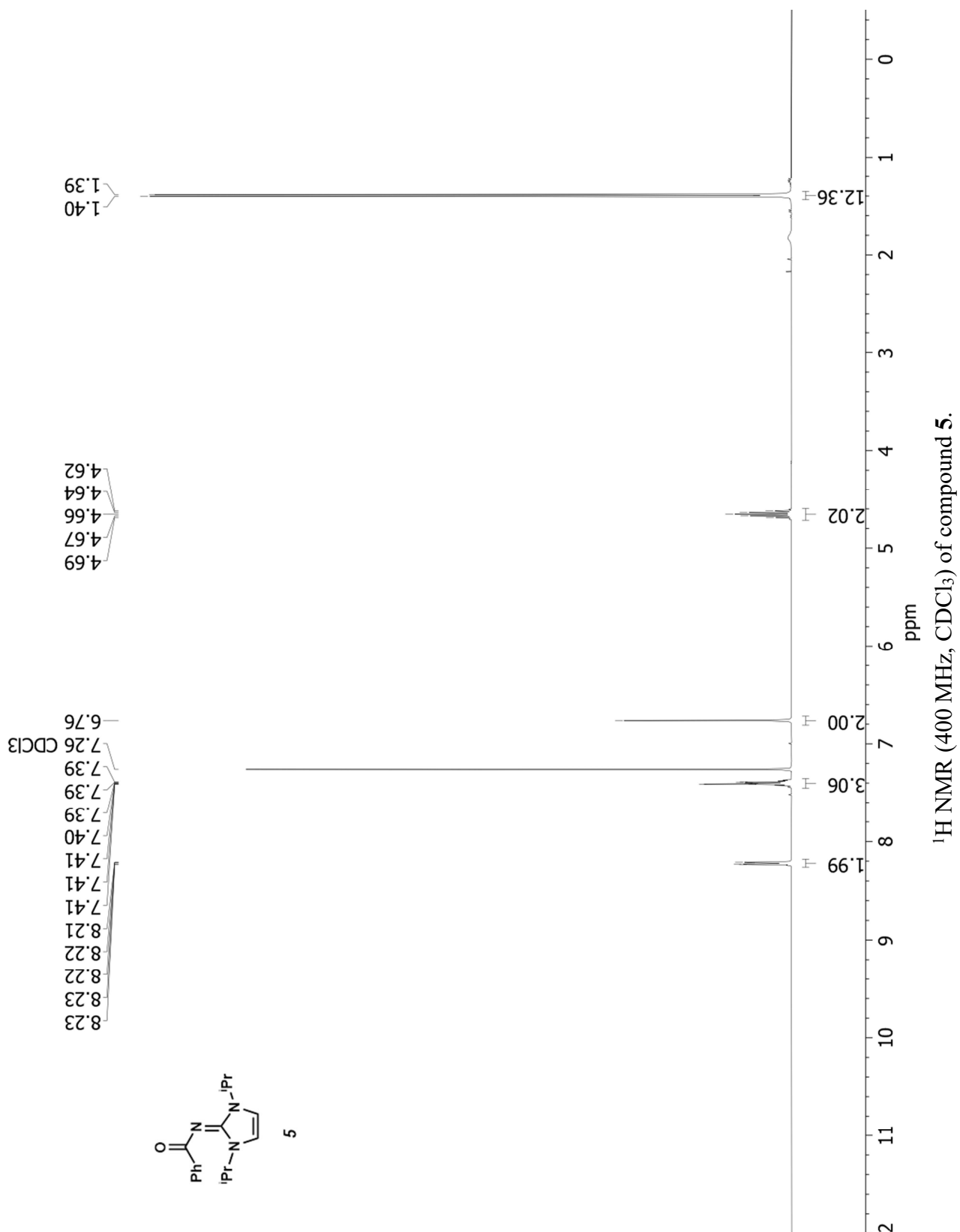
Infrared spectrum (Thin Film, NaCl) of compound **6j**.<sup>13</sup>C NMR (100 MHz, CDCl<sub>3</sub>) of compound **6j**.

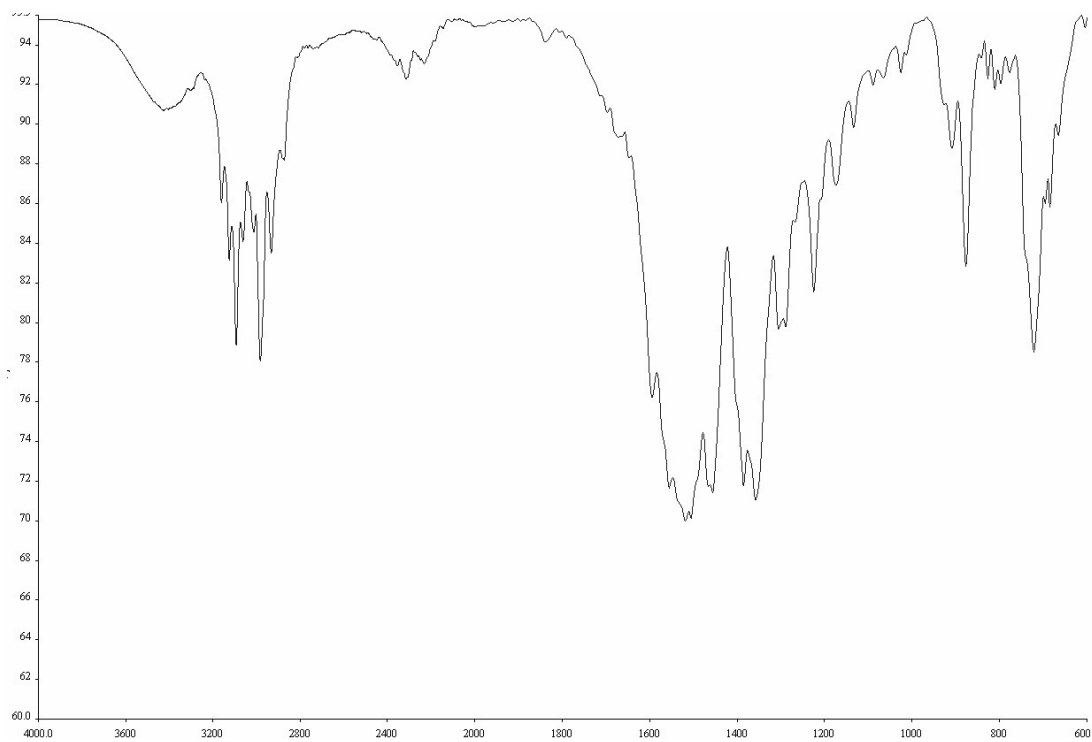
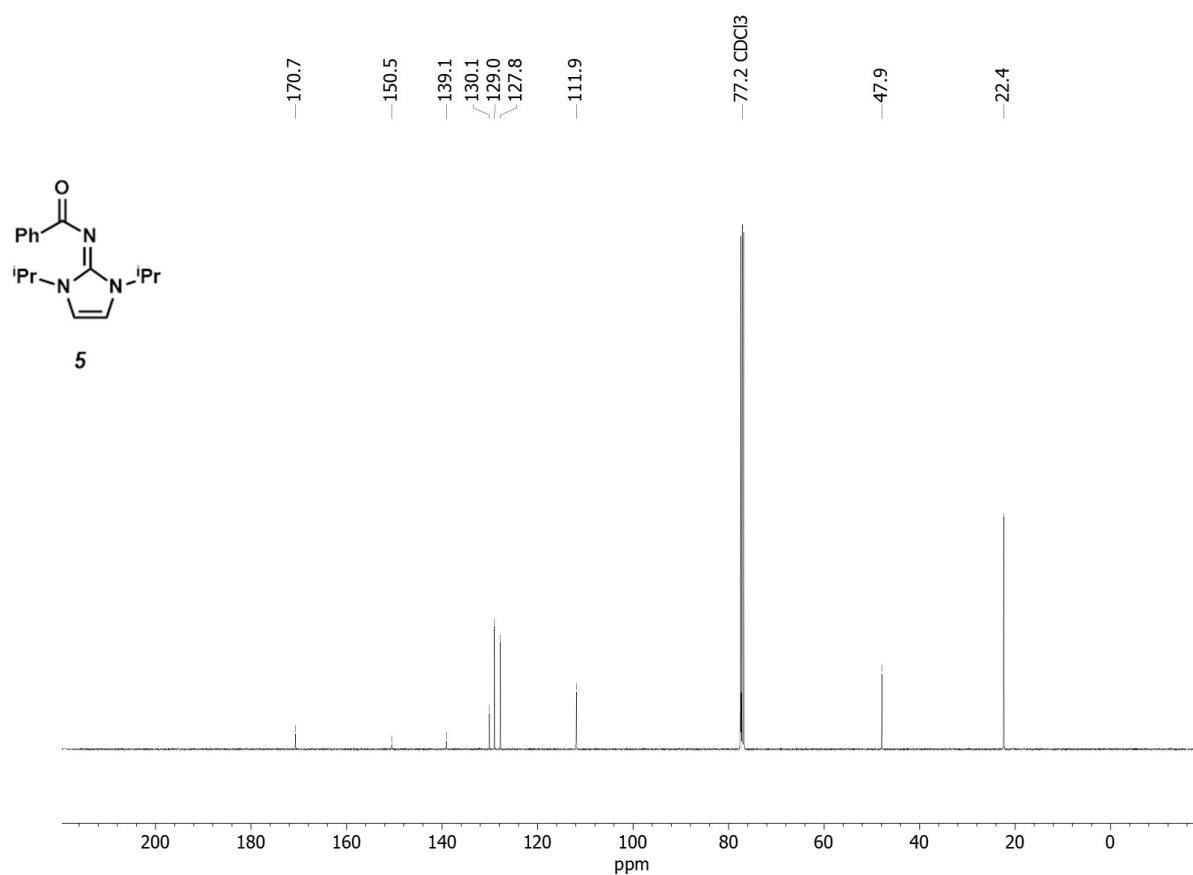


Infrared spectrum (Thin Film, NaCl) of compound **6k**.<sup>13</sup>C NMR (100 MHz, CDCl<sub>3</sub>) of compound **6k**.



Infrared spectrum (Thin Film, NaCl) of compound **6I**.<sup>13</sup>C NMR (100 MHz, CDCl<sub>3</sub>) of compound **6I**.



Infrared spectrum (Thin Film, NaCl) of compound **5**.<sup>13</sup>C NMR (100 MHz, CDCl<sub>3</sub>) of compound **5**.

